Software Systems Development

AS 2

Event Driven Programming

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Introduction / Background

I was approached by Mr Denvir, Mrs Murphy, Mr Clearly and Mrs Breaky, All of whom are head of departments, at Saint Patrick's Grammar School, for (In order) Maths, History, Geography and Science, these teachers have asked to create a fun, educational quiz in order to help students better understand maths, history, geography and all 3 sciences.

The teachers have come to me because the students that they teach are struggling in their class and don't seem to understand the content that is currently being taught and so they have asked me if I can help these students by creating a fun addicting game which leads them to a quiz to learn the content of the subject and to have a better understanding of the subjects.

The addicting game that I will design will meet the needs of teachers as if a the student losses a life they will be prompted a question, the subject of such question will be decided on what subject they are struggling on if the answer has been entered correctly then the student will resume his game until he loses another life, on the other hand if the student answers incorrectly then the student will lose another life and a new question will be given, if this continues then the student will lose the game.

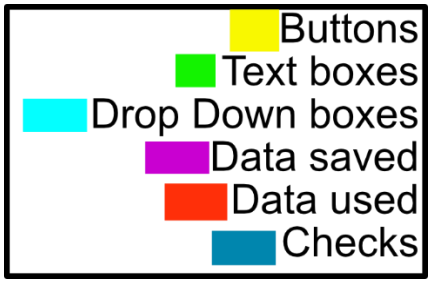
My addicting game will help the students understand the subjects that they are struggling with by giving different questions each time they play the game they will lose a life to make the game a bit more challenging but more brain storming anytime. Additionally the answers for the question will switch positions making the student think more before they answer.

For the quiz it will be based on what subject has been chosen when the user has first registered, this is so the user can select the subject that they are struggling with most rather than having a single subject based game. Along with the difficulty levels for the game, the difficulty of questions will become easier or harder along with more difficult with more confusing answers, and once the user is answering a question I do not plan to have any sort of image or video on the question form.

The user would interact with my addicting game and would be able to move in all 4 directions as well as shooting. The quiz, which follows, will have 4 possible answers present with only 1 of them being correct.

The mechanics of my addicting game will include moving about using WASD or the arrow keys and shooting using the space bar. For the quiz element you'll be able to using 1 of the 4 buttons provided to answer a question that appears.

User Requirements



Functional Requirements

1.Buttons must bring the user to a new form

2.buttons must bring the user back to a previous form

3.There must be a button for the user to log out.

4.Whilst in the game, the user must be able to pause the game

5.The user must be able to return to the main menu.

6.1 of 3 difficulties must be chosen via buttons before playing the game

7.1 of 4 Buttons must answer questions

8.There must be a submit button for maths questions

9.The user must be able to enter the answers for the maths questions

10. The user must be able to enter a password and or re-enter password

11. The user must be able to enter their username.

12.The user must be able to choose the year group that they are currently in.

13.The user must be able to choose their favourite subject.

14.The users account details must be saved.

15.The game and quiz score details and about the player must be saved.

16. with the saved user details, the user must be able log in

17.with the saved game and quiz details and about the player must be shown for the scoreboard

18.The top 3 players must be on the podium.

19.pre-saved questions(for subject and difficulty) must be shown

20.the password and re-entered passwords must match or else the user will be told to that they do not match

21.Each username must be unique and if not the user will be told that they must enter a different username

22. Check each account must be unique and if not the user will be told that the account already exists.

23.check that everything In the register form must not empty and has been filled out.

24. check that the username must contains no capitals, be more than 6 letters and less 30.

25. check that the password must contains at least 1 capital, a number and have a length of 10 and less than 30.

26.the log in fields must be checked to ensure that they are not empty.

27.Game must check that there is a difficulty being used.

28.Quiz must check that there is a difficulty being used.

29.Quiz must check that there is subject to use for the questions.

30.A loading bar must actively increase/decrease.

31.A timer that must take away 1 second.

32.each enemy killed must earn 1 game score point. Score must update.

33.if the user has ran out of ammo, ammo must spawn. Ammo must update.

34.if the user is below a certain health, health packs must spawn. Health must update.

35.enemies must spawn until they meet a certain number(dependent on difficulty)

36.user must use WASD or the arrow keys to move about

37.user must use the space bar to shoot in the current direction they are in.

38.Help forms for all forms must play a video upon loading into the form.

39.The user must see if they'd like to see easy, normal or hard player states via pressing buttons

Non-Functional

40.Each form title must be presented at the top in a big font size

41.For entering data the text for each box must be accurately labelled.

42.help buttons must be in the corners.

43. there must be sprite images for moving images

44.everything must start centred

45.there must be image(s) of the enemy

46.there must be image(s) of the player

47.must be an image on the quiz for each subject.

48.there must be an image on the game form for the heals, ammo and bullets

49.there must be an image for the podium on the scoreboard form.

50.there must be an image of an arrow pointing to a button.

51.Simple instructions must be underlined

52.There must be Instructions on how to play the game

53.There must be image of game controls

54.There must be GIF as an image of the game

55.There must be Difficulty Images

56.there must be description of difficulty

57.there must be correct naming of buttons.

Storyboards draft 1

Key

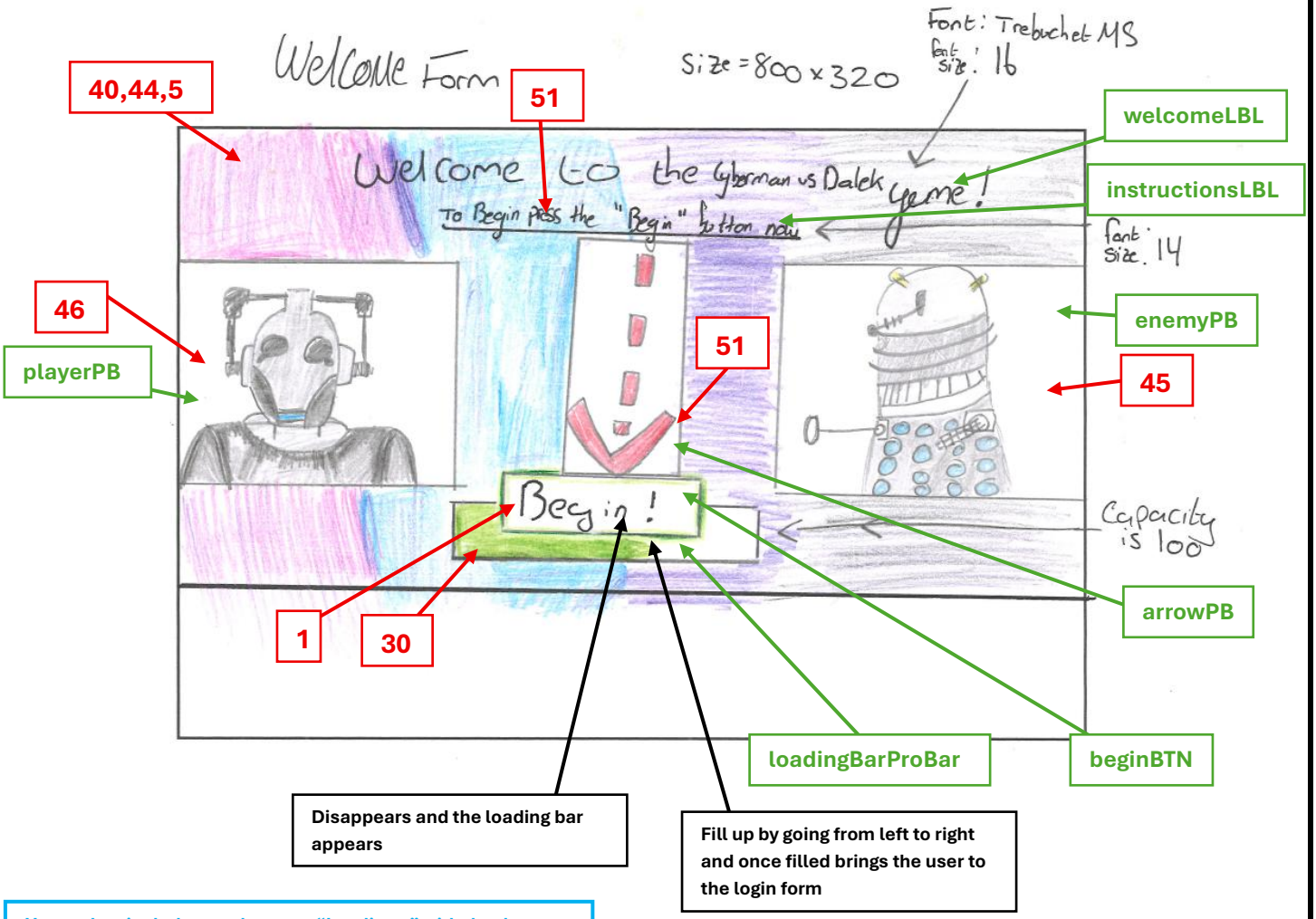
User Requirements

Names of elements

Function

User Feedback

Welcome Form

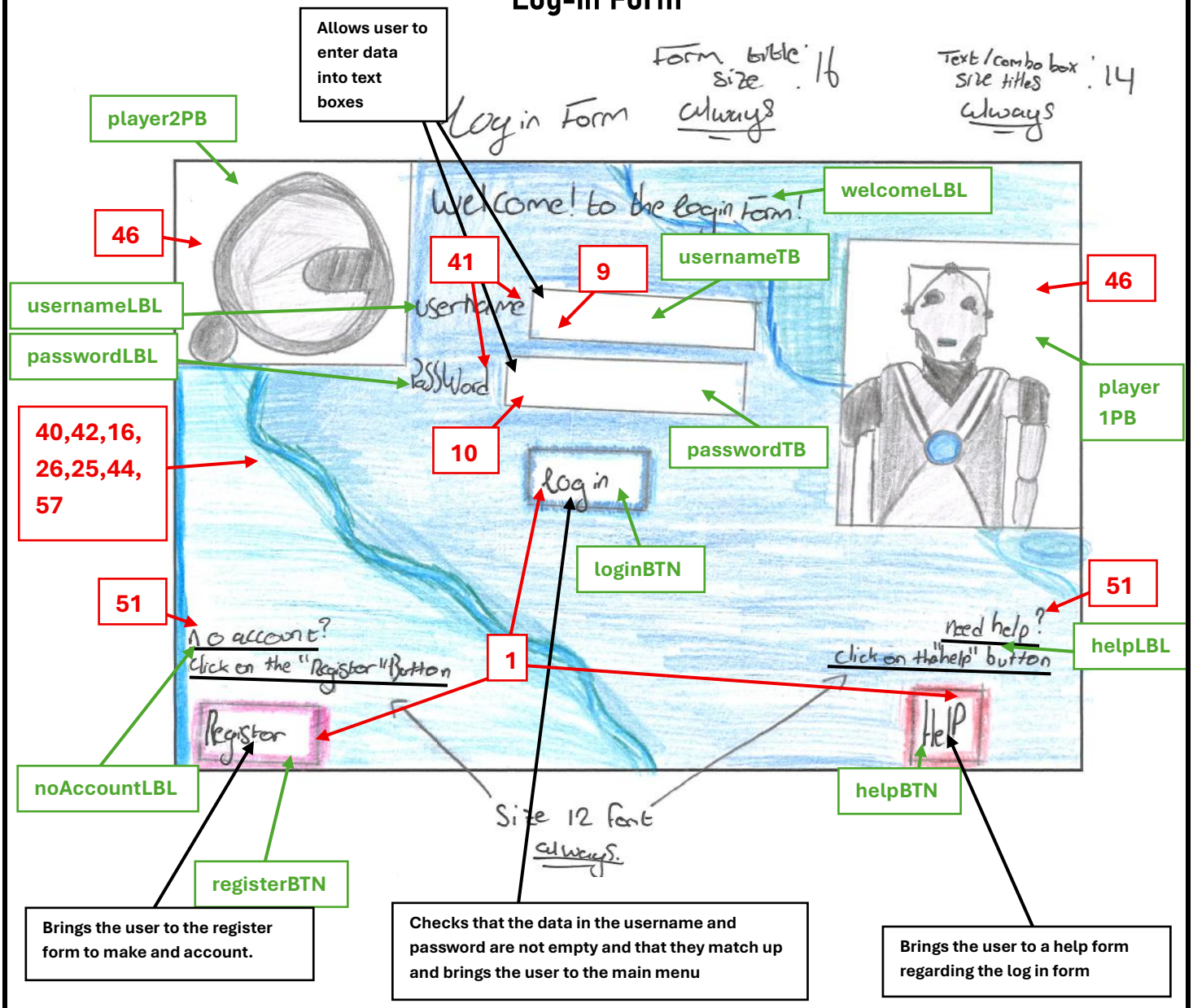


User ask to include text that says “Loading…” with the dots appearing and disappearing as they said it would feel more like a game

The user has said that the text looks plain and has asked to make the text more accurate towards the game

The user likes the use of showing who you will be playing the game as and who your enemy is.

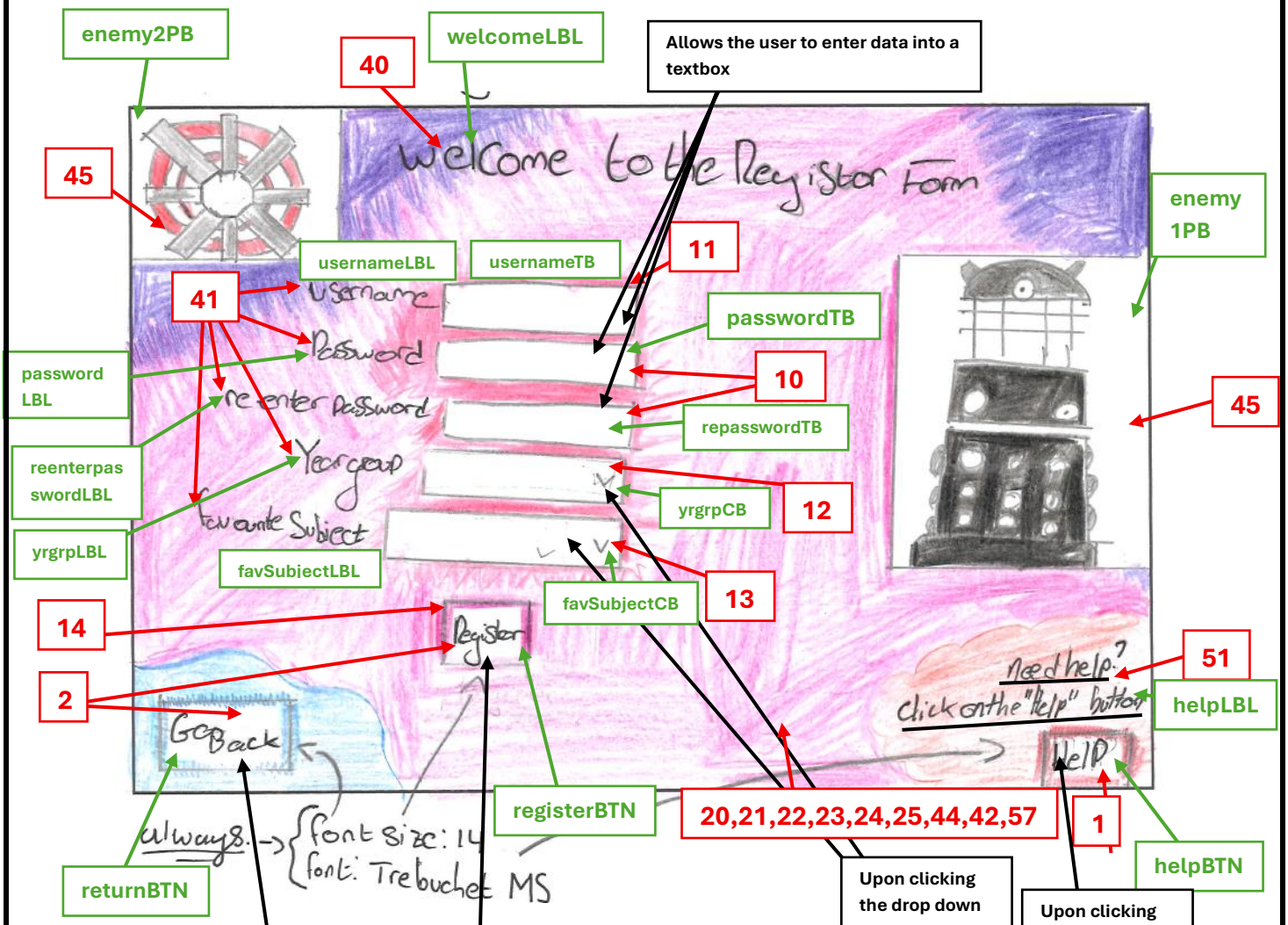
Log-in Form



User ask to include privacy for entering in their password

The user has found it helpful that the boxes for the password and username have been labelled

Sign up Form

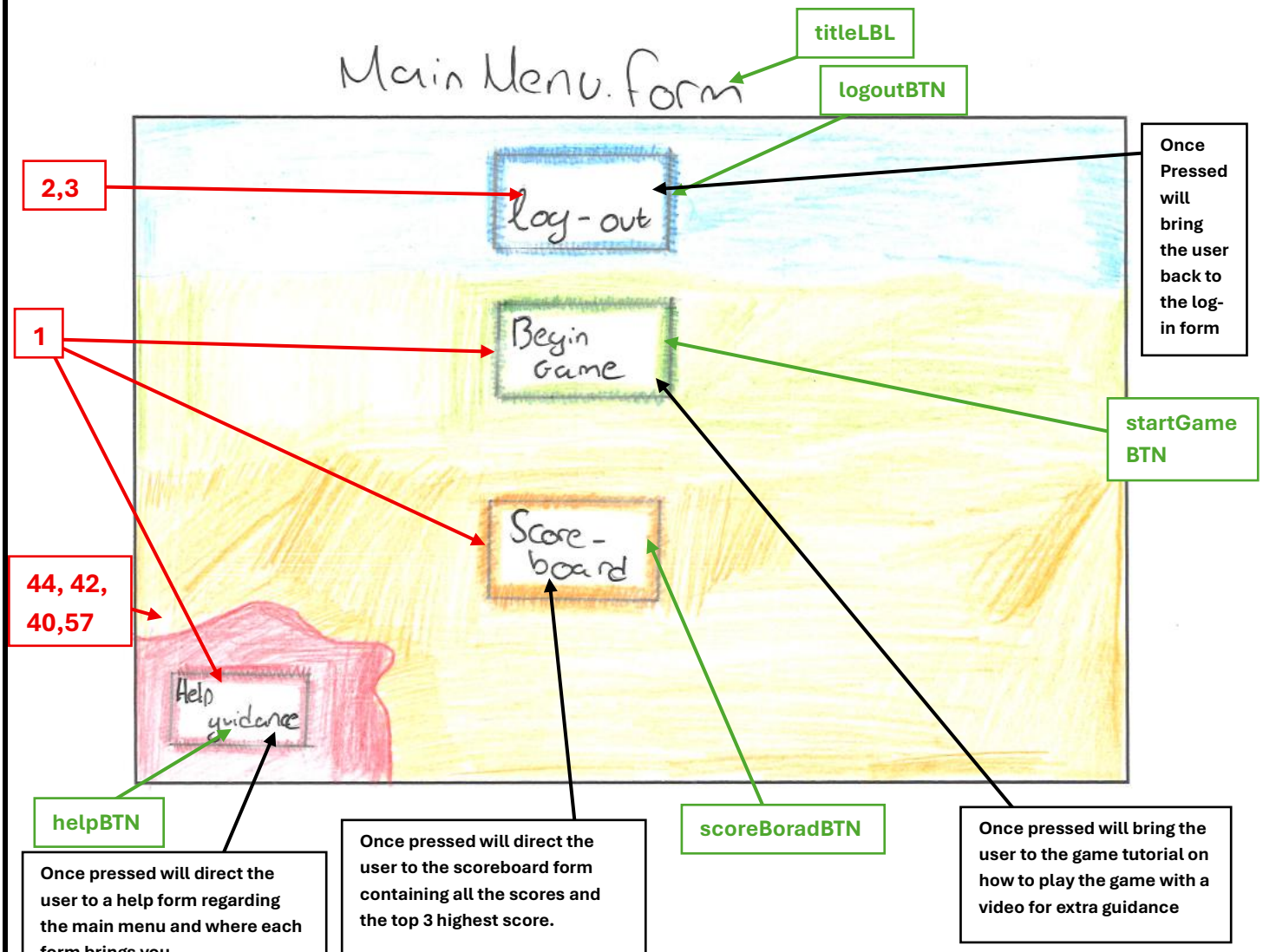


User did not understand why after signing up it would bring you to the log in form and so, they have asked that it brings you to the main menu

The user liked that I have included an option to go back to the log in form and found it helpful

Similar to the feedback from the log in form, the user has asked that the password and re-enter password be hidden for privacy

Main Menu



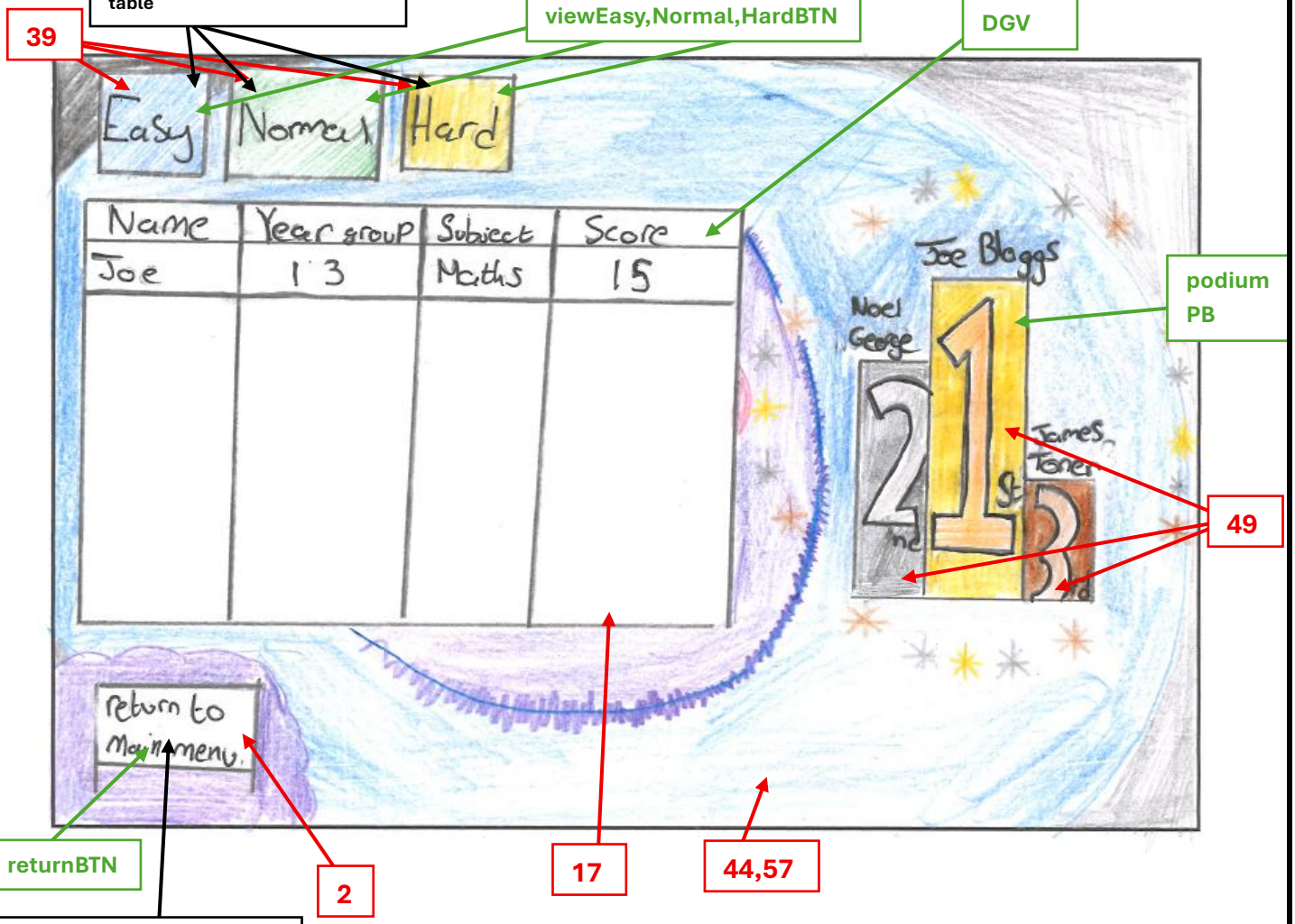
The user has asked, that the name they used to login, to appear on the main menu indicating that they have logged into the correct account.

The user has said that they found it thoughtful that there were was a confirmation to log-out to prevent accidently logging out.

The user has asked that on the button for the scoreboard also says "& leader board" they said that they got confused thinking that they were entering the scoreboard, but instead were entering the scoreboard and leaderboard.

Once one of these buttons are pressed, that button will be disabled and only that buttons difficulty will be shown on the table

Scoreboard & Leaderboard



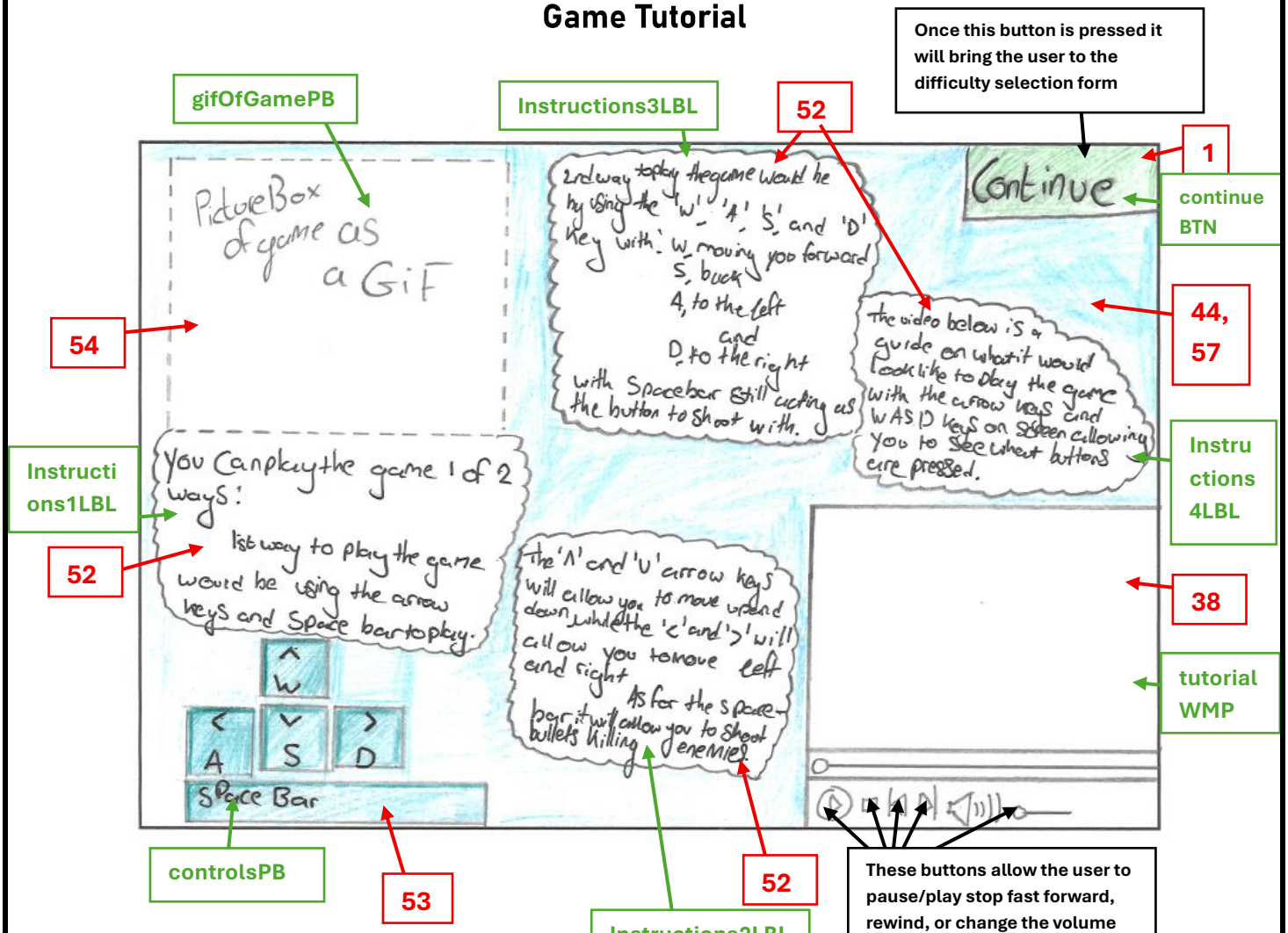
Once this button is pressed it will bring the user back to the main menu

The user has asked I implement a system to individually search each column by pressing a button which indicates each column, after which it disables, and by searching up something relating to the column it will appear. E.G. if name was pressed and "M" was entered it would display names beginning with M.

The user asked me to include a button which showed all the difficulties as go out of the form and returning seemed inefficient.

The liked how I have included a leaderboard to rank the top 3 best players and it also gave the user determination to play the game, the user also did not like how only 3 players were shown and has asked that all the players ranked from highest to lowest are shown on the table which can be shown by pressing a button called leaderboard.

Game Tutorial

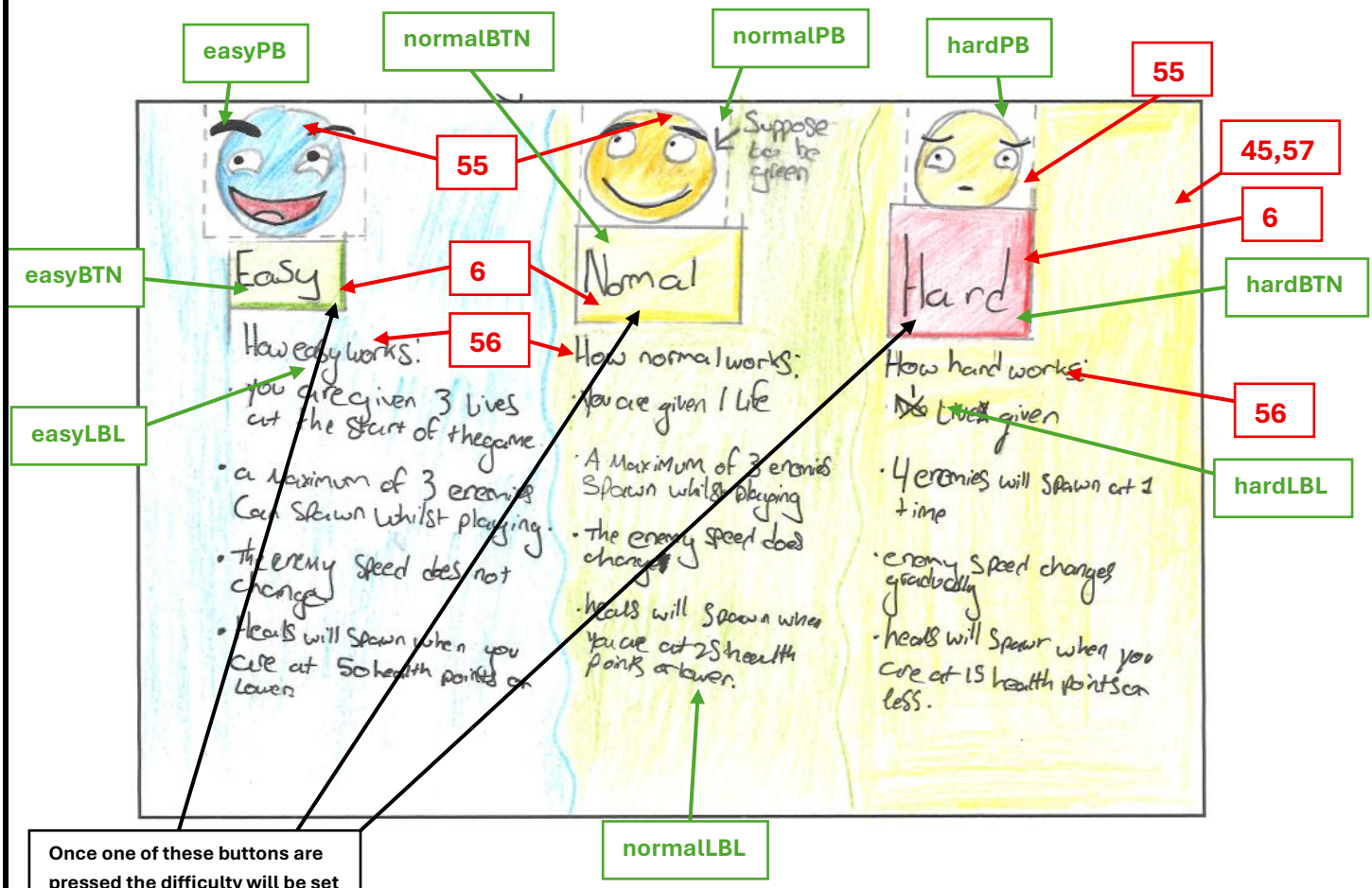


The user found it helpful that I had included notes on how to play but also a video of the game being played with each key being shown and pressed to the correct movement of the player.

The user found it use that I had included a gif of the game of the players movement and shooting mechanic, this allows the user to study the layout of the game before playing.

The user asked if I could add in a return button in case of accidently clicking on the game button with the only way of getting back via playing the game.

Difficulty Selection



Once one of these buttons are pressed the difficulty will be set depending on the difficulty selected.

When the user is selecting a difficulty they found it that there was no confirm button for the difficulty to help this I will add in a start game button will be disabled until a difficulty is selected, this will allow them to change difficulty if they so wish to do so.

The user liked how I included images of how I described each difficulty, as well as including what the difficulty changes in the game.

The user found it odd that I didn't include the title of form even though it is visually knowledgeable what the form is, so they have asked to add in the title to keep the over all program neat and tidy.

The user had suggested for me to give easy 5 lives and normal 3 lives, as well as giving 5 enemies on normal and 7 on hard.

The daleks will spawn until a certain condition has been met, the condition depends on the difficulty selected

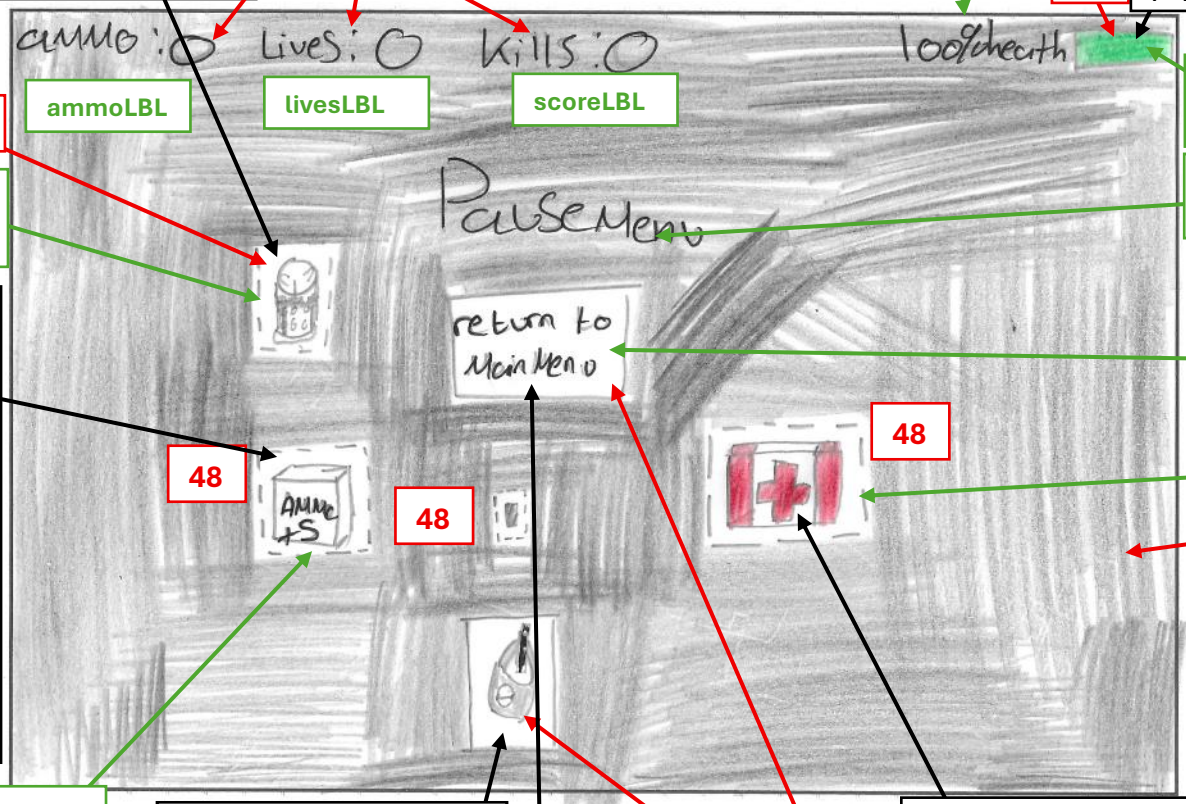
32,33,34

Game

healthLBL

30

The health will decrease when the dalek attacks the player



ammoLBL

livesLBL

scoreLBL

43,45

enemy PB

health Bar

pause LBL

return BTN

Once the user drops to 0 ammo then this ammo pack will spawn adding 5 to the ammo

48

48

48

healPB

27,32, 33,34, 35,44, 57

ammoPB

User can pressed the WASD or Arrow keys to navigate and play the game, with pressing the space bar to shoot.

36,37,43,46

5,4

Once pressed will bring the user back to the main menu. Any score gained WILL NOT be saved

Once the user drops below 50 health then these heals will spawn till the users health is more than 50.

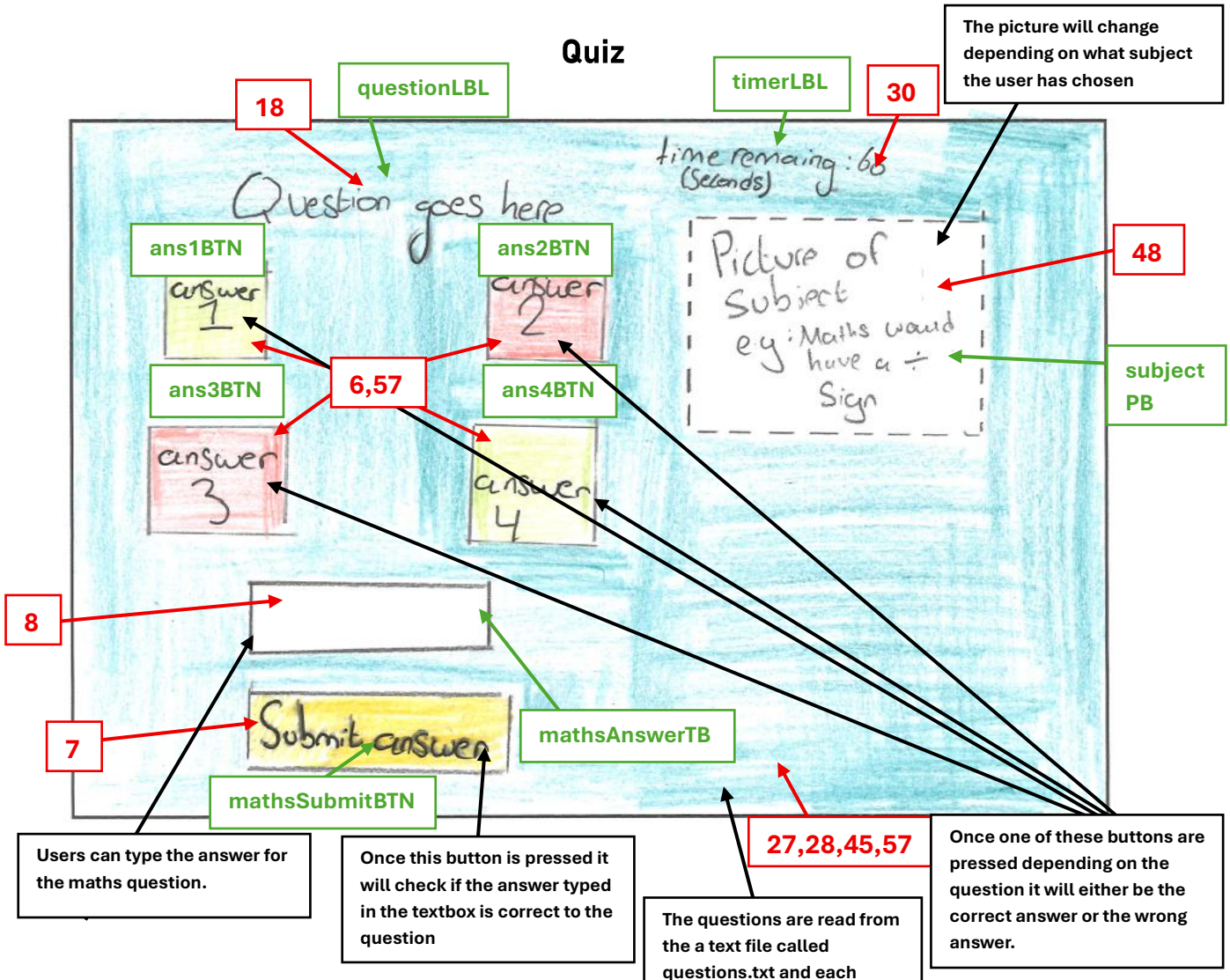
The user did not like how plan the background/ ground of the game was as it seemed boring with not a lot of colour in, they have asked me to change the background/ ground to a city with building in the way restricting the players space as a way of adding obstacles in the game.

The user did not like how plain the game looks and has asked that I can in some colour, they have suggested adding colour round the ammo, lives and kills text to give the game some sort of a vibrant colour scheme rather than just a plan grey background.

The user liked having the ability to shoot bullets but they have asked me if it was possible that I could change the texture of the bullet and give it a laser 'futuristic' look,

The user had also recommended that instead of spawning health packs I could instead spawn some sort of shield for the player to use with the same visual identification as the health, they said I did not have to add it if I didn't want to and that it was just suggestion.

The user has asked if I could add in a laser shooting sound when a bullet is fired this is done so it further gives the user the experience of playing a game of which is enjoyable and educational.



The user found that when they finished the quiz that did not know if they had won or lost so to fix this I will implement a win lose form that will notify the user if they have won or lost.

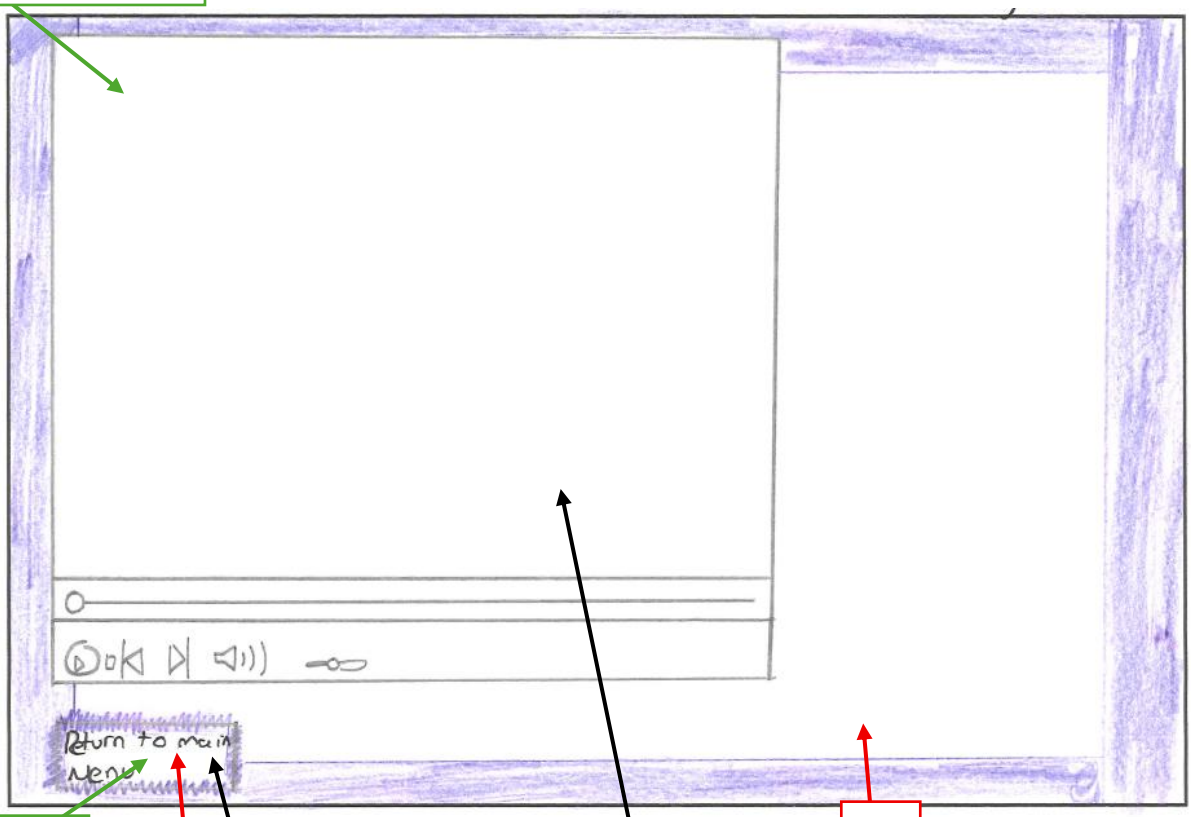
The user liked how I have included multiple choice and typing questions, they have asked if I can make other subject questions typeable for a more variety of question types.

The user liked how I have included a picture of the subject what the questions are currently being based of off, as they said that it helps them figure how what subject questions are currently being asked.

Although the user knew where to answer the maths questions they have asked to include some text inside of the textbox that tells the user where to answer maths questions.

Template of all Help forms

helpVideoWMP



mmBTN

2

Once this button is pressed it will direct the user back to the previous form, in this case its going back to the main menu.

Displays a video on how to use the form, in this case its main menu, and it will take you through where each button brings you.

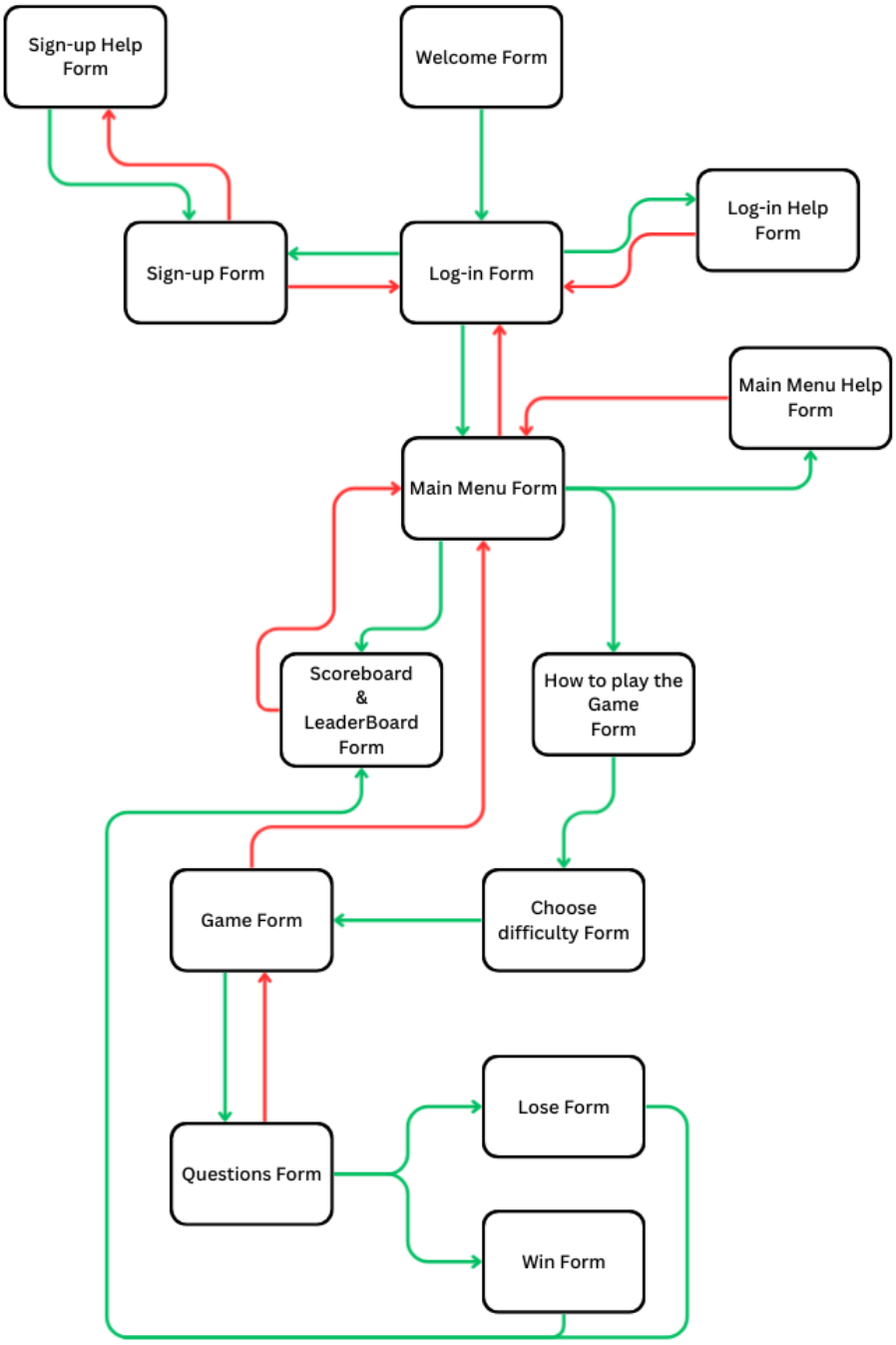
37

The user liked how I had included a video on how to navigate/ use the form and said they will find it helpful.

The user did not like how plain the form looked as all there is, is a single colour border, they have asked for me to include a photo that can be relevant to the game and or include a character that is part of the show of which the game is based of off.

The user has asked for me to include music in my whole program to give it more of a 'game' type feeling rather than just an application that is a game.

Wire Frame of Program

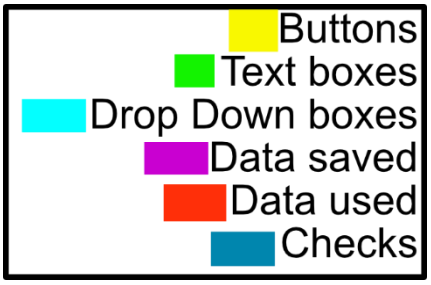


Green indicates bringing the user to a new form

Red indicates brings the user back to a previous form

Final User Requirements

Functional User Requirements



1.There must be a buttons to navigate to and from forms

~~Buttons must bring the user to a new form~~

~~Buttons must bring the user back to a previous form~~

~~There must be a button for the user to log out.~~

~~The user must be able to return to the main menu.~~

2.1 of 3 difficulties must be chosen via buttons before you play the game.

3.1of 4 buttons must answer questions.

4.There must be a submit button for math's questions

5.user must be able to choose one of 5 buttons for searching through the scoreboard and leaderboard

6.The user must be able to type what they want to search in the scoreboard and leaderboard.

7.The user must be able to enter the answers for the math's questions

~~The user must be able to enter a password and or re-enter password~~

8.The user must be able to enter their username and password

9.The user must be able to enter their password again.

~~The user must be able to choose the year group that they are currently in.~~

~~The user must be able to choose their favorite subject.~~

10.The user must choose their year group and favorite subject from a drop down box.

11.The users account details must be saved.

~~The game and quiz score details and about the player must be saved~~

12.The game and quiz score details and about the player must be saved upon wining the game or losing the game

13.Any previous data only the scores, difficulty and number of questions will be overwritten.

14.The saved user details must be used when logging in.

15.The saved user details, quiz and game score must be shown for the scoreboard.

16.using the saved quiz and game score it must display the leaderboard.

~~The top 3 players must be on the podium.~~

17.The pre-saved questions (for subject and difficulty) must be shown correctly.

18.The password and re-enter password must match or else the user will be told that they do not match.

19.The username must be unique and if not the user will be told that the username must be unique.

20.each account must be unique and if not the user will be told that the account already exists.

21.The data required in the registration form must not be empty.

~~the username must contain no capitals, be more than 6 letter and less than 30.~~

22.The username must contain no capitals, only 3 numbers, and have more than 6 letters and less than 30.

23.The password must contain 1 capital, a number and have a length of 10 and less than 30.

24.data required for the log in must not be left empty.

~~Game must check that there is a difficulty being used~~

~~Quiz must check that there is a difficulty being used.~~

25. A difficulty must be used for the game and quiz

26.A loading bar that actively increases and decreases

27.A timer in the quiz must count down by 1.

28.The length of the timer will be dependent on the difficulty selected, Easy is 30 seconds, Normal is 60 seconds and Hard is 90 seconds.

29.each enemy killed earns 1 game score, the game score must be updated correctly.

30.is the ammo reaches 0 then ammo must appear and once collected gives the user 5 ammo.

31.Health packs must spawn once the user reaches 50 or below health.

32.Enemies must spawn until they meet the correct condition set.

33.User must either use WASD or the arrow keys to move around.

34.The user must use the space bar to shoot in the direction they are currently in.

35.Help forms must play a video on the login, sign up, main menu and game and quiz form.

36.The user can press 3 buttons to see what level of difficulty they can choose from to view in the score board

37.upon completing the game, depending on the difficulty selected, you must be brought to the quiz.

38.upon completing the quiz, depending on if you achieved the correct score, you will either be brought to a win form or a lose form

39.if the user has lost a life, upon completing the quiz you must be brought back to game form where your life is 1 less than the original.

40.In the quiz, each difficulty must have their own amount of questions, easy will have 5, normal will have 10, and hard will have 15 questions

Non-Functional User Requirements

41.All forms must a have a title in big fonts

42.all boxes for entering data must be labelled correctly.

43.help buttons must be positioned in the corner of each form

44.there must be a up, down, left and right image for the player.

45.there must be a up, down, left and right image for the enemy.

46.the form must start centred

47.there must be image(s) of the player

48.there must be image(s) of the enemy

49.each subject for the quiz must have their own dedicated image

50.there must be images for heals, ammo and bullets

~~**There must be an image for the podium on the scoreboard**~~

~~**There must be an image of an arrow pointing to a button**~~

51.Any instructions must be underlined

52.There must be adequate and correct instructions on how to play the game.

53.there must be an image of the game controls.

54.there must be GIF of the game

55.there must be difficulty images

56.each difficulty must have a description

57.all the buttons must be correctly named.

58.music must be played in every form except for help forms

59.There must be text that says loading on the welcome form.

60.there must be an image on each help form.

61.All text must have a font that is suited to the game

62.There must be text for the questions.

63.text in boxes saying what to enter

Final User Requirements

Ranked from Most Important – Least important

1. all buttons must be correctly named.

Difficulty

2. 1 of 3 difficulties can be chosen via buttons before you play the game

3. A difficulty must be chosen for the game and quiz

4. The length of time timer will be dependent on the difficulty selected, easy is 30 seconds, Normal is 60 seconds and Hard is 90 seconds.

5. in the quiz, each difficulty must have their own amount of questions, easy will have 5, normal will have 10, and hard will have 15 questions.

6. there must be difficulty images.

7. each difficulty must have a description

Saving and displaying data

8. the users account details must be saved

9. the game and quiz score details and about the player must be saved upon winning the game or losing the game.

10.Any previous data only the scores, difficulty and number of questions will be overwritten

11.The saved user details must be used when logging in

12.The saved user details, quiz and game score must be shown for the scoreboard

13.using the saved quiz and game score it must display the leaderboard

14.the pre-saved questions (for subject and difficulty) must be shown correctly.

15.user must be able to choose of 5 buttons for searching through the scoreboard and leaderboard.

16.the user must press 1 of 3 buttons to see what level of difficulty they can choose from to view in the score board.

17.The user must be able to type what they want to search in the scoreboard and leaderboard.

Playing the game and quiz

18.1 of 4 buttons must answer questions

19.there must be a submit button for math's questions

20.The user must be able to enter the answers for the math's questions

21.a timer in the quiz must count down by 1.

22.each enemy killed earns 1 game score, the game score must be updated correctly.

23.if the ammo reaches 0 then ammo must appear and once collected gives the user 5 ammo

24.Health packs must spawn once the user reaches 50 or below health

25.Enemies must spawn until they meet the correct condition

26.User must either use WASD or the arrow keys to move around

27.the user must use the space bar to shoot in the direction they are currently in.

28.upon completing the game, depending on the difficulty selected, you must be brought to the quiz.

29.upon completing the quiz, depending on if you achieved the correct score, you must either be brought to a win form or a lose form.

30.if the user has lost a life, upon completion of the quiz you must be brought back to the game form where your life is 1 less than the original.

31.A loading bar that actively increases and decreases.

32.There must be adequate and correct instructions on how to play the game

33.there must be an image of the game controls

34.there must be a up, down, left and right image for the player

35.there must be a up, down, left and right image for the enemy

36.there must be images for heals, ammo and bullets

37.each subject for the quiz must have their own dedicated images

38.there must be text for the questions

Login and Sign up

39.The user must be able to enter their username and password

40.The user must be able to enter their password again.

41.The user must choose their year group and favourite subject from a drop down box

42.The password and re-enter password must match or else the user will be told that they do not match

43.The username must be unique and if not the user will be told that the username must be unique.

44.each account must be unique and if not the user will be told that the account already exists.

45.the data required in the registration form must not be empty

46.the username must contain no capitals, only 3 numbers and have more than 6 letters and have less than 30

47.the Password must contain 1 capital, a number and a length of 10 and less than 30.

48.the data required for the login must not be left empty.

49.text in box to say what to enter

50.there must be image(s) of the player

51.there must be image(s) of the enemy

Help forms

52.Help forms must play a video on the login, sign up, main menu and game and quiz forms

53.there must be an image on each help form.

54.There must be a GIF of the game

Visual Appearance and sounds

55.There must be buttons to navigate to and from forms

56.all forms must have a title in big fonts

57.all boxes for entering fata must be labelled correctly

58.help buttons must be positioned in the corner of each form

59.the forms must be centred.

60.Any instructions must be underlined

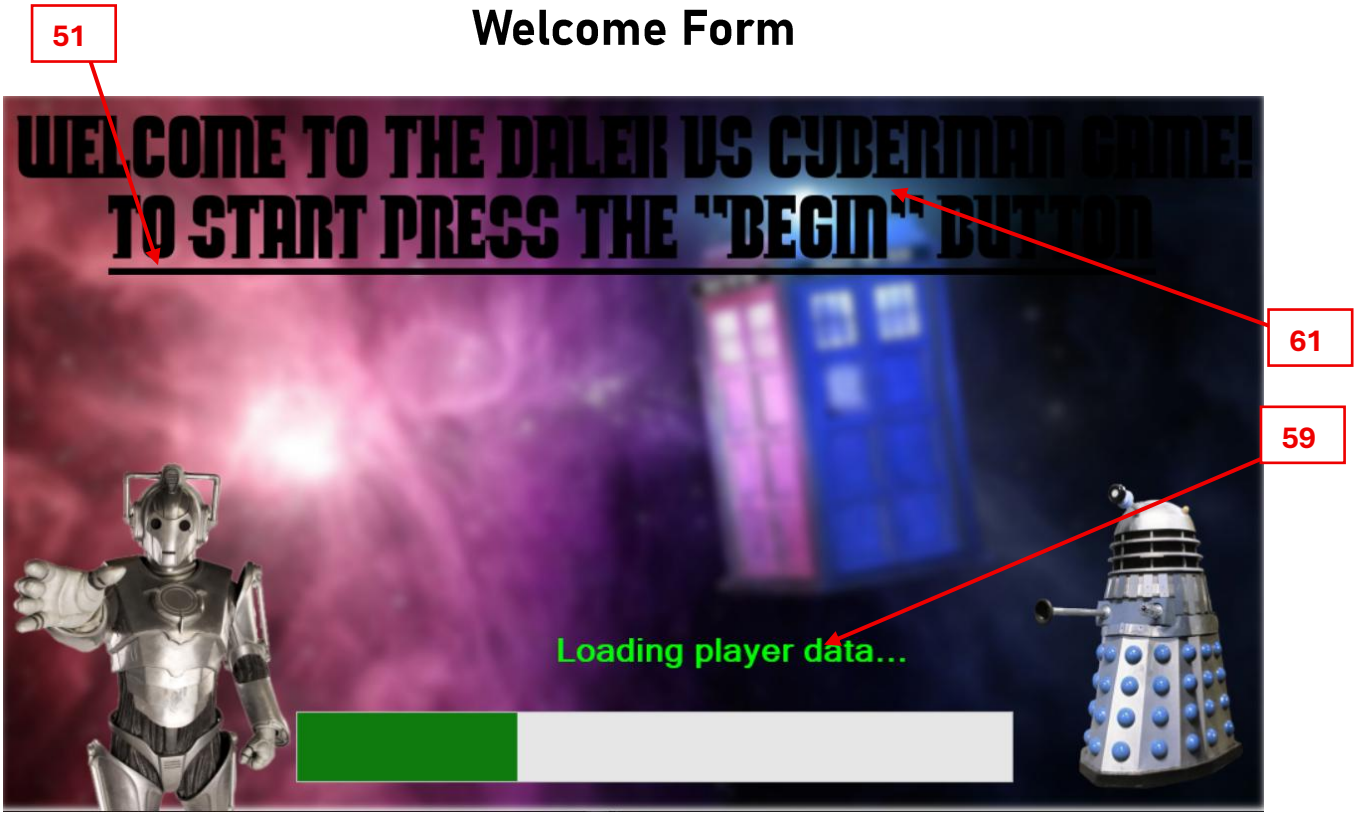
61.music must be played in every form except for help forms

62.There must be text that says loading on the welcome form

63.All test must have a font that is suited to the game.

Final Story Boards

Welcome Form

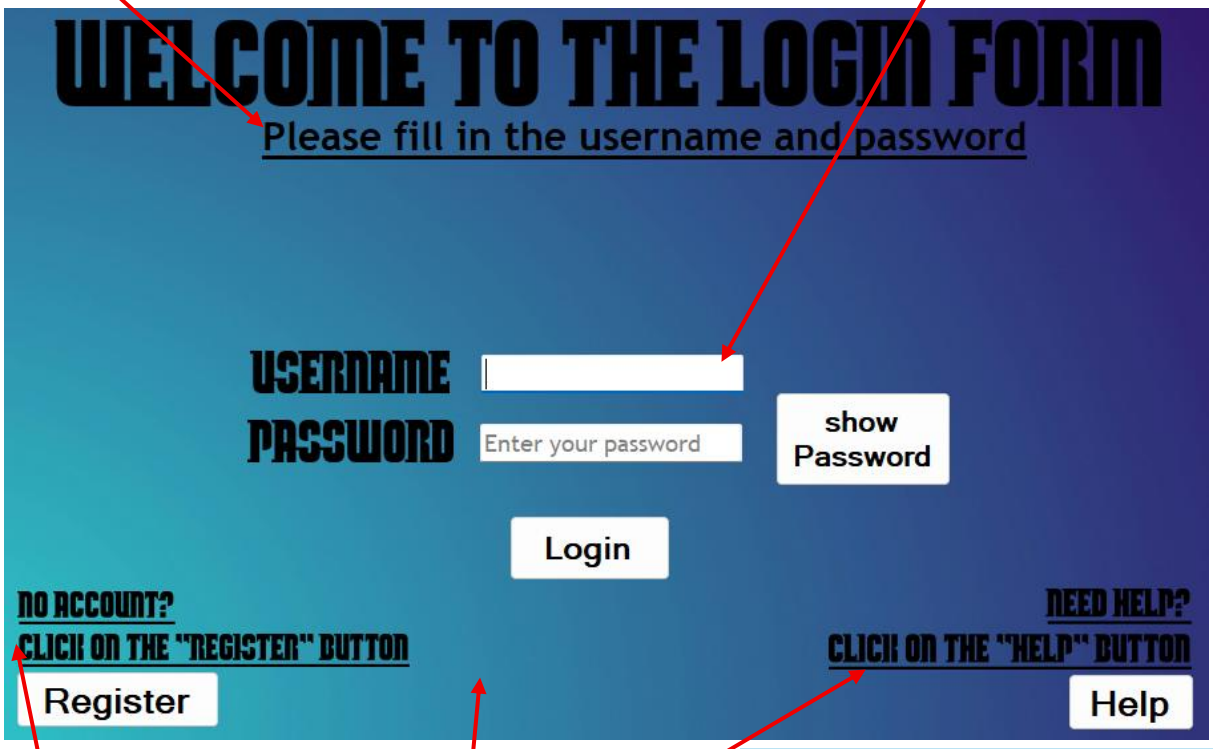



```
PSEUDOCODE
START
        REPEAT
                loadingWelcomeProBar.Progress + 5
        UNITL loadingWelcomeProBar.Progress = 100
        IF buttonClicked = TRUE THEN
                Show loading bar
                Hide start button
        ENDIF
        IF loadingWelcomeProBar.Progress = 100 THEN
                OPEN login Form
        END IF
END
```

CHANGES FROM 1ST DRAFT

- BACKGROUND
- ARROW REMOVED

Log in Form



CHANGES FROM 1ST DRAFT

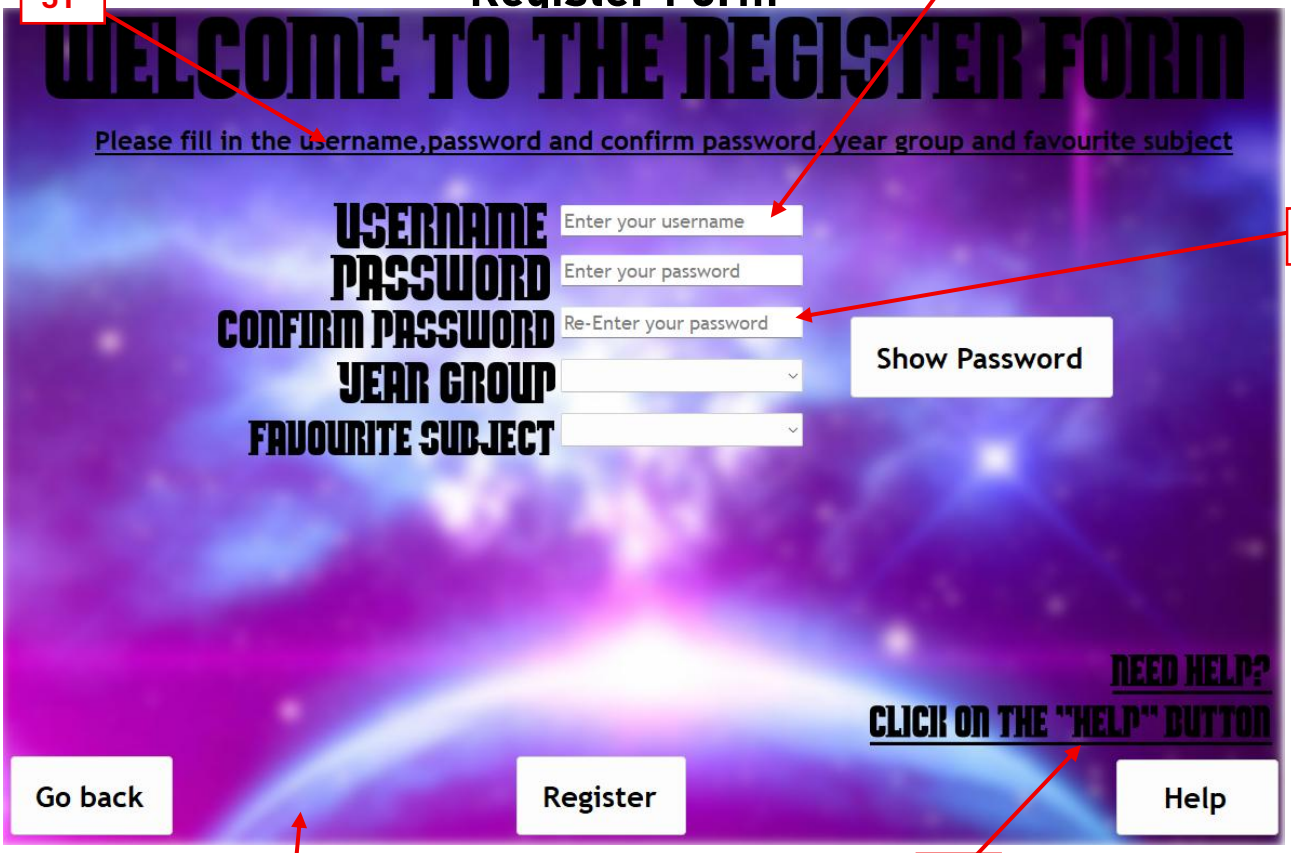
- HIDE PASSWORD BUTTON
- REMOVED IMAGES OF PLAYER
- FONT CHANGED
- ADDED INSTRUCTIONS
- ADDED WHAT TO TYPE IN TEXT BOXES

PSEUDOCODE

```
START
Bool usernameValid
Bool passwordValid
Bool validLogin
INPUT username
INPUT password
OUTPUT loginDetails From playerLoginDetails.txt
WHEN loginButton Pressed
        IF username != null THEN
                usernameValid = true
        ENDIF
        IF password != null THEN
                passwordValid = true
        ENDIF
        IF usernameValid == true AND passwordValid == true THEN

                IF username CONTAINS capitals THEN
                        OUTPUT “username cannot contain capitals”
                ENDIF
                IF username CONTAINS spaces OR password CONTAINS spaces THEN
                        OUTPUT “no spaces allowed”
                ENDIF
               IF username.length > 6 AND username.length < 31 THEN
                        IF password CONTAINS number AND password CONTAINS capital THEN
                                IF loginDetails CONTAINS username and CONTAINS passoword THEN
                                        validLogin = true
                                        OPEN main menu
                                ENDIF
                        ENDIF
                ENDIF
                IF validLogin = false THEN
                        OUTPUT "Incorrect details, please try again"
                END IF
        ENDIF
ENDWHEN
END
```

Register Form



CHANGES FROM 1ST DRAFT

- HIDE PASSWORD BUTTON
- REMOVED IMAGES OF ENEMY
- ADDED INSTRUCTIONS
- BRINGS USER TO MAIN MENU AFTER SIGNING IN
- RE ENTER PASSWORD NAME CHANGED
- ADDED WHAT TO TYPE IN TEXT BOXES

PSEUDOCODE

```
START

Bool isUsernameUnique
Bool isUsernameValid
Bool isPasswordValid
Bool isPasswordMatching


INPUT username
INPUT password
INPUT reEnterPassword
INPUT yearGroup
INPUT favSubject

OUTPUT loginDetails FROM playerLoginDetails.txt


WHEN RegisterButton Pressed

  IF LoginDetails.txt CONTAINS username THEN
    isUsernameUnique = false
  ENDIF

  IF username CONTAINS capitals THEN
    OUTPUT "username cannot contain capitals"
  ENDIF

  IF username CONTAINS numbers THEN
    OUTPUT "username cannot contain numbers"
  ENDIF

  IF username CONTAINS spaces OR password CONTAINS spaces THEN
    OUTPUT "no spaces allowed"
  ENDIF

  IF username Contains >= 7 characters AND <= 30 characters AND isUsernameUnique = true THEN
    isUsernameValid = true
  ENDIF

  IF password CONTAINS >= 1 number AND >= 1 capital AND >= 10 characters AND <= 30 characters THEN
    isPasswordValid = true
  ENDIF

  IF password == reEnterPassword THEN
    isPasswordMatching = true
  ENDIF

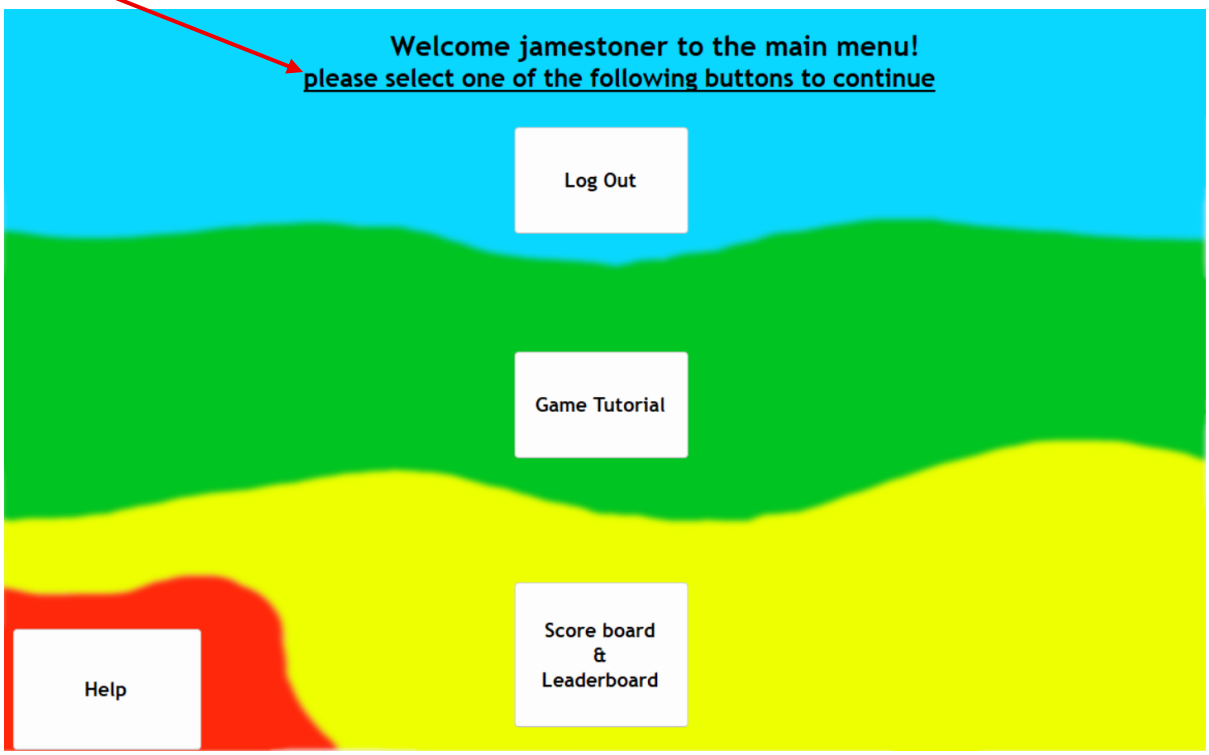
  IF isUsernameValid == true AND isPasswordValid == true AND isPasswordMatching == true THEN
    OUTPUT username AND password AND yearGroup AND favSubject TO playerLoginDetails.txt
    OUTPUT "New user has been added"
    OPEN Login Form
  ENDIF

ENDWHEN

END
```

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Main Menu



CHANGES FROM 1ST DRAFT

- USERNAME APPEARING
- ADDED INSTRUCTIONS
- CHANGED BOTTOM BUTTON NAME

PSEUDOCODE

```
START

OUTPUT "Welcome " + username + " to the main menu!"
OUTPUT playerList FROM scoreBoard.xml



WHEN startGameButton Pressed
    OPEN Game Help
ENDWHEN

WHEN scoreboardButton Pressed
    OPEN Scoreboard
ENDWHEN

WHEN helpButton Pressed
    OPEN Main Menu Help
ENDWHEN

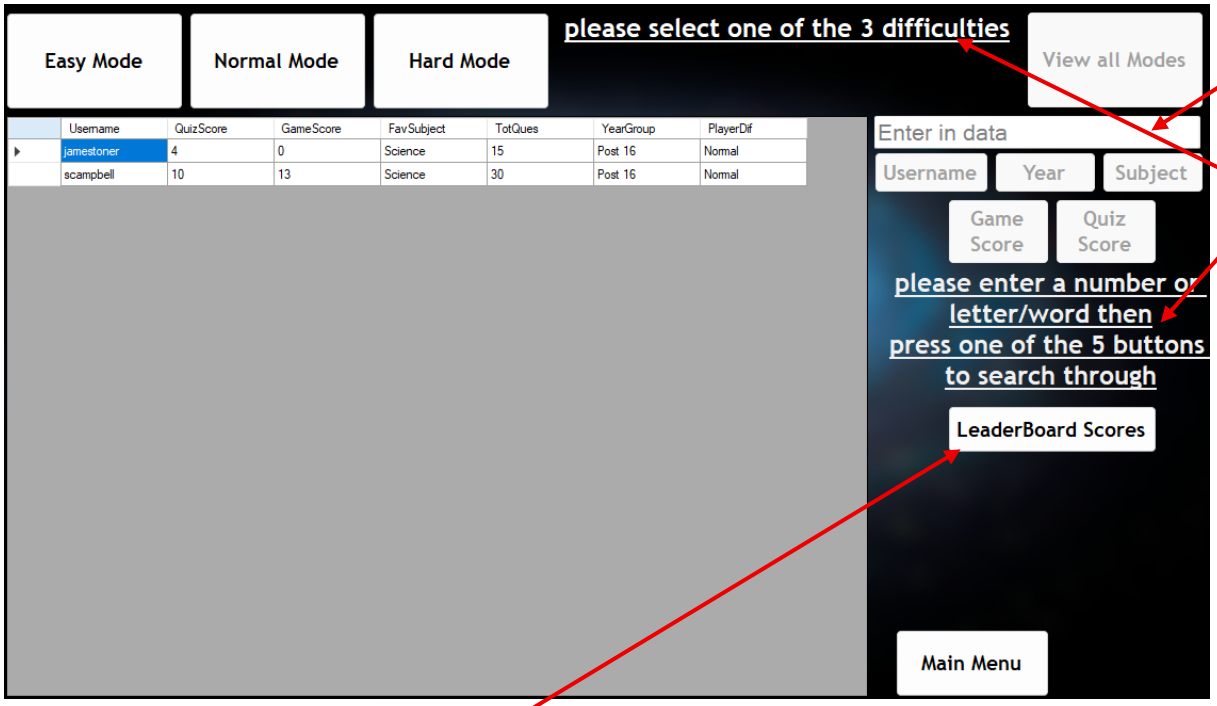
WHEN logoutButton Pressed
    OUTPUT "Are you sure you want to log out?"
    INPUT userAnswer

    IF userAnswer == true THEN
        OPEN Login Form
        OUTPUT "Log out successful, log in again soon " + username
    ENDIF

ENDWHEN

END
```

Scoreboard & Leaderboard



CHANGES FROM 1ST DRAFT

- REMOVED THE PODIUM
- ADDED LEADER BOARD TO THE TABLE
- INSTRUCTIONS
- SEARCH COLUMNS
- ADDED VIEW ALL MODE BUTTON
- LAYOUT CHANGED
- ADDED MORE COLUMNS
- BACKGROUND
- ADDED WHAT TO TYPE IN TEXT BOXES

PSEUDOCODE

```
START

INPUT playerList FROM scoreBoard.xml
INPUT searchQuery


WHEN MainMenuButton Pressed
    OPEN Main Menu
ENDWHEN

WHEN EasyModeButton Pressed
    REMOVE Rank column IF EXISTS
    OUTPUT players WHERE difficulty == "Easy" ORDERED BY gameScore
ENDWHEN

WHEN NormalModeButton Pressed
    REMOVE Rank column IF EXISTS
    OUTPUT players WHERE difficulty == "Normal" ORDERED BY gameScore
ENDWHEN

WHEN HardModeButton Pressed
    REMOVE Rank column IF EXISTS
    OUTPUT players WHERE difficulty == "Hard" ORDERED BY gameScore
ENDWHEN

WHEN ViewAllButton Pressed
    REMOVE Rank column IF EXISTS
    OUTPUT all players ORDERED BY gameScore
ENDWHEN

WHEN LeaderboardButton Pressed
    REMOVE Rank column IF EXISTS
    ORDER players BY gameScore AND quizScore DESCENDING
    ADD Rank column TO scoreboard
    OUTPUT rankedPlayerList
ENDWHEN

WHEN SearchTextChanged
    REMOVE Rank column IF EXISTS
    IF searchQuery != null THEN
        ENABLE search buttons
    ENDIF
ENDWHEN

WHEN SearchButton Pressed
    REMOVE Rank column IF EXISTS
    IF searchColumn == "Username" THEN
        OUTPUT players WHERE username CONTAINS searchQuery
    ELSEIF searchColumn == "Year" THEN
        OUTPUT players WHERE yearGroup CONTAINS searchQuery
    ELSEIF searchColumn == "Subject" THEN
        OUTPUT players WHERE favSubject CONTAINS searchQuery
    ELSEIF searchColumn == "Game Score" THEN
        OUTPUT players WHERE gameScore CONTAINS searchQuery
    ELSEIF searchColumn == "Quiz Score" THEN
        OUTPUT players WHERE quizScore CONTAINS searchQuery
    ENDIF
ENDWHEN

END
```

Game Help Form

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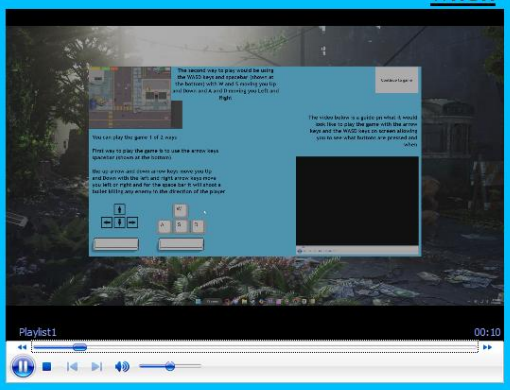


The second way to play would be using the WASD keys and spacebar (shown at the bottom) with W and S moving you Up and Down and A and D moving you Left and Right

Continue to game

Return to main menu

The video below is a guide on what it would look like to play the game with the arrow keys and the WASD keys on screen allowing you to see what buttons are pressed and when

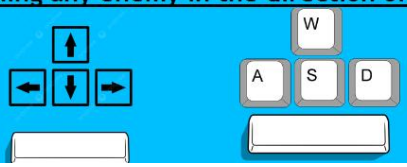


You can play the game 1 of 2 ways

First way to play the game is to use the arrow keys spacebar (shown at the bottom)

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the up arrow and down arrow keys move you Up and Down with the left and right arrow keys move you left or right and for the space bar it will shoot a bullet killing any enemy in the direction of the player



PSEUDOCODE

```
START

INPUT videoPath FROM "game tutorial.mp4"


WHEN Form Loads
    PLAY videoPath
ENDWHEN

WHEN ContinueButton Pressed
    STOP video
    OPEN Difficulty Selection
ENDWHEN

WHEN MainMenuButton Pressed
    STOP video
    PLAY intro sound
    OPEN Main Menu
ENDWHEN

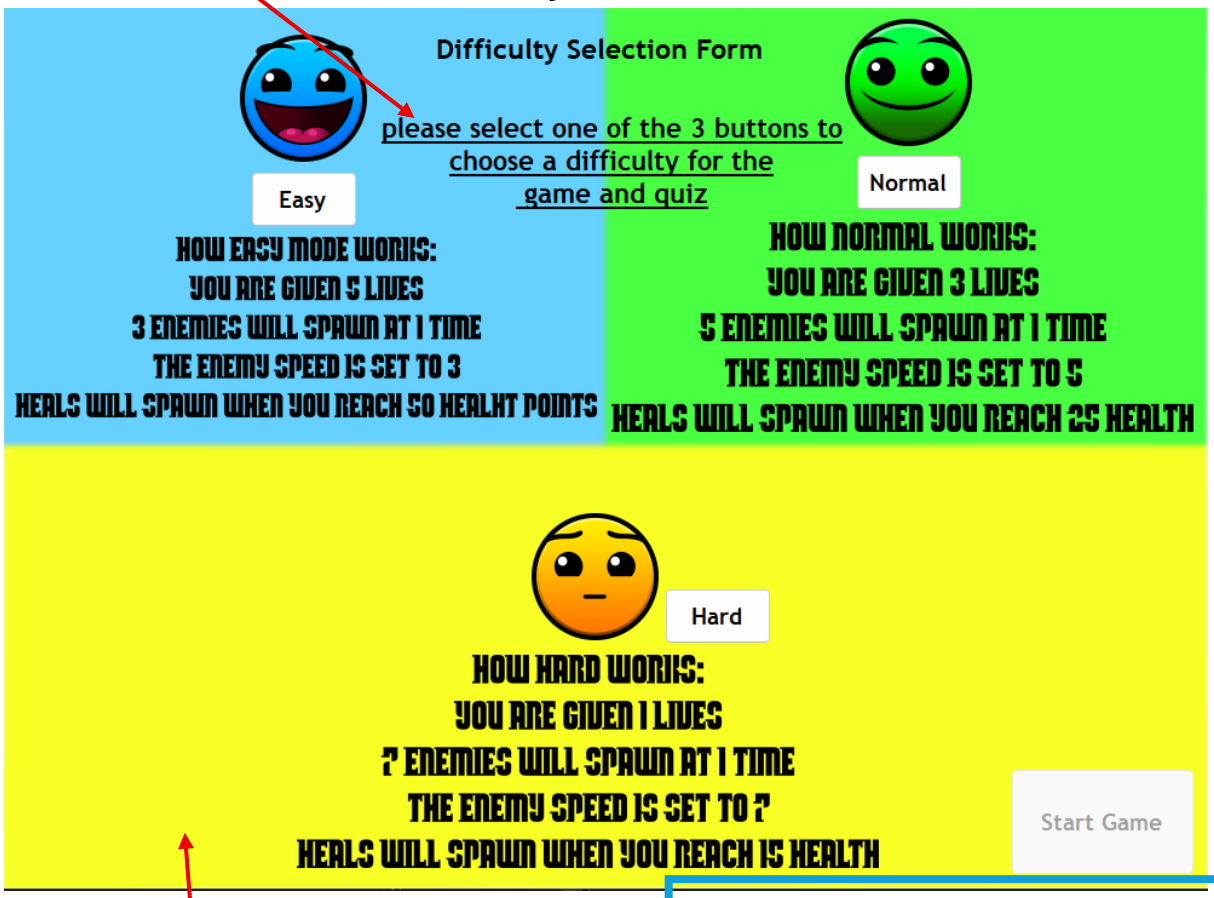
END
```

CHANGES FROM 1ST DRAFT

- ADDED A RETURN BUTTON
- ADDED IN THE ARROW KEYS AND WASD KEYS
- REMOVED THE CLOUDS AROUND THE INSTRUCTIONS
- UNDERLINED THE INSTRUCTIONS
- LAYOUT

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Difficulty Select form



25,61

CHANGES FROM 1ST DRAFT

- ADDED INSTRUCTIONS
- UNDERLINED THE INSTRUTIONS
- ADDED A START GAME BUTTON
- LAYOUT

PSEUDOCODE

```
START

INPUT difficulty
startButton DISABLED

WHEN EasyButton Pressed
    SET lives, enemies, questions, timer TO easy values
    OUTPUT "Lives: " + lives + " Enemies: " + enemies + " Time: " + timer + " Questions: " + questions
    INPUT userAnswer

    IF userAnswer == Yes THEN
        ENABLE startButton
    ELSE
        Reset easy difficulty TO null
    ENDIF
ENDWHEN

WHEN NormalButton Pressed
    SET lives, enemies, questions, timer TO normal values
    OUTPUT "Lives: " + lives + " Enemies: " + enemies + " Time: " + timer + " Questions: " + questions
    INPUT userAnswer

    IF userAnswer == Yes THEN
        ENABLE startButton
    ELSE
        Reset normal difficulty TO null
    ENDIF
ENDWHEN

WHEN HardButton Pressed
    SET lives, enemies, questions, timer TO hard values
    OUTPUT "Lives: " + lives + " Enemies: " + enemies + " Time: " + timer + " Questions: " + questions
    INPUT userAnswer

    IF userAnswer == Yes THEN
        ENABLE startButton
    ELSE
        Reset hard difficulty TO null
    ENDIF
ENDWHEN

WHEN StartButton Pressed
    IF difficulty == null THEN
        OUTPUT "Please select a difficulty"
    ELSE
        OPEN Game
    ENDIF
ENDWHEN

END
```

Game form



37,39

CHANGES FROM 1ST DRAFT

- ADDED COLOUR BEHIND THE TEXT
- ADDED A BACKGROUND
- CHANGED BULLET COLOUR TO BLUE

PSEUDOCODE

```
START

INPUT difficulty
INPUT lives, enemies, speed FROM difficulty settings
INPUT playerHealth = 100, ammo = 10, gameScore = 0


WHEN Game Loads
    SET player position
    SPAWN enemies
    START game timer
    PLAY game sound
ENDWHEN


EVERY Game Timer Tick

    SPAWN enemies IF current enemies < required amount

    UPDATE health bar, ammo, kills, lives labels

    MOVE player based on direction keys pressed

    MOVE enemies TOWARDS player

    IF player COLLIDES WITH enemy THEN
        REDUCE playerHealth
    ENDIF

    IF player COLLIDES WITH ammo pickup THEN
        INCREASE ammo
    ENDIF

    IF player COLLIDES WITH heal pickup THEN
        INCREASE playerHealth
    ENDIF

    IF playerHealth <= 50 AND heal NOT already spawned THEN
        SPAWN heal pickup
    ENDIF

    IF ammo == 0 THEN
        SPAWN ammo pickup
    ENDIF

    IF playerHealth <= 0 THEN
        REDUCE lives by 1
        OPEN Quiz Form
    ENDIF

    IF gameScore == target score THEN
        STOP timer
        OPEN Quiz Form
    ENDIF

ENDTIMER


WHEN Key Pressed
    IF W OR Up THEN move player up ENDIF
    IF S OR Down THEN move player down ENDIF
    IF A OR Left THEN move player left ENDIF
    IF D OR Right THEN move player right ENDIF

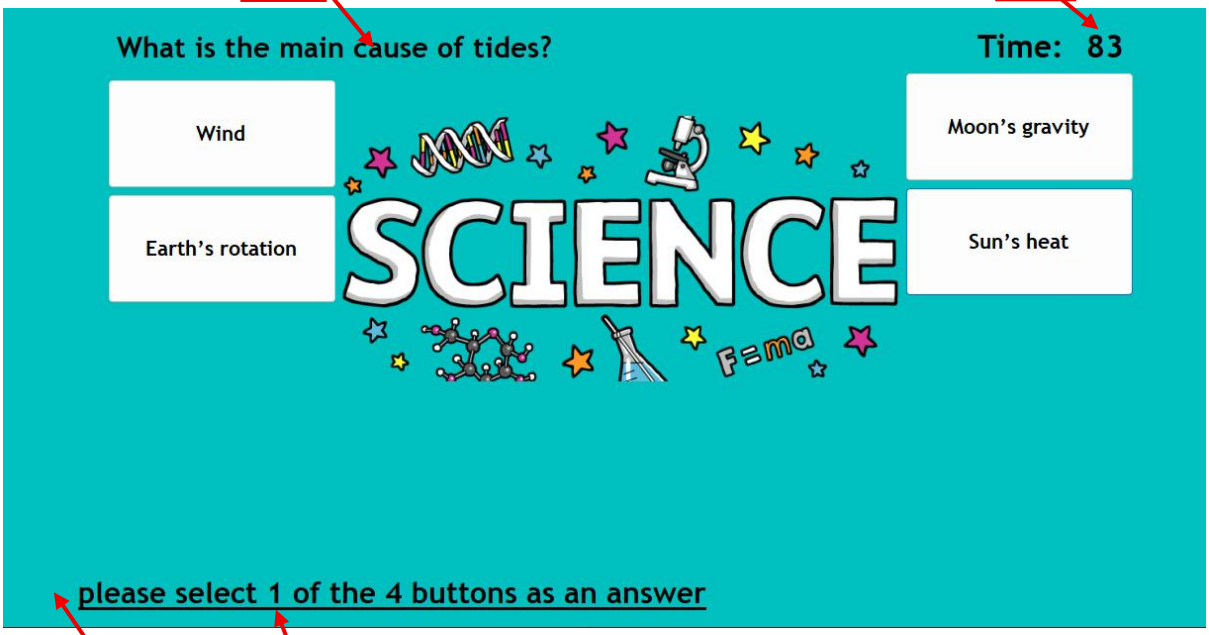
    IF Space THEN
        REDUCE ammo by 1
        SHOOT bullet in facing direction
    ENDIF

    IF Escape THEN
        IF game NOT paused THEN
            STOP timer
            SHOW pause menu
        ELSE
            HIDE pause menu
            START timer
        ENDIF
    ENDIF
ENDWHEN


WHEN MainMenuButton Pressed
    OPEN Main Menu
ENDWHEN

END
```

Quiz form



CHANGES FROM 1ST DRAFT

- ADDED INSTRUCTIONS
- LAYOUT

PSEUDOCODE

```
START

INPUT difficulty
INPUT questionDifficulty
INPUT playerFavSubject


WHEN Form Loads
    SET totalQuestions, questionType, timer FROM questionDifficulty

    IF questionDifficulty == Easy THEN
        SET totalQuestions, time TO easy values
    ELSEIF questionDifficulty == Normal THEN
        SET totalQuestions, time TO normal values
    ELSEIF questionDifficulty == Hard THEN
        SET totalQuestions, time TO hard values
    ENDIF

    INPUT questions FROM questionsComplete.txt

    IF no questions available THEN
        OUTPUT "No questions available"
        CLOSE form
    ENDIF

    DISPLAY first question
    START question timer
ENDWHEN


EVERY Timer Tick
    REDUCE timeLeft by 1

    IF timeLeft == 0 THEN
        STOP timer
        MOVE to next question
    ENDIF
ENDTIMER


WHEN Answer Button Pressed
    IF answer == correct answer THEN
        INCREASE quizScore
    ENDIF
    DISPLAY next question
ENDWHEN


WHEN Loading questions FROM file
    FOR each line IN questionsComplete.txt
        SPLIT line INTO difficulty, isMath, subject, question, answer, options

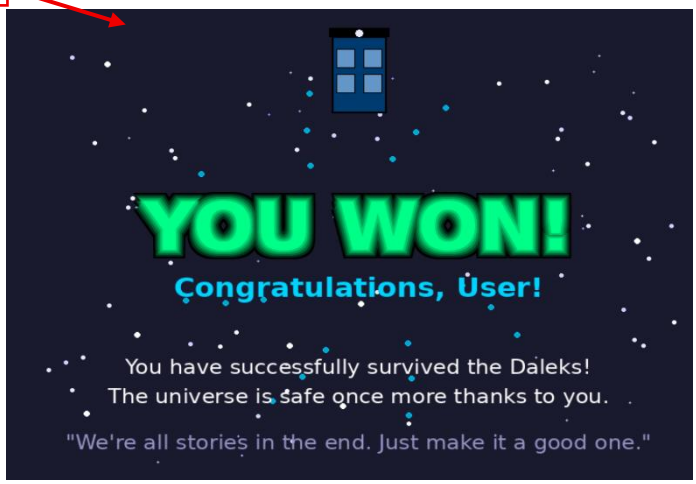
        IF subject == playerFavSubject AND difficulty == questionType THEN
            ADD question TO questionsList
        ENDIF
    ENDFOR
ENDWHEN

END
```

Win/Lose form

13





PSEUDOCODE

```
START

INPUT currentPlayer
INPUT playerList FROM scoreBoard.xml


WHEN Form Loads
    PLAY defeat/win sound
    START loading bar timer

    IF playerList CONTAINS currentPlayer THEN
        UPDATE existing player gameScore, quizScore, totalQuestions
    ELSE
        ADD currentPlayer TO playerList
    ENDIF

    OUTPUT playerList TO scoreBoard.xml
ENDWHEN


EVERY Timer Tick
    INCREASE loading bar by 1

    IF loading bar == 100 THEN
        STOP timer
        OPEN Scoreboard
    ENDIF
ENDTIMER

END
```

CHANGES FROM 1ST DRAFT

- WASN'T INCLUDED IN 1ST DRAFT

Testing

Testing against Final Draft User Requirements

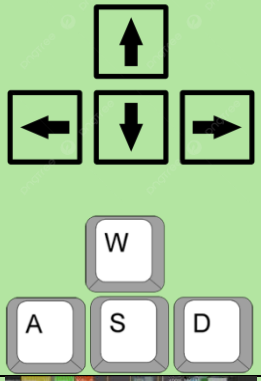
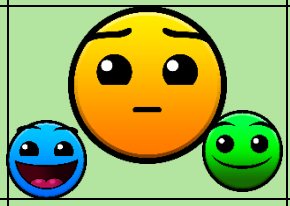
Test No.	User Requirement	Test Data	Test reason	Expected Outcome	Actual Outcome	Action Required
1	1	Once pressed closes the current form and opens a new form	Brings the user to a new form	Shows a new form	Shows a new form	No action Required
2	1	Once pressed closes the current form and brings the user to a previous/ old form	Brings the user to a old/previous form	Shows a old/previous form	Shows a old/previous form	No action Required
3	**2**	**Assigns the difficulty related methods**	**uses the correct difficulty**	**Only the selected difficulty methods are used**	**IF a difficulty is chosen and a new one is chosen it would use a bit of both**	**Action Required See Picture 1**
4	2	The same button can't be pressed again	No more than 1 difficulty can be chosen	Greys out or disables the button pressed	Greys out or disables the button pressed	No action Required
5	3	Can be pressed to answer the questions	Answer the questions presented	A new question appears once pressed	A new question appears once pressed	No action Required
6	4	Can be pressed to submit the data in the text box as an answer	Answer the math's questions by submitting the answer	A new question appears once pressed	A new questions appears once pressed	No action Required
7	5	1 of 5 buttons can be pressed to search through that specific column	Searches through the specified column	The data entered in the text box, will be searched in the specific column and displayed	The data entered in the text box, will be searched in the specific column and displayed	No action Required
8	6	"S", "5", "b"	The data entered in the text box will enable the 5 buttons	Data entered will make the 5 buttons useable	Data entered will make the 5 buttons useable	No action Required
9	7	"2", "6.7", "67"	Data can be entered to answer math's questions	Data can be entered	Data can be entered	No action Required
10	8	"michael123" and "Password123"	User can enter a username and password to login to sign in	Data can be entered for username and password	Data can be entered for username and password	No action Required
11	9	"Password123"	User can enter in their password again to continue signing in	Data can be entered to continue signing in	Data can be entered to continue signing in	No action Required

12	10	"Year 10" "Science"	The users year can be chosen along with their favorite subject for the questions	A year group and favorite subject can be chosen	A year group and favorite subject can be chosen	No action Required
13	11	michael123 Password123 Year10 History	Data is saved to the text file	The data entered is saved to the loginDetails.txt file and the main menu is shown	The data entered is saved to the loginDetails.txt file and the main menu is shown	No action Required
14	12	Michael123 15 5 Science 5 Year10 Easy	Data saved once the user completes the game	Data is saved	Data is saved	No action Required
15	**13**	**Game score: 15 Quiz score: 5 Number of Quest: 5 Difficulty: Easy**	**So that the same users data doesn't get written more than once and only the necessary data is over written**	**The Game score is overwritten, the quiz score is overwritten, the number of questions is overwritten and the difficulty is overwritten**	**IF the same user plays more than once then all their data is recorded again regardless if they've already played**	**Action Required See Picture 2**
16	14	The txt is read and the data entered is compared to the data saved	The user can log in using the saved user details	The user is brought to the main menu	The user is brough to the main menu	No action Required
17	15	The XML file is read and is displayed on the scoreboard	The data is displayed when the form loads	The scoreboard is displayed	The scoreboard is displayed	No action Required
18	16	The XML file is read and is displayed on the scoreboard	The data is displayed for the leaderboard once the form loads	The leader board is displayed along with each users place ment	The leader board is displayed along with each users place ment	No action Required
19	**17**	**The variable subject is assigned to the place where subject is in the txt file**	**So that the correct questions, answers etc is displayed**	**Questions, answers etc is displayed**	**Questions not displayed**	**Action Required See Picture 3**
20	18	"Password123" "Passsword123"	The user cannot continue to the main menu with the incorrect password	The user will be told that the passwords do not match	The user will be told that the passwords do not match	No action Required
21	19	Saved username: "michael123" New username: "michael123"	The user cannot create a new account with an	The user will be told that the username	The user will be told that the username	No action Required

			existing username	is already in use	is already in use	
22	20	Username: "michael123" Password:"Password123"	The same account can't be created twice	The user will be told that the account already exists	The user will be told that the account already exists	No action Required
23	21	Nothing entered in the registration form	Data must be entered in the registration form	The user will be told that they must enter data in the reg form	The user will be told that they must enter data in the reg form	No action Required
24	22	"michael123"	The username is unique	The username will be allowed	The username will be allowed	No action Required
25	23	"Password123"	The password is long, secure and unique	The Password will be allowed	The password will be allowed	No action Required
26	24	Nothing entered in the Login form	Data must be entered in the log in form	The user will be told that they must enter data in the log in form	The user will be told that they must enter data in the log in form	No action Required
27	25	The difficulty selected class will be assigned to a variable	So the game and quiz can be played	The user will continue to the game/quiz	The user will continue to the game/ quiz	No action Required
28	26	The bar fills up	Accurately displays the correct amount	The bar will fill up or decrease	The bar will fill up or decrease	No action Required
29	27	The timer counts down	Accurately displays the timer that is remaining	The timer ticks down by 1 second	The timer ticks down by 1 second	No action Required
30	28	Easy Mode Normal Mode Hard Mode	Correct timer is used and displayed	30 seconds 60 seconds 90 seconds	30 seconds 60 seconds 90 seconds	No action Required
31	29	1 being added to the game score variable	The user attaining the correct score	1 is added to the game score	1 is added to the game score	No action Required
32	30	5 is added to the ammo variable	The player can shoot more than 10 times	+5 ammo is added	+5 ammo is added	No action Required
33	31	The player gains back 5 health each time the user collects one	The user doesn't have a set amount of health	+5 is added to the players health	+5 is added to the players health	No action Required
34	**32**	**Enemies will spawn until they equal the number 5, which is chosen depending on the difficulty selected by the user**	**The correct number of enemies are spawning**	**The enemies spawn until they meet the condition**	**No enemies spawn**	**Action Required See Picture 4**
35	33	The user pressing W,A,S,D or the arrow Keys	The user can move about on the game	The Player moves about	The Player moves about	No action Required

36	34	The user pressing the space	The user can shoot at the enemies	A bullet appears in the direction the player is facing	A bullet appears in the direction the player is facing	No action Required
37	35	Opening a help form and the video plays automatically	The user can understand how to navigate/use/play the form	Video plays upon loading into the form	Video plays upon loading into the form	No action Required
38	36	The user can press 1 of 3 buttons on the scoreboard/leaderboard form	The user can see all the players who played easy, normal and hard	The button is disabled upon being pressed	The button is disabled upon being pressed	No action Required
39	**37**	**The user reaches the score 15**	**So the user HAS to do the questions and cant opt out of doing the questions**	**Once the correct score has been achieved the user will be brought to the quiz**	**The user is brought to the quiz after NOT achieving the correct score**	**Action Required See Picture 5**
40	**38**	**Achieving the correct score in the game**	**The user can see if they've won or lost**	**Being brought to a win or a lose form**	**Being brought to a win or a lose form when achieving the incorrect score**	**Action Required See Picture 5**
41	39	The player loses a life and after completing the quiz their lives will be 1 less	The user can only play the game a certain amount of times in one go	The user is brought back to form with 1 life less	The user is brought back to form with 1 life less	No action Required
42	40	Easy Question Mode Normal Question Mode Hard Question Mode	Each difficulty will have their own number of questions relevant to their difficulty	5 questions used and Displayed 10 questions used and Displayed 15 questions used and Displayed	5 questions used and Displayed 10 questions used and Displayed 15 questions used and Displayed	No action Required
NON-FUNCTIONAL						
43	41	The title is big and visible	The user can identify what form they are currently on	"Log in Form" In big font	"Log in Form" In big font	No action Required
44	42	Each box is labeled correctly	The user knows where to input data	"Confirm password" Beside a text box	"Confirm password" Beside a text box	No action Required
45	43	The help form button is positioned in the bottom corner of every form	There are help forms for all forms and that they are functional	Help button in the corner of form	Help button in the corner of the form	No action Required

46	44	The player has visible up, down, left and right images	Without the images the user would have no idea of where they are moving during the game play	All 4 images are seen and being used	All 4 images are seen and being used	No action Required
47	45	The enemy has visible up, down, left and right images	Without the images the user would have no idea of where the enemies are	All 4 images are seen and being used	All 4 images are seen and being used	No action Required
48	**46**	**The form is in the middle of the screen**	**It would be easier and be more natural to interact with if it is centered**	**The form is in the middle when the program is being ran**	**The form off to the left**	**Action Required see Picture 6**
49	47	An image of the person of who the player is playing as	The user can see who they will be playing as in the game	An image ,different from what is used in the actual game, of the player	An image ,different from what is used in the actual game, of the player	No action Required
50	48	An image of the enemy	The player will be able to see who the enemy of the game is	An image ,different from what is used in the actual game, of the enemy	An image ,different from what is used in the actual game, of the enemy	No action Required
51	49	An image of the word for their chosen subject: History	The player will be able to identify they are being questioned on the subject history	An image of the word of the subject: History	An image of the word of the subject: History	No action Required
52	49	An image of the word for their chosen subject: Geography	The player will be able to identify they are being questioned on the subject geography	An image of the word of the subject: Geography	An image of the word of the subject: Geography	No action Required
53	49	An image of the word for their chosen subject: Science	The player will be able to identify they are being questioned on the subject science	An image of the word of the subject: Science	An image of the word of the subject: Science	No action Required
54	49	An image of the word for their chosen subject: Math's	The player will be able to identify they are being questioned on the subject math's	An image of the word of the subject: Math's	An image of the word of the subject: Math's	No action Required
55	50	Images for the med kit(healing), ammo supply and bullet	So that all 3 images actually exist in the game	An image of the Med kit, Ammo and bullet	An image of the Med kit, Ammo and bullet	No action Required

56	51	"Don't have an account? Press the sign up button"	The user can identify when the program is providing help	The text is underlined	The text is underlined	No action Required
57	52	Instructions on how to play the game, e.g. "use WASD or the Arrow keys to move about"	The user knows how to play the game	The instructions tell the user how to play the game with a video to help	The instructions tell the user how to play the game with a video to help	No action Required
58	53		The user can see what controls to use to play the game	The keys and space bar is visible and is understandable	The keys and space bar is visible and is understandable	No action Required
59	54		The user will have an understanding of what the game looks like	A gif of the game that loops the video	A gif of the game that loops the video	No action Required
60	55		The user will be able to have a visual identification of the difficulty of the level	The images are used and are in the correct place	The images are used and are in the correct place	No action Required
61	56	"easy will have 3 lifes, there will be 3 enemies etc"	The user will know what each difficulty contains	Under each image there will be a description of what each difficulty will do to the game	Under each image there will be a description of what each difficulty will do to the game	No action Required
62	57	"Login", "Help", "Hide password", "Show password"	The user will be able to identify what each button does	The buttons will have the correct names	The buttons will have the correct names	No action Required
63	**58**	**Music is played in each form**	**So that the user can have a fun experience with the music playing**	**Music being played in each form except the help forms**	**Error when the form that contains the music loads**	**Action Required see Picture 7**
64	59	"Loading..."	This give the game a more game like feel	The text says loading	The text says loading	No action Required

65	60		The help form(s) contain the image	The image appears on the help forms	The image appears on the help forms	No action Required
66	61	Enter font name here	Make the program feel more relevant towards the game	The font has been correctly used	The font has been correctly used	No action Required
67	62	“What is 6 + 7”, “what is the name of the planet we live on”	The questions appear so the user can answer them	The questions show on the quiz form	The questions show on the quiz form	No action Required
68	63	“Enter your username”	So the user knows what the program is expecting the user to enter	The text is displayed and disappears upon clicking on the text box	The text is displayed and disappears upon clicking on the text box	No action Required

Testing rate success is 60/68 = **88.23%**

The testing plan for my program was highly successful with 60/68 of the tests I conducted being successful. Almost all the functioning requirements were met as expected such as logging and registering. The failed tests were either big issues for my program or small minor issues that were caused by my miss-types. Overall, this testing has shown that my program is stable and can function according to needs of teachers, with some adjustments needed for the program to be fully reliable.

Picture 1

Error's

```
1 reference
private void easyBTN_Click(object sender, EventArgs e)
{

    //disables the button that has been clicked and enables the other 2
    easyBTN.Enabled = false;
    normalBTN.Enabled = true;
    hardBTN.Enabled = true;


    //assign new object to declare reference type variable
    easyMode = new Easy(); // object created from derived
    QuesEasy = new EasyQues();

    //calling override methods from derived class to assign lives and amount of enemies
    easyMode.setLives();
    easyMode.setEnemies();

    //caling override methods from derived class to assign number of questions, timer and difficulty
    QuesEasy.SetNoOfQues();
    QuesEasy.setTimerSet();
    QuesEasy.setDifficulty();

    DialogResult result = MessageBox.Show("The number of lives are: " + easyMode.Lives
        + "\nThe amount of enemies are: " + easyMode.Enemies + "\nThe time given for the questions are: " + QuesEasy.TimeSet
        + "\nThe amount of questions are: " + QuesEasy.NoOfQuestions, "Ready to Game?",MessageBoxButtons.YesNo);
    if (result == DialogResult.Yes)
    {
        //button will be enabled if the user has confirmed they want to play on easy mode
        startBTN.Enabled = true;
    }
    else if (result == DialogResult.No)
    {
        easyBTN.Enabled = true;
    }
}
```

```
1 reference
private void normalBTN_Click(object sender, EventArgs e)
{
    //disables the button that has been clicked and enables the other 2
    easyBTN.Enabled = true;
    normalBTN.Enabled = false;
    hardBTN.Enabled = true;

    //assign new object to declare reference type variable
    normalMode = new Normal(); //objected from base class
    QuesNormal = new NormalQues();

    //calling virtual methods from base class to assign lives and amount of enemies
    normalMode.setLives();
    normalMode.setEnemies();

    QuesNormal.SetNoOfQues();
    QuesNormal.setTimerSet();
    QuesNormal.setDifficulty();

    DialogResult result = MessageBox.Show("The number of lives are: " + normalMode.Lives
        + "\nThe amount of enemies are: " + normalMode.Enemies + "\nThe time given for the questions are: " + QuesNormal.TimeSet
        + "\nThe amount of questions are: " + QuesNormal.NoOfQuestions, "Ready to Game?", MessageBoxButtons.YesNo);
    if (result == DialogResult.Yes)
    {
        startBTN.Enabled = true;
    }
    else if (result == DialogResult.No)
    {
        normalBTN.Enabled = true;
    }
}
```

```
1 reference
private void hardBTN_Click(object sender, EventArgs e)
{
    easyBTN.Enabled = true;
    normalBTN.Enabled = true;
    hardBTN.Enabled = false;

    //assign new object to declare reference type variable
    hardMode = new Hard(); //object created from derived class
    QuesHard = new HardQues();

    //calling override methods from derived class to assign lives and amount of enemies
    hardMode.setLives();
    hardMode.setEnemies();

    QuesHard.SetNoOfQues();
    QuesHard.setTimerSet();
    QuesHard.setDifficulty();

    DialogResult result = MessageBox.Show("The number of lives are: " + hardMode.Lives
        + "\nThe amount of enemies are: " + hardMode.Enemies + "\nThe time given for the questions are: " + QuesHard.TimeSet
        + "\nThe amount of questions are: " + QuesHard.NoOfQuestions, "Ready to Game?", MessageBoxButtons.YesNo);
    if (result == DialogResult.Yes)
    {
        startBTN.Enabled = true;
    }
    else if (result == DialogResult.No)
    {
        hardBTN.Enabled = true;
    }
}
```

Fixes

```
1 reference
private void easyBTN_Click(object sender, EventArgs e)
{
    //disables the button that has been clicked and enables the other 2
    easyBTN.Enabled = false;
    normalBTN.Enabled = true;
    hardBTN.Enabled = true;


    //set normal mode and question to null so only easy will be used
    normalMode = null;
    QuesNormal = null;

    //set hard mode and question to null so only easy will be used
    hardMode = null;
    QuesHard = null;

    //assign new object to declare reference type variable
    easyMode = new Easy(); // object created from derived
    QuesEasy = new EasyQues();

    //calling override methods from derived class to assign lives and amount of enemies
    easyMode.setLives();
    easyMode.setEnemies();

    //caling override methods from derived class to assign number of questions, timer and difficulty
    QuesEasy.SetNoOfQues();
    QuesEasy.setTimerSet();
    QuesEasy.setDifficulty();

    DialogResult result = MessageBox.Show("The number of lives are: " + easyMode.Lives
        + "\nThe amount of enemies are: " + easyMode.Enemies + "\nThe time given for the questions are: " + QuesEasy.TimeSet
        + "\nThe amount of questions are: " + QuesEasy.NoOfQuestions, "Ready to Game?",MessageBoxButtons.YesNo);
    if (result == DialogResult.Yes)
    {
        //button will be enabled if the user has confirmed they want to play on easy mode
        startBTN.Enabled = true;
    }
    else if (result == DialogResult.No)
    {
        easyBTN.Enabled = true;
        easyMode = null;
        QuesEasy = null;
    }
}
```

```
1 reference
private void normalBTN_Click(object sender, EventArgs e)
{
    //disables the button that has been clicked and enables the other 2
    easyBTN.Enabled = true;
    normalBTN.Enabled = false;
    hardBTN.Enabled = true;

    //set easy mode and question to null so only normal will be used
    easyMode = null;
    QuesEasy = null;

    hardMode = null;
    QuesHard = null;

    //assign new object to declare reference type variable
    normalMode = new Normal(); //objected from base class
    QuesNormal = new NormalQues();

    //calling virtual methods from base class to assign lives and amount of enemies
    normalMode.setLives();
    normalMode.setEnemies();

    QuesNormal.SetNoOfQues();
    QuesNormal.setTimerSet();
    QuesNormal.setDifficulty();

    DialogResult result = MessageBox.Show("The number of lives are: " + normalMode.Lives
        + "\nThe amount of enemies are: " + normalMode.Enemies + "\nThe time given for the questions are: " + QuesNormal.TimeSet
        + "\nThe amount of questions are: " + QuesNormal.NoOfQuestions, "Ready to Game?", MessageBoxButtons.YesNo);
    if (result == DialogResult.Yes)
    {
        startBTN.Enabled = true;
    }
    else if (result == DialogResult.No)
    {
        normalBTN.Enabled = true;
        normalMode = null;
        QuesNormal = null;
    }
}
```

```
1 reference
private void hardBTN_Click(object sender, EventArgs e)
{
    easyBTN.Enabled = true;
    normalBTN.Enabled = true;
    hardBTN.Enabled = false;

    easyMode = null;
    QuesEasy = null;

    normalMode = null;
    QuesNormal = null;

    //assign new object to declare reference type variable
    hardMode = new Hard(); //object created from derived class
    QuesHard = new HardQues();

    //calling override methods from derived class to assign lives and amount of enemies
    hardMode.setLives();
    hardMode.setEnemies();

    QuesHard.SetNoOfQues();
    QuesHard.setTimerSet();
    QuesHard.setDifficulty();

    DialogResult result = MessageBox.Show("The number of lives are: " + hardMode.Lives
        + "\nThe amount of enemies are: " + hardMode.Enemies + "\nThe time given for the questions are: " + QuesHard.TimeSet
        + "\nThe amount of questions are: " + QuesHard.NoOfQuestions, "Ready to Game?", MessageBoxButtons.YesNo);
    if (result == DialogResult.Yes)
    {
        startBTN.Enabled = true;
    }
    else if (result == DialogResult.No)
    {
        hardBTN.Enabled = true;
        hardMode = null;
        QuesHard = null;
    }
}
```

What happened?

When selecting a difficulty then a new difficulty is selected the program is recognizing that both of the classes are being used

How was it fixed

This error was fixed by setting the other 2 difficulty classes to null which is not allowing the other 2 classes to be used and allowing the selected class to be used ONLY

Picture 2

Error's

```
1 reference
private void Win_Load(object sender, EventArgs e)
{
    loadingBarTimer.Enabled = true;


    soundDrPlayer.playVictorySound();



        Player_Class newPlayerClass = new Player_Class(currentPlayer.Username, currentPlayer.QuizScore, currentPlayer.GameScore, currentPlayer.FavSubject, currentPlayer.TotQues, currentPlay
        Main_Menu_Form.playerList.Add(newPlayerClass);



    Player_Class[] playerArray = Main_Menu_Form.playerList.ToArray();

    try
    {
        using (StreamWriter writer = new StreamWriter("scoreBoard.xml"))
        {
            serializerObject.Serialize(writer, playerArray);

            writer.Close();
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("An error occurred while saving player data: " + ex.Message);
    }


}
```

```
1 reference
private void Lose_Load(object sender, EventArgs e)
{
    loadingBarTimer.Enabled = true;

    soundDrPlayer.playDefeatSound();



        Player_Class newPlayerClass = new Player_Class(currentPlayer.Username, currentPlayer.QuizScore, currentPlayer.GameScore, currentPlayer.FavSubject, currentPlayer.TotQues, currentPlaye
        Main_Menu_Form.playerList.Add(newPlayerClass);

    //==============================Serialization of the list==============================
    Player_Class[] playerArray = Main_Menu_Form.playerList.ToArray();

    try
    {
        //writes the data to the file to be read
        using (StreamWriter writer = new StreamWriter("scoreBoard.xml"))
        {
            serializerObject.Serialize(writer, playerArray);

            writer.Close();
        }

        //MessageBox.Show("Player data saved successfully.");
    }
    catch (Exception ex)
    {
        MessageBox.Show("An error occurred while saving player data: " + ex.Message);
    }
```

info.txt | INSTALLATION_GUIDE.md | UCAS PASSWORD.txt | questionsComplete.txt | scoreBoard.xml | New Text Document.txt

File Edit View

```
<?xml version="1.0" encoding="utf-8"?>
<ArrayOfPlayer_Class xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <Player_Class>
    <Username>jamestoner</Username>
    <QuizScore>4</QuizScore>
    <GameScore>0</GameScore>
    <FavSubject>Science</FavSubject>
    <TotQues>15</TotQues>
    <YearGroup>Post 16</YearGroup>
    <PlayerDif>Normal</PlayerDif>
  </Player_Class>
  <Player_Class>
    <Username>scampbell</Username>
    <QuizScore>10</QuizScore>
    <GameScore>13</GameScore>
    <FavSubject>Science</FavSubject>
    <TotQues>30</TotQues>
    <YearGroup>Post 16</YearGroup>
    <PlayerDif>Normal</PlayerDif>
  </Player_Class>
<Player_Class>
    <Username>scampbell</Username>
    <QuizScore>5</QuizScore>
    <GameScore>15</GameScore>
    <FavSubject>Science</FavSubject>
    <TotQues>30</TotQues>
    <YearGroup>Post 16</YearGroup>
    <PlayerDif>Easy</PlayerDif>
  </Player_Class>
</ArrayOfPlayer_Class>
```

Ln 30, Col 23 | 962 characters | Plain text | 100% | Windows (CRLF) | UTF-8

Fixes

```
1 reference
private void Win_Load(object sender, EventArgs e)
{
    loadingBarTimer.Enabled = true;


    soundDrPlayer.playVictorySound();


    bool playerExists = false;

    for (int i = 0; i < Main_Menu_Form.playerList.Count; i++)
    {
        if (Main_Menu_Form.playerList[i].Username == currentPlayer.Username && Main_Menu_Form.playerList[i].FavSubject == currentPlayer.FavSubject
            && Main_Menu_Form.playerList[i].YearGroup == currentPlayer.YearGroup)
        {
            // Update existing player's scores only
            Main_Menu_Form.playerList[i].GameScore = currentPlayer.GameScore;
            Main_Menu_Form.playerList[i].QuizScore = currentPlayer.QuizScore;
            Main_Menu_Form.playerList[i].PlayerDif = currentPlayer.PlayerDif;
            Main_Menu_Form.playerList[i].TotQues = currentPlayer.TotQues;
            playerExists = true;
            break;
        }
    }

    // Only add new player if they don't exist
    if (!playerExists)
    {
        Player_Class newPlayerClass = new Player_Class(currentPlayer.Username, currentPlayer.QuizScore, currentPlayer.GameScore, currentPlayer.FavSub
        Main_Menu_Form.playerList.Add(newPlayerClass);
    }


    Player_Class[] playerArray = Main_Menu_Form.playerList.ToArray();

    try
    {
        using (StreamWriter writer = new StreamWriter("scoreBoard.xml"))
        {
            serializerObject.Serialize(writer, playerArray);

            writer.Close();
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("An error occurred while saving player data: " + ex.Message);
    }


}
```

```
1 reference
private void Lose_Load(object sender, EventArgs e)
{
    loadingBarTimer.Enabled = true;

    soundDrPlayer.playDefeatSound();



    bool playerExists = false;
    for (int i = 0; i<Main_Menu_Form.playerList.Count; i++)
    {
        //if the players details have been saved before then onlt the 2 scores difficulty and number of questions will be replaced
        if (Main_Menu_Form.playerList[i].Username == currentPlayer.Username && Main_Menu_Form.playerList[i].FavSubject == currentPlayer.FavSubject
            && Main_Menu_Form.playerList[i].YearGroup == currentPlayer.YearGroup)
        {
            // Update existing player's scores only
            Main_Menu_Form.playerList[i].GameScore = currentPlayer.GameScore;
            Main_Menu_Form.playerList[i].QuizScore = currentPlayer.QuizScore;
            Main_Menu_Form.playerList[i].TotQues = currentPlayer.TotQues;
            playerExists = true;
            break;
        }
    }

    // Only add new player if they don't exist
    if (!playerExists)
    {
        Player_Class newPlayerClass = new Player_Class(currentPlayer.Username, currentPlayer.QuizScore, currentPlayer.GameScore, currentPlayer.FavSub
        Main_Menu_Form.playerList.Add(newPlayerClass);
    }

    //============================Serialization of the list============================
    Player_Class[] playerArray = Main_Menu_Form.playerList.ToArray();

    try
    {
        //writes the data to the file to be read
        using (StreamWriter writer = new StreamWriter("scoreBoard.xml"))
        {
            serializerObject.Serialize(writer, playerArray);

            writer.Close();
        }

        //MessageBox.Show("Player data saved successfully.");
    }
    catch (Exception ex)
    {
        MessageBox.Show("An error occurred while saving player data: " + ex.Message);
    }
```

info.txt | INSTALLATION_GUIDE.md | UCAS PASSWORD.txt | questionsComplete.txt | scoreBoard.xml | New Text Document.txt

File Edit View

```
<?xml version="1.0" encoding="utf-8"?>
<ArrayOfPlayer_Class xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <Player_Class>
    <Username>jamestoner</Username>
    <QuizScore>4</QuizScore>
    <GameScore>0</GameScore>
    <FavSubject>Science</FavSubject>
    <TotQues>15</TotQues>
    <YearGroup>Post 16</YearGroup>
    <PlayerDif>Normal</PlayerDif>
  </Player_Class>
  <Player_Class>
    <Username>scampbell</Username>
    <QuizScore>10</QuizScore>
    <GameScore>13</GameScore>
    <FavSubject>Science</FavSubject>
    <TotQues>30</TotQues>
    <YearGroup>Post 16</YearGroup>
    <PlayerDif>Normal</PlayerDif>
  </Player_Class>
</ArrayOfPlayer_Class>
```

Ln 20, Col 18 | 705 characters | Plain text | 100% | Windows (CRLF) | UTF-8

What Happened

Every time a player finishes the game, weather they win or lose, all of the data collected and stored in the player class would always be written again even if they had won or lost before

How was this fixed?

This was fixed by including an if statement as well as a Boolean variable to check if the player first exists and if the player does exists it will check if the data already exists, I.E username, year group etc. and if so it will only overwrite the scores and difficulty. On the other hand if the player does not exist then it will write all the data

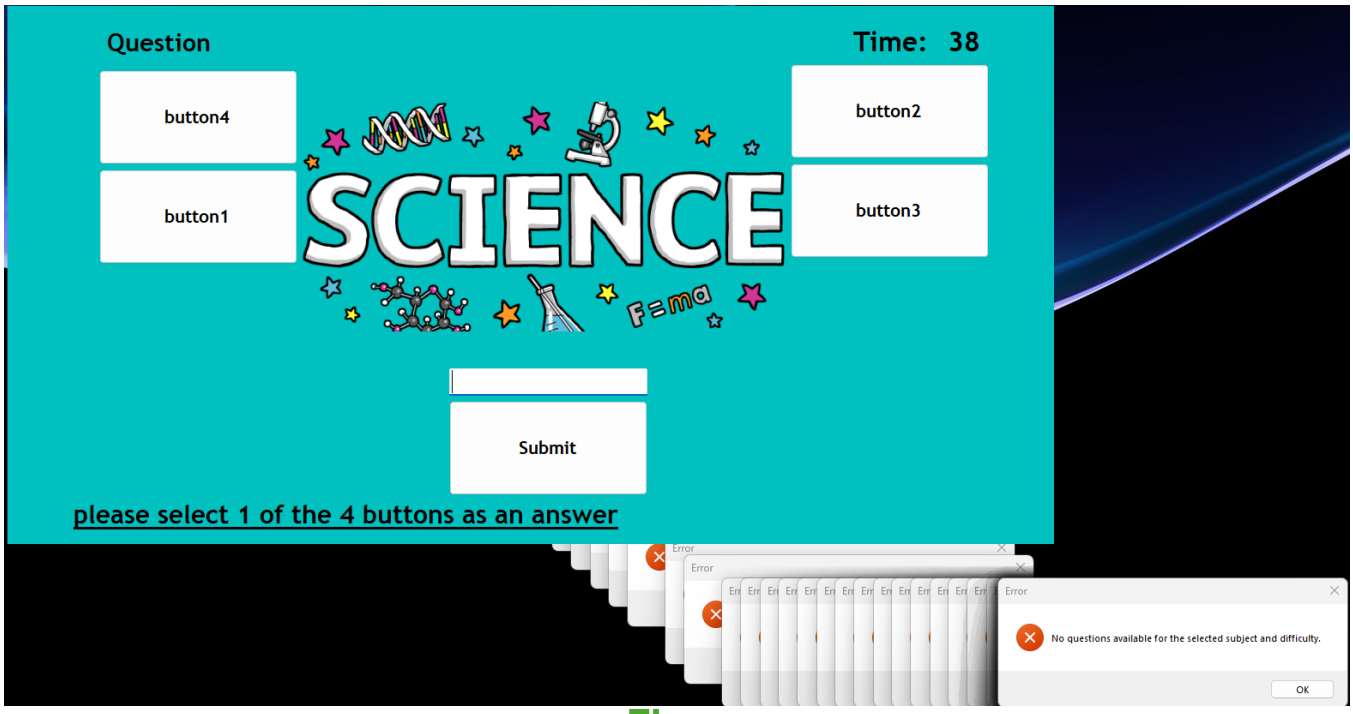
Picture 3

Error's

```
var lines = File.ReadAllLines(filePath);

foreach (var line in lines)
{
    var parts = line.Split('~');
    if (parts.Length < 5)
        continue;


    //assigns each part to a varible
    string subject = parts[2].ToLower();
    bool isMath = parts[1].ToLower() == "yes";
    string diff = parts[0].ToLower();
    string ques = parts[3];
    string answer = parts[4];
    List<string> option = !isMath && parts.Length > 5
                          ? parts[5].Split(',').ToList()
                          : new List<string>();
```



Fixes

```
foreach (var line in lines)
{
    var parts = line.Split('~');
    if (parts.Length < 5)
        continue;


    //assigns each part to a varible
    string subject = parts[0].ToLower();
    bool isMath = parts[1].ToLower() == "yes";
    string diff = parts[2].ToLower();
    string ques = parts[3];
    string answer = parts[4];
    List<string> option = !isMath && parts.Length > 5
                          ? parts[5].Split(',').ToList()
                          : new List<string>();
```

What happened?

When assigning the variables to the parts in the txt file that contains the data for the questions the parts were numbered in correctly causing an error to appear saying no questions were available.

How was it fixed?

It was a simple fix as it was just replacing the parts number to ensure that they where in the correct order.

Picture 4

Error's

```
// if there is less than 3 daleks make more daleks till i (daleks) is equal to 3

for (int i = 0; i < amountOfEnemies; i++)
{
    MakeDaleks();
}
```



Fixes

```
1 reference
private void MainTimerEvent(object sender, EventArgs e)
{
    CheckScore();

    // if there is less than the amount of daleks make more daleks till i (daleks) is equal to the amount of enemies
    while (daleksList.Count < amountOfEnemies)
    {
        MakeDaleks();
    }
```



What happened?

In the Restart game method (called in the form constructor) the if statement was stored there and this if statement would not be executed because the amountOfEnemies variable would be set to 0.

How was it fixed?

This was fixed by removing that if statement and putting it in the timerEvent and making it a while loops so it consistently spawns enemes

Picture 5

1st image is UR 37 and 2nd image is UR 38

Error's

```
1 reference
private void CheckScore()
{
    foreach(var diff in difficulty)
    {
        if (diff.GetType() == typeof(Hard) && gameScore == 35)
        {
            gameTimer.Stop();
            this.Hide();
            TheGameQuiz qForm = new TheGameQuiz(SoundPlayer,difficulty, currentPlayer, difficultyBasedQuestions, lives);
            qForm.Show();
            return;
        }
        else if (diff.GetType() == typeof(Normal) && gameScore == 25)
        {
            gameTimer.Stop();
            this.Hide();
            TheGameQuiz qForm = new TheGameQuiz(SoundPlayer,difficulty, currentPlayer, difficultyBasedQuestions, lives);
            qForm.Show();
            return;
        }
        else if (diff.GetType() == typeof(Easy) && gameScore == 15)
        {
            gameTimer.Stop();
            this.Hide();
            TheGameQuiz qForm = new TheGameQuiz(SoundPlayer,difficulty, currentPlayer, difficultyBasedQuestions, lives);
            qForm.Show();
            return;
        }
    }
}
```

```
//if the player wins
if (livesLeft > 0)
{

    foreach(var diff in Difficulty)
    {
        if (diff.GetType() == typeof(Hard) && player.GameScore == 45)
        {
            this.Hide();
            Win youHaveWon = new Win(player, soundmeplay);
            youHaveWon.Show();
        }
        else if (diff.GetType() == typeof(Normal) && player.GameScore == 25)
        {
            this.Hide();
            Win youHaveWon = new Win(player, soundmeplay);
            youHaveWon.Show();
        }
        else if(diff.GetType() == typeof(Easy) && player.GameScore == 15)
        {
            this.Hide();
            Win youHaveWon = new Win(player, soundmeplay);
            youHaveWon.Show();
        }
        else
        {
            this.Hide();
            Form1 game = new Form1(soundmeplay, Difficulty, QuestionDifficulty, player, livesLeft);
            game.ShowDialog();
        }
    }
```

Fixes

```
1 reference
private void CheckScore()
{
    foreach(var diff in difficulty)
    {
        if (diff.GetType() == typeof(Hard) && gameScore == 40)
        {
            gameTimer.Stop();
            this.Hide();
            TheGameQuiz qForm = new TheGameQuiz(SoundPlayer,difficulty, currentPlayer, difficultyBasedQuestions, lives);
            qForm.Show();
            return;
        }
        else if (diff.GetType() == typeof(Normal) && gameScore == 25)
        {
            gameTimer.Stop();
            this.Hide();
            TheGameQuiz qForm = new TheGameQuiz(SoundPlayer,difficulty, currentPlayer, difficultyBasedQuestions, lives);
            qForm.Show();
            return;
        }
        else if (diff.GetType() == typeof(Easy) && gameScore == 15)
        {
            gameTimer.Stop();
            this.Hide();
            TheGameQuiz qForm = new TheGameQuiz(SoundPlayer,difficulty, currentPlayer, difficultyBasedQuestions, lives);
            qForm.Show();
            return;
        }
    }
}
```

```
//if the player wins
if (livesLeft > 0)
{

    foreach(var diff in Difficulty)
    {
        if (diff.GetType() == typeof(Hard) && player.GameScore == 40)
        {
            this.Hide();
            Win youHaveWon = new Win(player, soundmeplay);
            youHaveWon.Show();
        }
        else if (diff.GetType() == typeof(Normal) && player.GameScore == 25)
        {
            this.Hide();
            Win youHaveWon = new Win(player, soundmeplay);
            youHaveWon.Show();
        }
        else if(diff.GetType() == typeof(Easy) && player.GameScore == 15)
        {
            this.Hide();
            Win youHaveWon = new Win(player, soundmeplay);
            youHaveWon.Show();
        }
        else
        {
            this.Hide();
            Form1 game = new Form1(soundmeplay, Difficulty, QuestionDifficulty, player, livesLeft);
            game.ShowDialog();
        }
    }
```

What happened?

The if statement for hard was set to detect if the players game score was set to 45 in the quiz and set to 35 in the game making the loop unintentionally

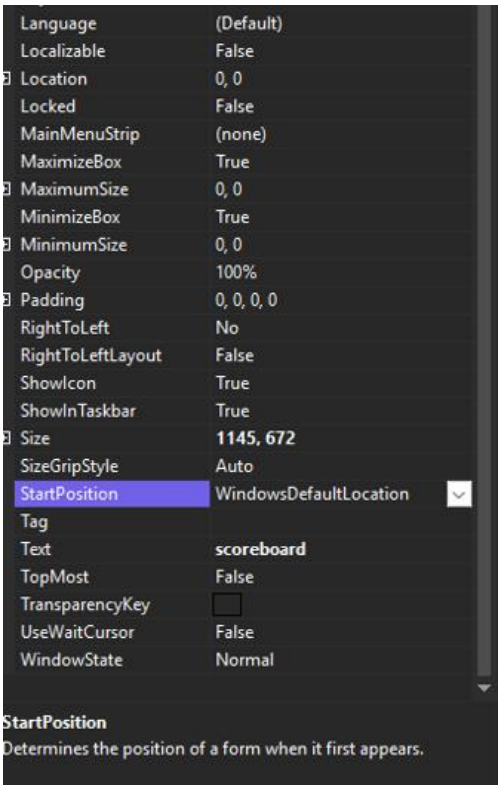
How was this fixed?

This was a simple fix as it was fixed by setting both the if statements to detect if the game score reaches 40.

Picture 6

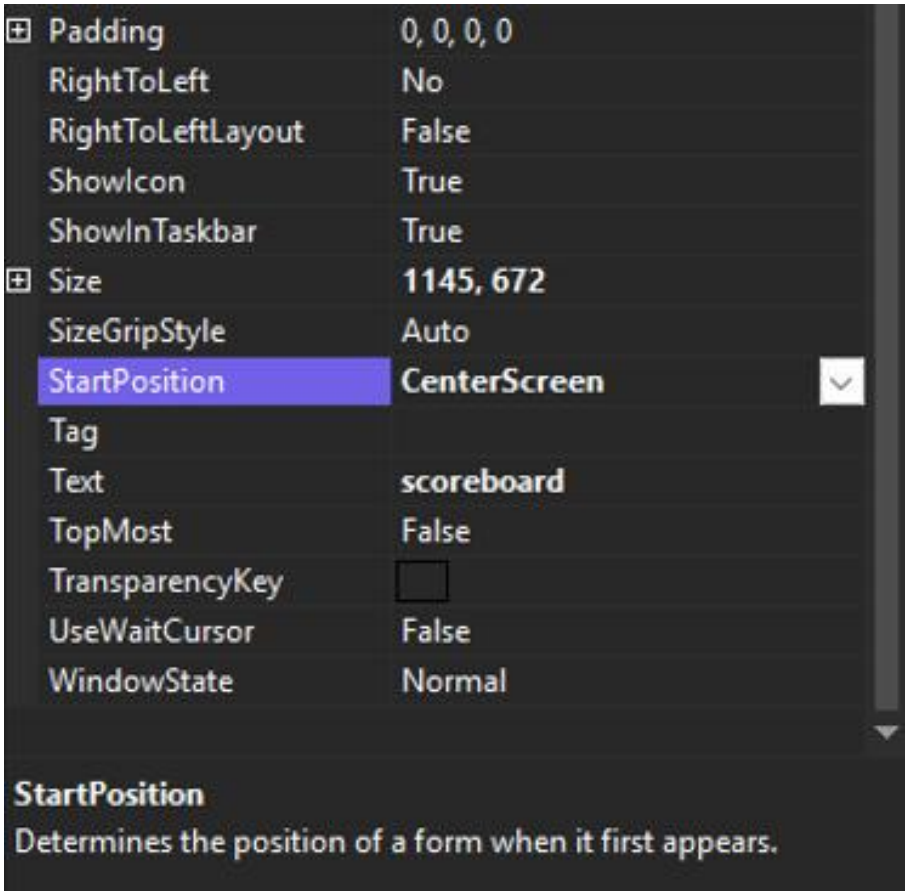
Error's




Language (Default)
Localizable False
Location 0, 0
Locked False
MainMenuStrip (none)
MaximizeBox True
MaximumSize 0, 0
MinimizeBox True
MinimumSize 0, 0
Opacity 100%
Padding 0, 0, 0, 0
RightToLeft No
RightToLeftLayout False
ShowIcon True
ShowInTaskbar True
Size 1145, 672
SizeGripStyle Auto
StartPosition WindowsDefaultLocation
Tag
Text scoreboard
TopMost False
TransparencyKey
UseWaitCursor False
WindowState Normal
StartPosition
Determines the position of a form when it first appears.

Fixes


Begin



What happened?

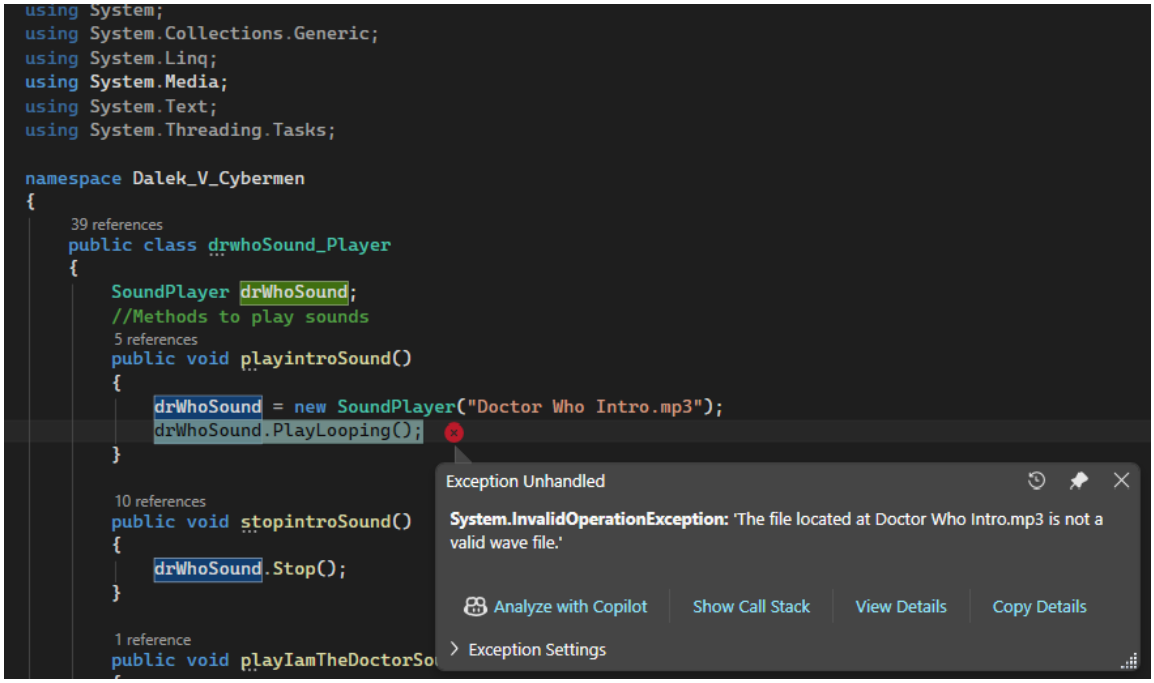
The start position for some of the forms was set to the default windows location

How was this fixed?

This was fixed by setting the start position to center screen

Picture 7

Error's



Fixes

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Media;
using System.Text;
using System.Threading.Tasks;

namespace Dalek_V_Cybermen
{
    39 references
    public class drwhoSound_Player
    {
        SoundPlayer drWhoSound;
        //Methods to play sounds
        5 references
        public void playintroSound()
        {
            drWhoSound = new SoundPlayer("Doctor Who Intro.wav");
            drWhoSound.PlayLooping();
        }
```

What happened?

The sound player was not recognizing the MP3 file as valid wave file and so outputted an exception

How was this fixed?

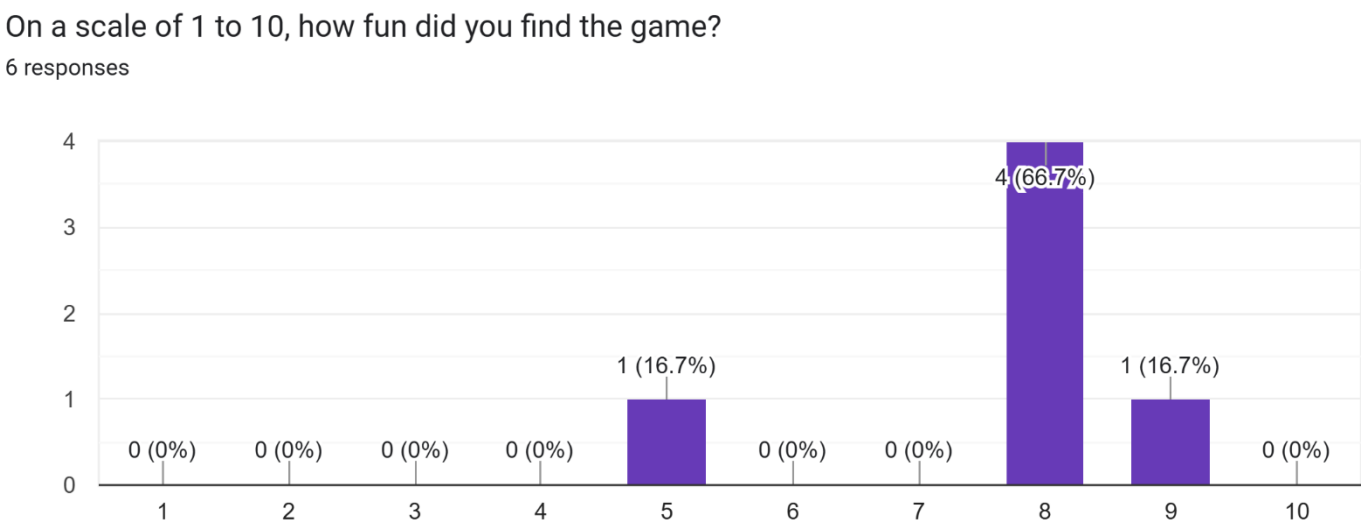
This issue was fixed by converting the MP3 audio file to a wav file and setting the sound player to recognize it as a .wav file

Evaluation

User Evaluation

My user evaluation will consist of 10 questions, all 10 will be closed end questions, closed end meaning yes, no, maybe etc. My users will have to answer all of the questions provided and will do so after completing the game and quiz.

Question 1: "On a scale of 1 to 10, How fun did you find the game?"

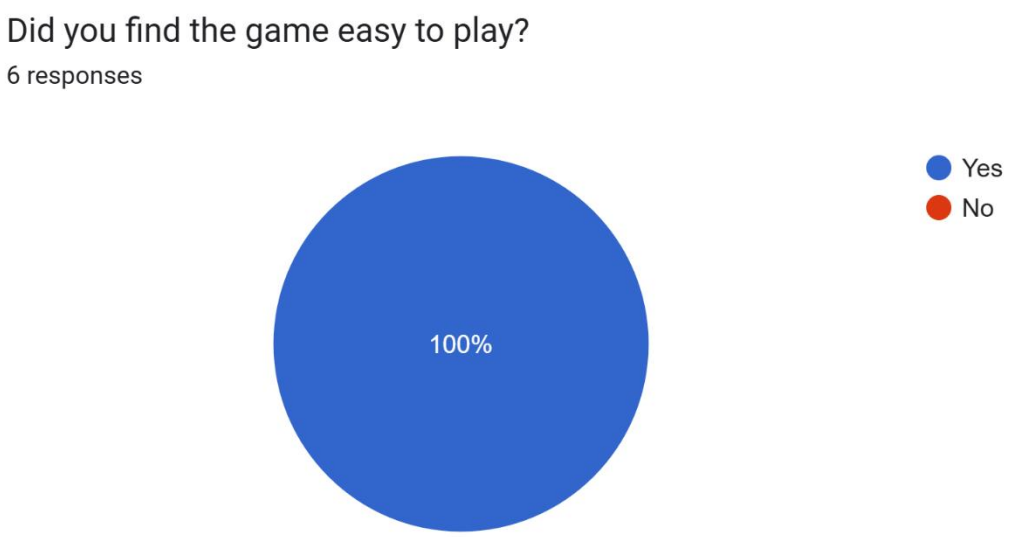


The response for how fun the user found the game were split into 5/10, 8/10 and 9/10 with most common response being 8/10 and the average rating being 7.67/10. The answer to this question shows that most or all of the players found the game enjoyable with most of users giving it a high score. The only concern that I have is that 1 user found the game fun but didn't at the same time, the single rating of 5.

With the single rating of 5/10 shows that the user had a mixed opinion of how enjoyable the game was, this shows that there is room for improvement as to make the game more enjoyable for all users. With no other user rating the fun of the below 5 shows that no other user found the game boring to play.

In summary, most of the users found the game enjoyable with rating like 8 and 9 out of 10 with the average rating being 7.67/10. The only limitation is the single 5/10 rating which could show that this user had a mixed experience. The next time this questions is asked again I would include a suggestion box along with this question as to see what most users would find fun in a game.

Question 2: "Did you find the game easy to play"

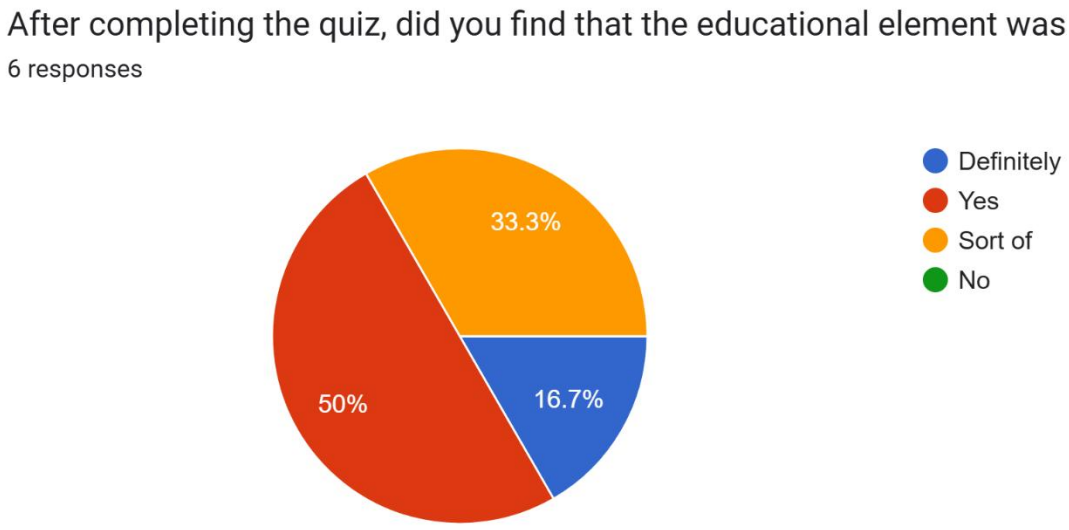


The response for did you find the game easy to play was seen to have a 100% answer for yes. The answer for this question shows that all of the users found the game easy to play. This indicates that the controls I have made for the game were easy to use, as the user could've used either the WASD buttons and space bar or the arrow keys and the space bar. The reason for the 100% pie chart could be because most shooter games use WASD.

Without any user answering the question with no it shows that no user found the game it shows that the movement I have implemented in my game has worked accordingly.

In summary, all of the users who answered the question said that the game was easy to play with a 100% pie chart. With this questions there is no concern as this question has provided no reason to raise concern. If I were to ask this question I would add a follow up question asking the user what method they used to move about, WASD or arrow keys.

Question 3: "After completing the quiz, did you find that the educational element was helpful?"

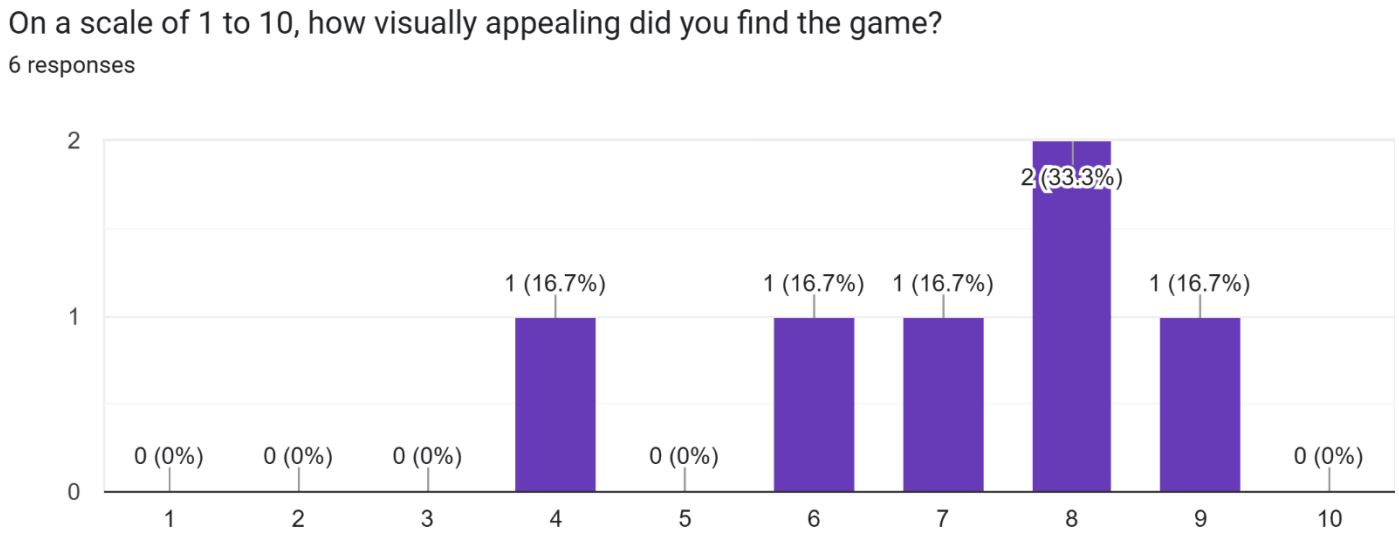


The response for after completing the quiz, did you find that the educational element was helpful has shown that the educational element was helpful with 16.7% of the users answered Definitely, 1 user, 50% of the users choose yes to the question, 3 users, and 33.3% of the users choose sort of, 2 users. These answers show that the educational element that I have implemented into my program has worked.

There is no drawback for this question as none of users who answered the question said that the educational element that I have implemented wasn't helpful. The only half exception there is that 2 of the users said it was sort of help, this shows that there can be room for improvement for the educational aspect of the game.

In summary, the users who said that the educational aspect of the game was helpful shows that the users are now starting to understand the subject than previously. Although 1 of the users said that the educational aspect of the game was sort of helpful this answer could mean that the user doesn't enjoy the subject, if I were to give this question again I would add a continuation onto this question asking what subject they had chosen.

Question 4: "On a scale of 1 to 10, how visually appealing did you find the game?"

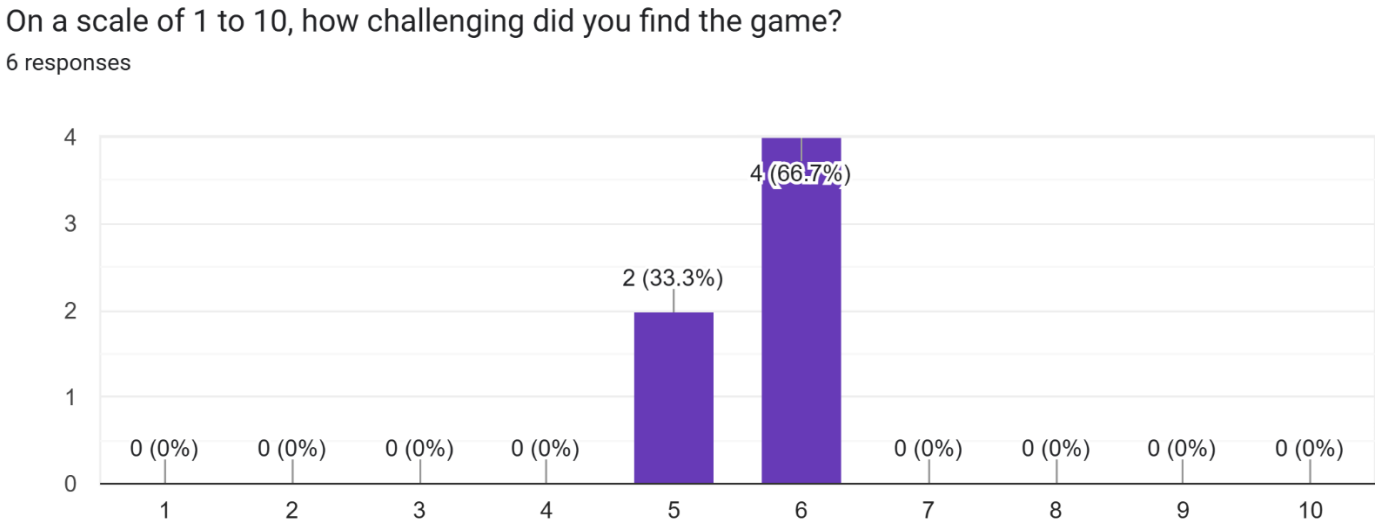


The response for how visually appealing the game was shown to have a spread variety of answers with the most common answer being 8/10 with 2 users choosing it, 33.3% and 1 user each choose 4/10, 6/10, 7/10 and 9/10, each user being 16.7%. These answers show that some users found the game visually pleasing with 1 user finding that the game wasn't visually pleasing.

With the single answer of 4/10 shows that the user found the visual appeal of game to not be appealing, with this single answer it shows that there is room for making the game more visually appealing for all users. With no other users giving a rating of below 4 shows that no user GREATLY disliked how visually appealing the game was.

In summary, most of the users who answered the questions seemed to give the visual appeal of the game with a score of 6 or more with the highest being 9/10 and the most common answer being 8/10. The one issue for this question is one user answered the question with 4/10, although there is room for improvement I will not know what to improve if I were tasked to make a program like this again. But if I were to ask this question again I would add a suggestion box asking the user what they liked and didn't like about the visuals of the game and what they would do to make it more visually appealing.

Question 5: "On a scale of 1 to 10, how challenging did you find the game?"

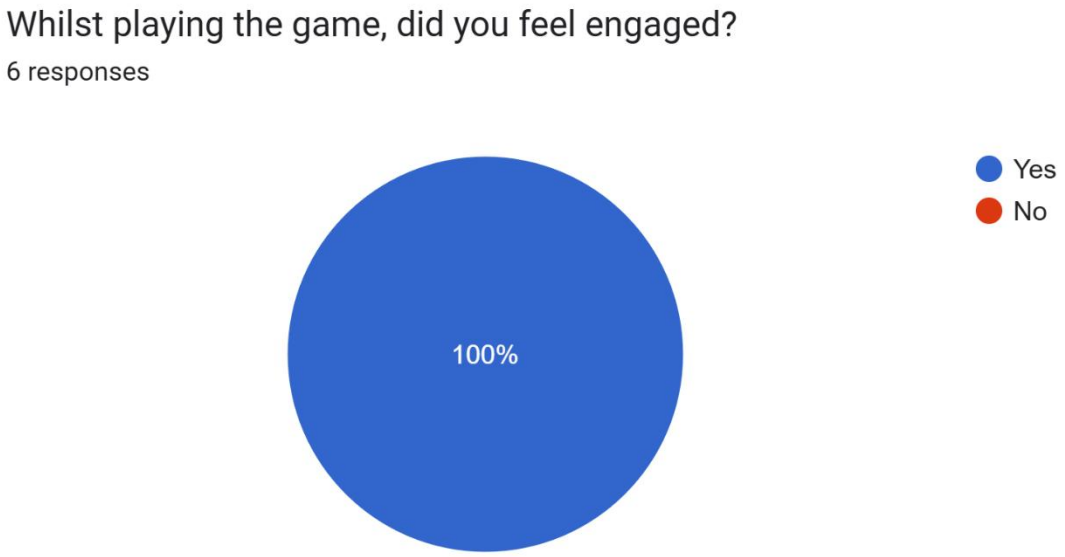


The response for how challenging the game was shown to almost have split set of answers, with the most answered being 6/10. From looking at the chart it shows 4 out of 6 users answered the question with 6/10, a percentage of 66.7. The other 2 users answered the question with 5/10, a percentage of 33.3. These answers show that the 4 users who answered the question with 6/10 found the difficulty of the game to be slightly harder than easier, whilst the other 2 users found that the challenging of the game to be both easy and hard.

With non of the users answering the question below 5, it shows that the difficulty system I have implemented into my game has worked, as well as that with no other users answering the question with more than 6/10 it shows that I have not made my game extremely difficult and unplayable.

In summary, the users who answered the question have shown that my difficulty system works accordingly with the user answers ranging from 5 and 6 out of 10, this shows that's my difficulty system that I have implemented into my game has worked to my standard. If I were to ask this question again I would asked the user what difficulty they chose and what made it seem more unfair than the other difficulties.

Question 6: "Whilst playing the game did you feel engaged?"

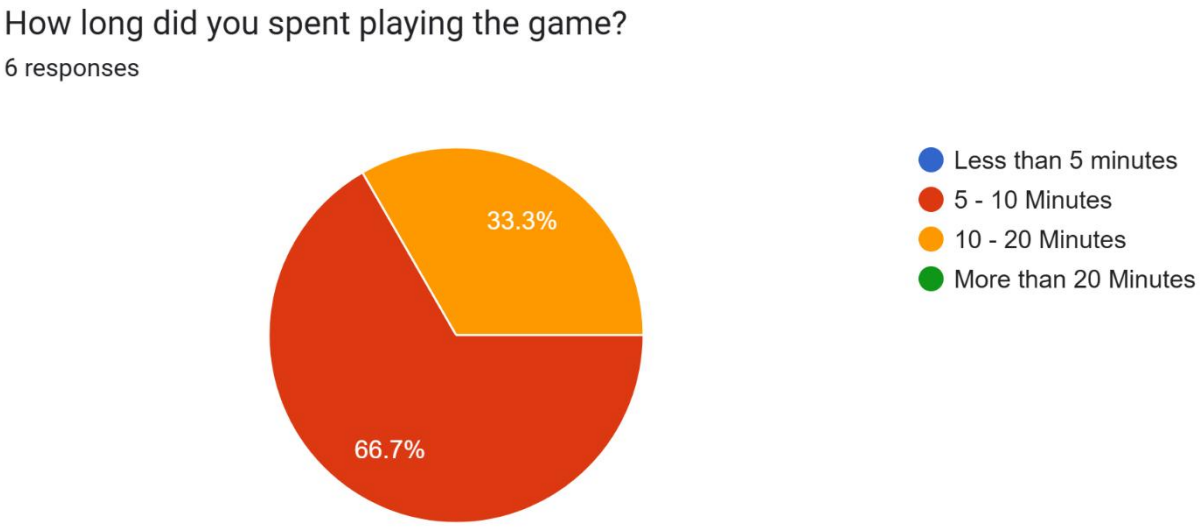


The response for whilst playing the game, did you feel engaged was shown to have a 100% answer response with all of the users answering yes for the question. These answers show that all the users found the game engaging, there is no concern with this question, but if even one of the users answered no it would cause a problem as I had planned to create a game that felt extremely engaging and by looking at the question answers it shows that the game I had created was engaging.

With no user answering the question with no it shows that the game I had designed, built and validated is engaging for the end user and thus there is no need for improvement as this has meet my expectations

In summary, the results I have received from this question have shown that my idea to create an engaging game for the users has been meet as the question I have asked has been answered with yes with, 100%. There isn't any concern with this question has the answers I have received have all met my expectations with no room for improvement needed.

Question 7: "How long did you spend playing the game?"

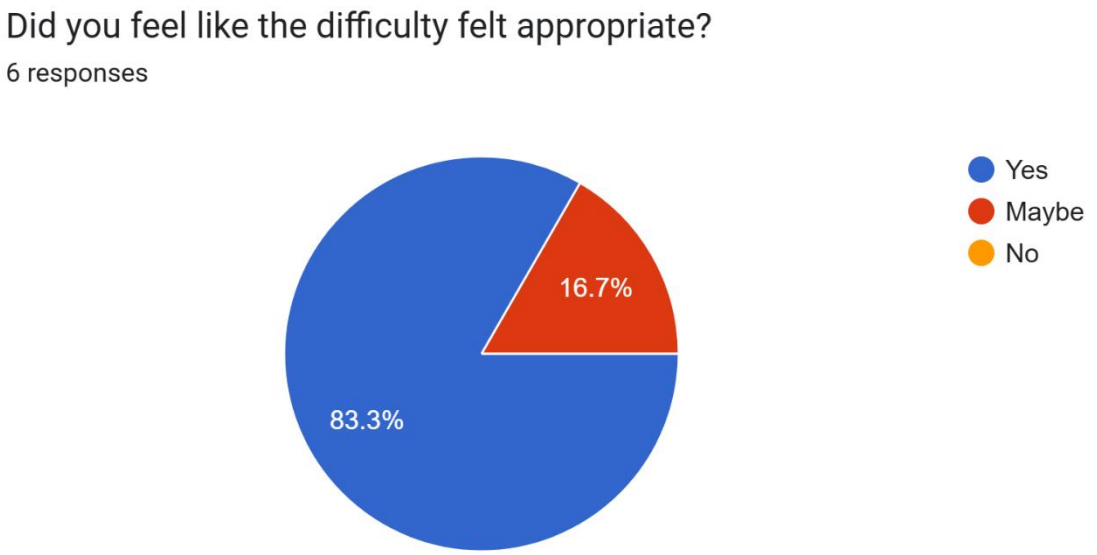


The response for how long did you spent playing the game was shown to have a majority answer of 5 – 10 minutes with 66.7% of the users answering this question, 4 users, and the other 33.3% of the users answering 10 – 20 minutes, 2 users. These answers show that 66.7% of the users didn't spent a long time playing the game, this suggests that the game I had created is a quick and well suited game. The possible reason for why most users only spent 5 – 10 minutes playing the game could be because these users have played a game that is similar but different. But with the other 33.3% of the users who spent 10 – 20 minutes playing the game the possible reason for why they took this long could be that the game I had developed was more challenging for them or it could be that these 2 users were less experienced in the game I had developed.

With this question there is no worry as the answers that the users had provided all seem to be within reason and as such there isn't a need for the play time of my game to be improved as it doesn't cause any worry.

In summary, The majority of players, 66.7% of the users, spent 5–10 minutes playing the game, while the remaining 33.3% of the users spent 10–20 minutes. The shorter play times suggest the game is quick and easy to pick up, which may be because players have experience with similar games. Those who took slightly longer may have found the game more challenging or were less familiar with this type of game. Overall, the play time results are satisfactory and suggest no changes need to be made to the game's length.

Question 8: "Did you feel like the difficulty felt appropriate?"



The result for did you feel the difficulty felt appropriate shows that 83.3% of the users, 5 users, said that the difficulty did feel appropriate, with the other 16.7% of the users, 1 user, said that the difficulty felt appropriate and not appropriate at the same time. These answers suggest that the difficulties that I have implemented into my game and quiz was shown to be well balanced and playable for the users.

With 1 user answering this question with 'Maybe' shows that some of the aspects of my game and quiz difficulty could have been too easy or hard for the difficulty that they have selected. Using this answer it shows that my difficulty system has room for improvement to have a based difficulty system. Although with no user answering the question with 'No' shows that my difficulty system was appropriate and wasn't biased at all.

In summary, this question has shown that 83.3% of the users found that the difficulty system I have implemented into my game and quiz was set appropriately, with only 1 user answering the question with 'Maybe' which could suggest that they found the difficulty appropriate but not appropriate at the same time. This answer could suggest that there is room for improvement, there isn't an issue for this question as no users selected 'No' this shows that my difficulty system wasn't unbalanced. These results show that my difficulty system is generally well balanced for the users. If I were to ask this question again though I would add a follow up question asking the user what difficulty they used as they answered anything but 'Yes' I would be able to make the adjustments necessary.

Question 9: "After completing the quiz, Do you feel more confident about the topic?"

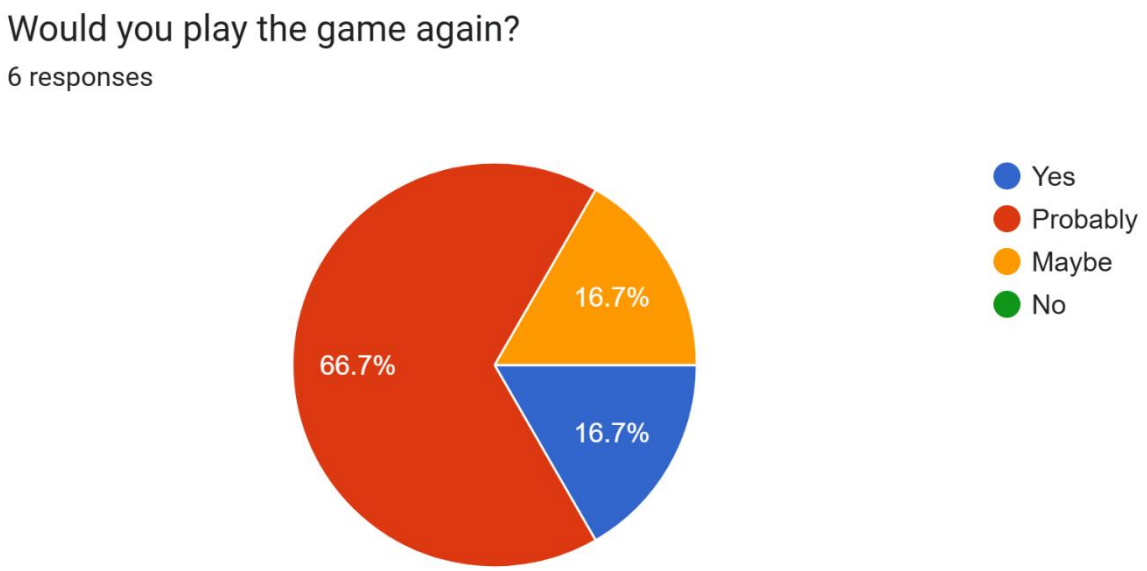


The response for do you feel more confident about the topic was shown to have a range of answers with the most common answer being that the user sort of felt more confident about the topic, with 50% of the users, 3 users, answering this question. Although only 2 users, 33.3%, said that they felt more confident about the topic, and only 1 user, 16.7%, said that they definitely felt more confident about the topic. With the 50% of the users who said sort of shows that there is room for improvement in the educational aspect of my quiz.

There is no GREAT concern for this question, but there is a weakness for the educational aspect of my quiz as with half of the users who answered the question said that they felt sort of more confident about the topic, although I do not know why they didn't feel more confident but I can suggest that they have a different way of learning which could be a good reason for why most of the users answered 'Sort of'. There isn't a GREAT concern for this question as no other user answered the question with no, this enforces that my educational aspect of my game has shown to be successful in a way.

In summary, 50% of the users answered 'sort of' when asked if they felt more confident about the topic, 33.3% of the users answered 'yes ' showing that they felt more confident about the topic and 1 user said that they definitely felt more confident about the topic. The possible reason for why 'sort of' is answered more could be that the users have different learning styles. With the highest response being 'sort of' shows that there is room for improvement in the educational aspect of the quiz. With no users answering 'no' suggests that the educational aspect of my quiz has been partially successful.

Question 10: "Would you play the game again?"



NOTE I WILL COMBINE PROBABLY AND MAYBE AS THEY ARE THE SAME THING

The response for would you play again has shown that the majority of the users said that they would probably play the game again with only 1 other user saying that they would play the game again. These answers could show that my game isn't as entertaining as I had previously thought, this shows that there is room for improvement for the entertainment of my game. There isn't a great concern for my game as no users said that they wouldn't play the game again.

There is an issue for this question as most of the users had said that they would probably play the game again, the reason for this majority could mean that my game isn't as engaging as I had hoped it to be or it could mean that I haven't included a strong reason to play it again.

In summary, most of the players, 83.3%, said that they would probably play the game again with only 1 user saying they would play the game again, these answers could show that my game isn't entertaining or engaging enough as it shows that there is room for improvement in my game. How ever a positive side to this question is that no user said that they wouldn't play the game again. There is an issue for this question but no major concern as the results have shown that players would return to the game.

User feedback summary

The overall feedback from the questions I have asked have shown to almost have a fully positive feedback, this is evident by the rating questions being answered with high ratings such as scores with 7, 8, 9 and 10 / 10, these can be evidenced by questions 1 and 4, or with reasonable answered ratings such as 5 or 6 / 10, this can be evidenced by question 5. The almost fully positive ratings are also evident by the multiple-choice questions with users answering yes and no for most of the questions with the occasional maybe for some of the questions. The user's response for these questions shows that I have created a fun game that has is a playable with excellent visual appeal that engages the user enough to make them play again.

The educational element that I have implemented has shown to be successful and partially successful at the same time with the users answering the questions about the educational element with answers like 'Yes' and 'Definitely', whilst the topics of the subjects haven't shown to be successful for making the user more confident

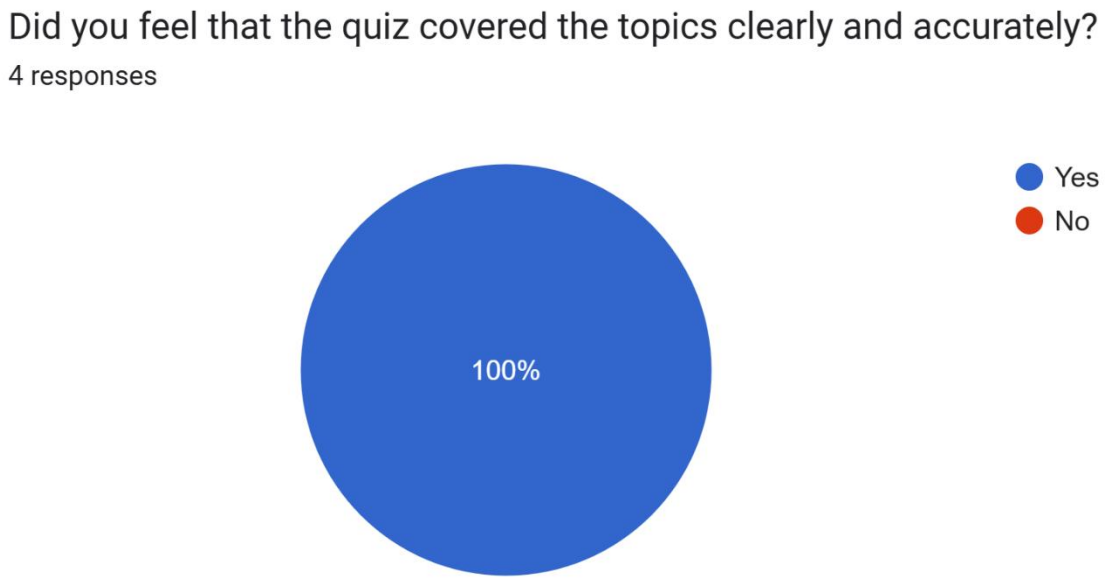
The feedback I received from the hand drawn UIs has shown to be a mix of positivity and negativity. These types of feedback can include changes to the UI such as changes to colours or changing names or adding in a new type of functionality, this feedback can also include improvements to the UI such as making enemies harder to defeat. Upon looking back at the user's feedback, it is clear that the game has successfully captured their interest and has now encouraged them to see the further development of the game.

In summary, I state that the game I have developed has shown to be an enjoyable and engaging experience with an educational learning aspect to it, with a few improvements needed so the experience of the game can be advanced.

Candidate Evaluation

My candidate Evaluation will consist of 10 questions which will be a mix of closed end questions as well as open end questions. The candidates will answer these questions after they have observed the users, their students, playing the game.
NOTE Before the client had answered the 10 questions, I had asked them which head of department they are so I could see what client answered/said what.

Question 1: "Did you feel that the quiz covered the topics clearly and accurately?"

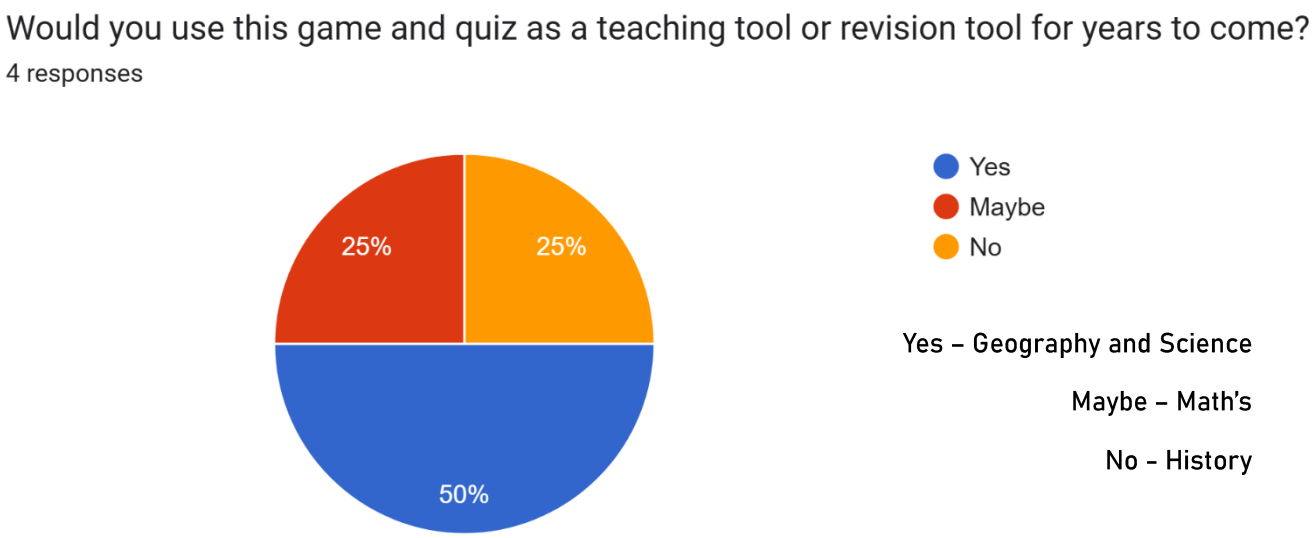


The responses for 'Did you feel that the quiz covered the topics clearly and accurately?' has shown to be a 100% single answer with all of clients answering this question with Yes. These answers reflect that my questions I have implemented and asked are shown to have successfully be accurate towards what the clients have been teaching. With these answers it shows that I have made no room for error for the educational aspect of my quiz.

With these question answers there isn't an issue present as all of clients have answered the question with no, further reflecting that my question topics and subjects are accurate to what content is being taught.

In summary, all of the clients have shown that my questions have been clearly asked and are specific to what the clients are teaching in class, along with no issue present further enforces that I have created a fully working educational quiz.

Question 2: "Would you use this game and quiz as a teaching tool or revision tool for years to come?"



The response for 'Would you use this game and quiz as a teaching or revision tool for years to come', has shown to be a range of 3 answers with the geography and science client answering this question with yes showing that I have created a quiz suitable for not only learning but also revision for when the time comes. The math's client answered this question with 'maybe' this answer reflects that, although it has covered the topics accurately and clearly, it mightn't be the best tool for teaching or revision. The history client answered this question with 'no' this answer reflects that my quiz isn't an excellent way to teach or revise this subject, this answer shows that there is an area for me to improve upon.

The area to improve upon is with the history client answering 'no' without knowing the proper reason why the client answered no I can assume a reason such as the curriculum might change as history is constantly looking into the past and my questions won't be accurate to what is being taught in class for years to come.

In summary, the geography, science and client's have all said that they would use my game and quiz as a teaching or revision tool, with math's answering with 'maybe'. The history client has said that they wouldn't use my game and quiz as a teaching or revision tool, I had assumed a reason for why the client answered no but without a the proper answer I will not know how to improve the quiz. If I were to ask this question again I would add along side a suggestion box so if any clients answer maybe or no they will be made to say what can be done so the quiz can better suit their subject.

Question 3: "What did your students enjoy most about the game and or quiz?"

What did your students enjoy most about the game and or quiz?

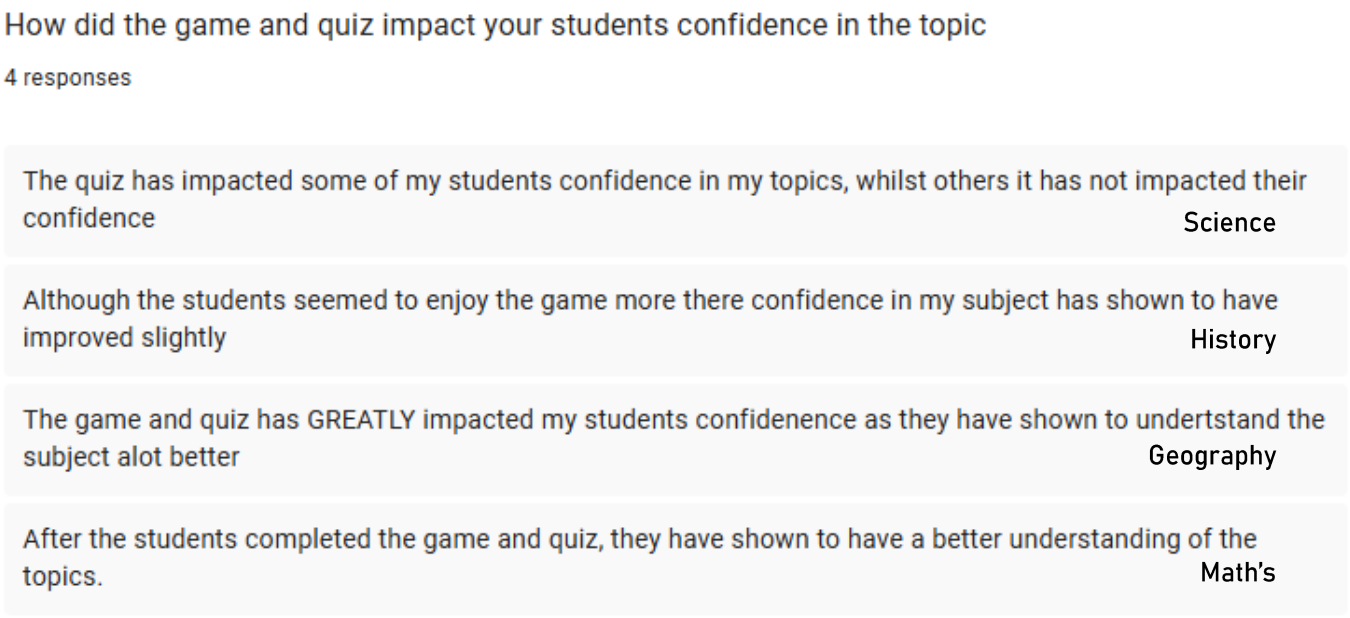
4 responses

My students have show to enjoy the quiz more so than the game	Science
The students seemed to enjoy the game side more than the quiz side	History
My students have shown to enjoy both the game and quiz	Geography
My mathematics students enjoyed the game part, but found the questions engaging	Math's

The response for 'what did your students enjoy most about the game and or quiz' has shown to have a variety of answers from all 4 clients with 1 of the clients saying that the users enjoyed the quiz, and the other 2 clients saying that the users enjoyed joyed the game, with the other client, the geography client, saying that their students enjoyed both the game and the quiz. There is no need for concern as the clients have said that the students have at least enjoyed either the game or quiz.

In summary, the science client said that their students had enjoyed the quiz, whilst the math's and history clients said that their students had enjoyed the game aspect with the geography client's students enjoying both the game and the quiz. There has been no concern raised as the clients have noted that their students have at least enjoyed the quiz and or game.

Question 4: "How did the game and quiz impact your students confidence in the topic?"

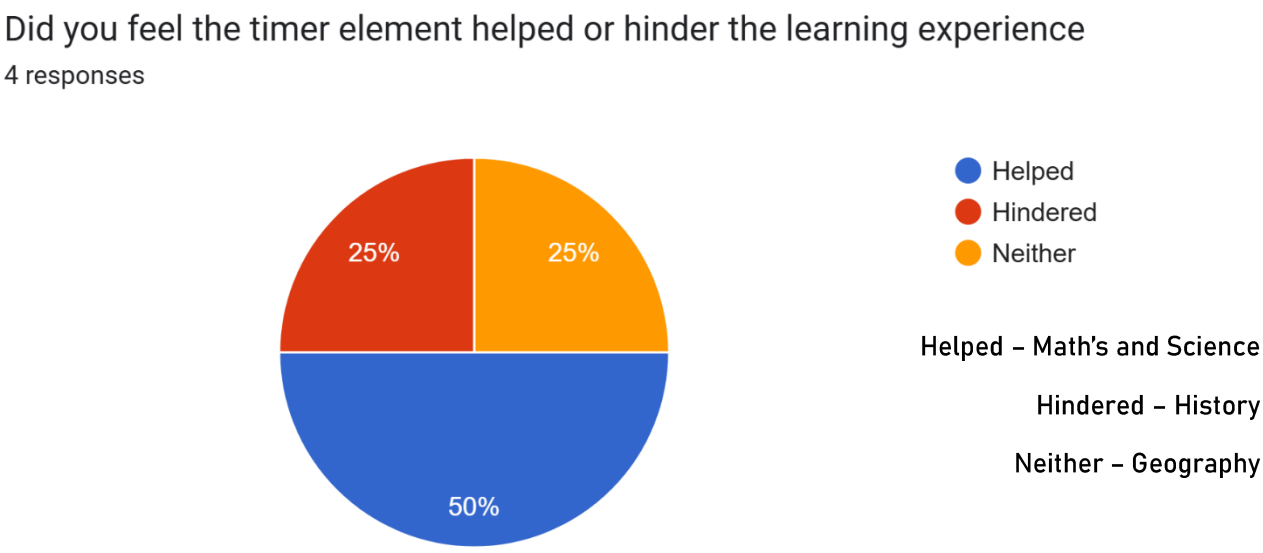


The response for 'how did the game and quiz impact your students confidence in the topic' has shown that all of the students, user, confidence in the subject has been impacted as all of the clients have at least said in their response that my game and quiz has impacted their confidence. For the science questions the answers have reflected that it has impacted some of their confidences, for the history questions it has impacted all of the students confidences but only slightly, for the geography questions it has greatly impacted all of the students confidences in the topic and for the math's questions it has impacted all of the students confidence in the topic to a better understanding. From looking at the answers, there is no indication of an issue with this question as my educational aspect of my game has been achieved.

There isn't an indication of an issue with this question as all of the clients have said that it has impacted their students confidence in some way or another.

In summary, all of the clients have shown that the game and quiz has impacted their students confidence in the topic in some sort of way with geography students being impacted the greatest, math's students were able to understand the topics better, history students were only impacted slightly and science students were only impacted slightly. From these answers I have been able to find out that my game and quiz works good for some subjects but not for others as different subjects may require different approaches to learning within my game and quiz. If I were to ask this question, I would reflect upon what I could improve such as adding in a suggestion box after the question asking the client what I could do to make the questions more accurate.

Question 5: "Did you feel the time element helped or hindered the learning experience?"



Helped – Math's and Science

Hindered – History

Neither – Geography

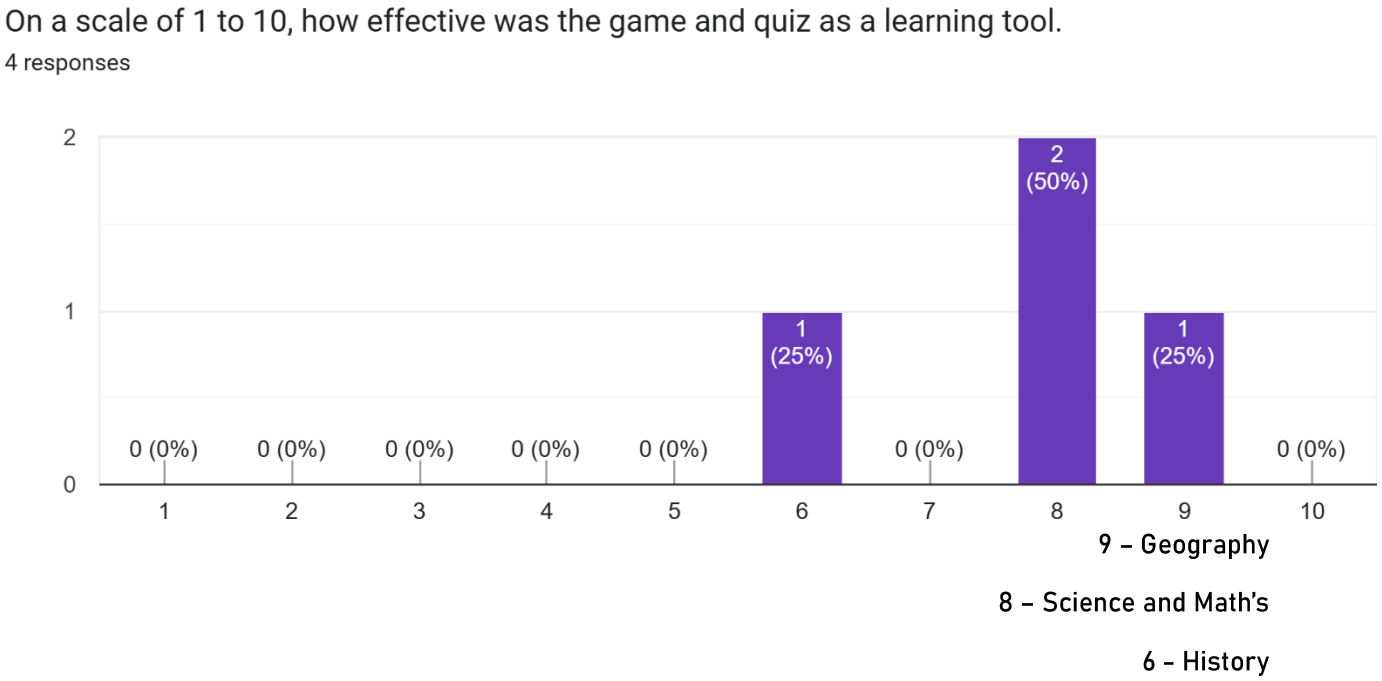
The response for 'did you feel the timer element helped or hinder the learning experience' as shown to have a majority of the clients answering that the timer element helped the learning experience, answered by the Math's and Science clients with the history client saying that it hindered the learning experience and Geography answering that the timer neither helped nor hindered, this and the history clients answer could suggest that there might be a need for some concern. From looking at the Math's and Science client answers it tells me that they felt that the timer helped their students learning experience as the timer could've added a sense of challenge and engagement, this was one of my intended goals when designing my program.

As I had previously said there might be need of concern as looking from the responses it tells me that the history client said that timer hindered the learning experience, looking back this answer makes sense as with history it requires more detail but with my questions for my users I had made them all multiple choice for history, reflecting upon this answer I should've accounted the time it would take for the users to answer the question and set it accordingly. From looking at the geography clients answer it tell me that the timer I had implemented has shown to make no difference in learning or hindering the users progress in learning their subject, reflecting upon this answer it is making me consider weather the timer I have implemented has shown to add any real value as the multiple choice questions might've made the questions very straightforward.

In summary, the majority of the answers have said that the timer element that I have implemented into mt program helped the learning experience especially with

math's and science as they are both challenging subjects along with it, it added a sense of challenge and engagement. With the history client they felt that the timer hindered the users experience, although the questions were multiple choice the history subject itself requires more reading and consideration time and upon reflection the timer should've been adjusted. With the Geography client they answered that the timer neither helped nor hindered, this raises a further question of weather my timer element added any real value for that subject, as the questions were multiple choice it might've made the questions straightforward enough.

Question 6: "On a scale of 1 to 10, how effective was the game and quiz as a learning tool?"



9 – Geography

8 – Science and Math's

6 - History

The response for 'on a scale of 1 to 10, how effective was the game and quiz as a learning tool' has shown to be widely effective with 3 of clients answering with ratings of 8 and 9 out of 10 those being answered by the geography, science and math's clients with the lowest rating being 6/10 by the history client this could cause a weakness as it shows the client didn't find the game and quiz as a learning aid. The high rating answers show that the game and quiz was an excellent learning tool for the client's students this reflects that my aim to have a educational game and quiz has been successful.

The weakness I had identified has shown to be a only higher than average rating of 6/10 this shows that my game and quiz has only achieved a slightly better than average rating as a learning tool by the history client. Although a good potential reason for why the client has answered with this score could be that history requires more thinking and reading.

In summary, the response for the question has shown to have achieved a high overall rating of 7.75/10 this tell me that my game and quiz has shown to be successful as a learning tool. With the weakness of the history rating, which most likely brought down the overall rating, the client answered with 6/10 this could be because that this subject requires more reading and thinking. If I were to ask this question again I would add a follow up question so that the clients could say what could be added to further improve the game and quiz as a learning tool.

Question 7: "What additional subjects would you like the quiz to cover in the future?"

What additional subjects would you like the quiz to cover in the future?

4 responses

having all 3 sciences as seperate subject	Science
Politics	History
English and Computers	Geography
None	Math's

The response for 'what additional subjects would you like the quiz to cover in the future' has shown to have a range of different answers. The science client said that they wished for each science subject as a separate stand alone subject so they are basically asking for individual questions for biology as a subject, chemistry as a subject and physics as a subject, although the client hasn't said why they would like these subjects to be included I can assume, that by looking at my science questions that I have only include 10 of each science subject, 10 biology, chemistry and physics, that I haven't included enough questions for each science subject. For the history client they have asked that I add in a politics subject this answer makes sense as not only is the history client the head of history she is also the head of politics so this answer makes sense for why she has asked for a politics subject. The Geography client had unexpectedly answered the question with English and computers which completely surprised me has I hadn't expected an answer like that from the geography client, when the geography client asked for English I could understand why they had said that as English is an important widely taught subject in schools so I can understand why the geography client has asked for English and will thus make my program more versatile and useful across the school and maybe other schools, as for the geography client asking for computers it is fitting that it is asked since my program it self is a computer application, the geography client could've asked for this subject as they have seen their students playing and engaging with the program and recognizing that they are already in a computer-like environment. Finally the math's client has answered the question with none this could mean that the math's client thought that I have included all of the relevant subjects, although I am questioning why the math's client didn't answer with English as I hadn't included English as a topic in my program.

In summary, the science client has said that they want each scientific subject as their own subject those being : Biology, Chemistry and Physics which is suggesting

that in my quiz I hadn't included enough questions for the science questions. The History client had asked for a politics subject which now makes sense as I later found out that she is also the head of the politics department at Saint Patrick's Grammar School, Armagh. I was surprised by the geography client asking for English and computers, how ever after looking into it I can understand why as English is a core subject, for Saint Patrick's Grammar School, Armagh and other schools, and my program it self is computer based. The Math's client answered none which I found question my self as I had not included English in my program as English and math's are both vital subjects, this answer makes me wonder if the client had fully considered their answer.

Question 8: "What improvements would you suggest to make the game and quiz more effective for learning"

What improvements would you suggest to make the game and quiz more effective for learning?

4 responses

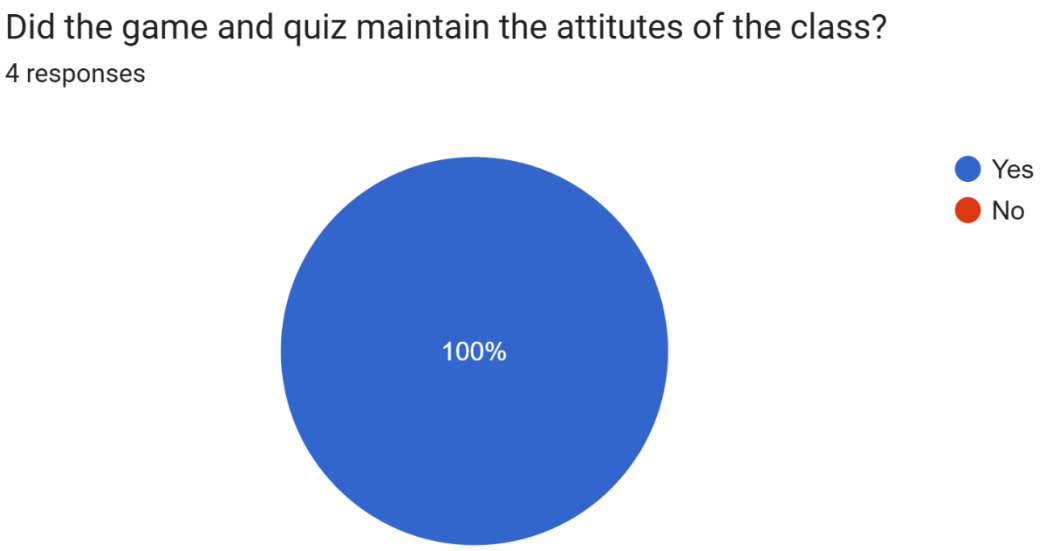
None, as far as i am concerned the game ad quiz is perfect and no improvements are needed	Science
Have a wide more range of questions, such as from different times like the troubles	History
None, the game and quiz is perfect as they are	Geography
the amount of lives you are given depends on how many questions you answered correctly	Math's

The response for 'what improvements would you suggest to make the game and quiz more effective for learning' has shown to a split set of answers with the science and geography clients saying that no improvements were needed and both the game and quiz works correctly, these answers show I have created and designed a fully operational game and quiz for my users and clients.

The responses from the history and math's clients have shown that they would make improvements to the game and quiz, the history client had asked for more questions that contain different topics with the client noting Northern Irelands own History the troubles and the math's client had asked that you gain lives through a different way instead of them being chosen for you, they suggested that you answer questions for your lives for every correct answer you get you are given 1 life. These answers suggest that the clients found ways of making the game and quiz more engaging or educational to suit their students.

In summary, the Science and geography clients found no need to add any improvements towards the game and quiz suggesting that u have created and designed an operational game and quiz to the standards I have set for my self and my clients and their students. As with the history and math's clients they saw that my game and quiz could be improved and thus gave suggestions like having a wider range of questions including Northern Irelands own history the troubles, and changing the way you obtain lives in my game. If I were to ask this question again I would have it be a closed end question and if yes was selected the clients would be able to type in the improvement that should be added.

Question 9: "Did the game and quiz maintain the attitudes of the class?"

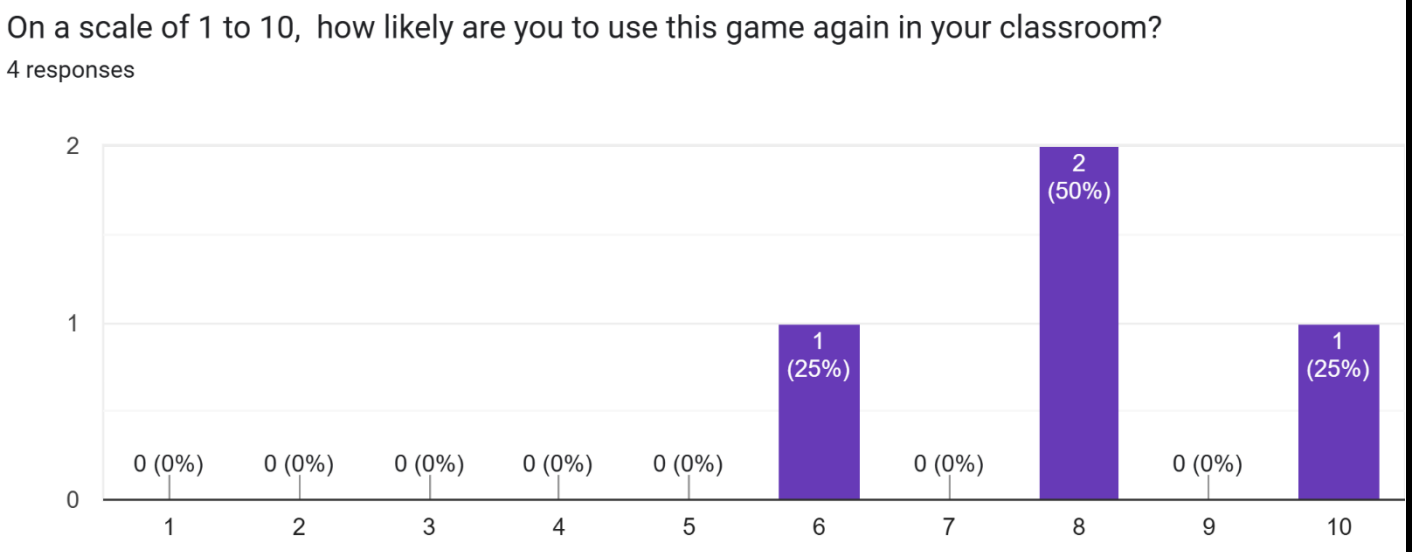


The response for 'Did the game and quiz maintain the attitudes of the class ' has shown to achieve a rating of 100% yes, these answers tell me that the attitudes of the clients students where maintained by them playing my game and quiz, although I do not know if the attitude was a well-behaved attitude I can only assume it was.

Without the absence of any negative answers tell me that I have created a an engaging game which is was one of my goals when starting this project.

In summary, the question has received a 100% answer of yes showing that I have designed and created an engaging game for the clients students, with the absence of the answer 'no' shows that all the students where well behaved whilst playing the game further showing I've created an engaging game. If I were to ask this question again I would add in a follow up question asking the client if the game maintained the attitudes better than your usual teaching methods.

Question 10: "On a scale of 1 to 10, how likely are you to use this game again in your classroom?"



10 – Geography

8 – Science and Math's

6 - History

The response for on a scale of 1 to 10, how likely are you to use this game again in your classroom has shown to have a spread range of answers ranging above the median (5/10). The most answered rating, 8/10, was answered by the science and math's client's these answer suggest to me that my program is effective in a learning classroom environment, reflecting upon these answers it shows me that there is still room for improvement as the final 2 rating tells me that there might be limitations that I am not aware of. For the geography client they answered the question with a rating of 10/10, this answer tells me that there is no limitations towards the program itself and thus showing that I have no room for improvement. For the History client they answered the question with 6/10, this answer tells me that they might use my program again but not as often as this client has answered the lowest out of all clients, this answer also shows that there is room for improvement.

The MAIN area to improve upon would be with the history clients answer as this answer tells me that there is great room for improvement towards the history side of the program and referring back to question 8 the history client suggests that there be more relevant questions to where their students currently are (questions about the troubles) The other area to improve upon would be with the Science and Math's clients although these clients have answered with a high rating this is still an area I believe to be improved upon, for the science client I believe a possible factor could be that in question 8 the science client had suggested that there be an

individual subject for biology, chemistry and physics as I only had 10 questions of each all put under the one subject, as for the math's client I believe that the factor affecting this rating is the timer I had set as I mightn't of set the correct allocated time for solving math's problems.

In summary, this question has received a range of high scores being 8 and 10 out of 10, with 1 lowish score being 6/10, other than that score the answers have shown me that the clients would use my program again in their classroom. I had identified an area of which to improve upon and found some reasons for why that area needed to be improved. If I was to ask this question again I would add a follow up question for why they choose that rating and if it was below 10 I would ask what would be needed to give it a rating of 10/10.

Candidate feedback summary

From reviewing the clients answers and their opinions all 4 clients have shown to be well pleased with my final program, as I received an overall positive feedback from the clients. The 4 clients had all said that my quiz had covered all of the topics accurately and clearly, this makes me feel that I have fully demonstrated the time and research that I carried out into each topic and their related subject. I am very glad that the Geography client rated their likelihood of reusing my program at a score of 10/10 and both science and math's clients rated it 8/10, these scores that I was given by the clients tells me that the program that I designed and developed will be useful tool in the classroom's of Saint Patrick's Grammar School, Armagh and hopefully other schools. I am also very satisfied that all of my 4 clients, Math's, Science, History and Geography, had agreed that my quiz aligned with what they were teaching in class, this to me feels like one of my most important goals when I was designing the educational aspect of my program.

However, upon reflecting on my clients answers I had noticed that they have identified a number of issues that I must acknowledge. The client that was the most notable area of concern was the history client, as they were the only client to not use my program as a long term teaching aid as well as a revision tool, the client had also rated the effectiveness and likelihood of reusing my program at a score of 6/10. Reflecting upon this I believe that this can be linked to the true nature of the history subject as the subject changes overtime this means that my questions I have made for the history subject could become outdated, as well as that my timer element I felt hindered the history clients students learning experience. I had also felt the my multiple choice format may not have suited the history subject as well as it did for other subjects, the reason why I believe this is because this subject typically requires more reading and thinking.

Reflecting upon the educational aspect of my program has show to be well received as looking at questions relating to subject content has shown that those questions have received a high rating score or majority rating.

In summary, I believe that I have designed and developed a game that has met all of my main aims I had set out to complete as well as achieving some side goals I hadn't initially planned as well as that the aims set by the client have also been achieved. If I were to recreate my program I would only change what I had based the game off of as I had based it off of a show only mainly know in England, and I would instead base it off of a show much more known such as the Irish show Derry Girls.

Evaluation of User Requirements

UR NO.	User Requirement	Degree of Fulfilment	Summary of outcome
1	Buttons must bring the user to a new form	Fully met	This original user requirement was able to be fully met as this is an important requirement as it is needed to bring you to new forms
2	Buttons must bring the user back to a previous form	Fully met	This original user requirement was able to be fully met as this is an important requirement as it is needed to bring you to back to previous forms
3	There must be a button for the user to log out	Fully met	This original user requirement was able to be fully met as all that was needed was to only bring over the soundPlayer class
4	Whilst in the game, the user must be able to pause the game	Fully met	This original user requirement was able to be fully met as this is important incase the user wanted to change difficulties.
5	The user must be able to return to the main menu	Fully met	This original user requirement was able to be fully met as all I had to do was pass the class containing the player's details back into the main menu
6	1 of 3 difficulties must be chosen via buttons before playing the game	Fully met	This original user requirement was able to be fully met as all I had to do was assign the correct difficulty selected.
7	1 of 4 buttons must answer questions	Fully met	This original user requirement was able to be fully met as this is an important requirement as all what was needed to was to assign each button to the 1 click event, allowing the user to answer the questions provided
8	There must be a submit buttons for math's questions	Fully met	This original user requirement was able to be fully met as this is also an important requirement as all what was needed was to create a simple click event to check the answer in the text box and move onto the net question
9	The user must be able to enter the answers for math's questions	Fully met	This original user requirement was able to be fully met as this is another important requirement as it is needed to answer questions
10	The user must be able to enter a password and or re-enter password	Fully met	This original user requirement was able to be fully met as all I had to do is check the contents of the password and re-enter password text box to see if they match
11	The user must be able to enter their username	Fully met	This original user requirement was able to be fully met as it allows the user to enter in a name to their liking such as a nickname, this is important as it allows the game and quiz to add points to the users score
12	The user must be able to choose the year group that they are currently in	Fully met	This original user requirement was able to be fully met as all that was needed was to give the user options to choose from, This is an important requirement as part of it is required for the questions.
13	The user must be able to choose their favourite subject	Fully met	This original user requirement was able to be fully met as all I had to do was give the user options to choose from. This is also an important requirement as it is required for the questions

14	The users account details must be saved	Fully met	This original user requirement was able to be fully met as all I had to do was save the details to a txt file
15	The game and quiz score details and about the player must be saved	Fully met	This original user requirement was able to be fully met as all I needed to do was save the data to a .xml file
16	With the saved user details, the user must be able to log in	Fully met	This original user requirement was able to be fully met as all I had to do was check against the data entered with the data that was previously saved
17	With the saved game and quiz details and about the player must be shown for the scoreboard	Fully met	This original user requirement was able to be fully met as all I had to do was display the saved data onto a data grid view
18	The top 3 players must be on the podium	Not met	This original user requirement was not able to be met as I had changed it from displaying the top 3 players to all of the players.
19	Pre-saved question(for subject and difficulty) must be shown	Fully met	This original user requirement was able to be fully met as all I did was read and display the questions
20	The password and re-entered passwords must match or else the user will be told that they do not match	Fully met	This original user requirement was able to be fully met as all I had to do was check the 2 entered password to check if they match
21	Each username must be unique and if not the user will be told that they must enter a different username	Fully met	This original user requirement was able to be fully met as all I had to do was check against the already saved accounts usernames with the entered username. This is an important requirement as it will prevent the duplication of accounts with the same username
22	Check each account must be unique and if not the user will be told that the account already exists	Fully met	This original user requirement was able to be fully met as all I had to do was check against the saved username and password with the entered username and password. This is an important requirement as it will prevent the duplication of accounts.
23	Check that everything in the register form must not be empty and has been filled out	Fully met	This original user requirement was able to be fully met as all I had to do was check that some form of data has been entered in all available slots.
24	Check that the username must contains no capitals, be more than 6 letters and less than 30	Partially met	This original user requirement was able to be partially met as any new username created must contain 3 numbers that ranges from 100 – 999 this is to further make every username unique.
25	Check that the password must contain at least 1 capital, a number and have a length of 10 and less than 30	Fully met	This original user requirement was able to be fully met as all I had to do was check that the password entered meets the criteria I have set.
26	The log in fields must be checked to ensure that they are not empty	Fully met	This original user requirement was able to be fully met as all I had to do was check that the log in fields were at least populated with data.
27	Game must check that there is a difficulty being used	Fully met	This original user requirement was able to be fully met as all I had do was check if any of the game difficulty classes were null. This is an important requirement as it is needed to play the game.
28	Quiz must check that there is a difficulty being used	Fully met	This original user requirement was able to be fully met as all I had to do was check if any of the quiz difficulty classes were null. This is an important requirement as it is needed to play the quiz.

29	Quiz must check that there is a subject to use for the questions	Fully met	This original user requirement was able to be fully met as all I had to do was import the questions into the list and check if the count was less than 1
30	A loading bar must actively increase/decrease	Fully met	This original user requirement was able to be fully met as all I had to do was increment/decrement the bar by 5 or 1
31	A timer that must take away 1 second	Fully met	This original user requirement was able to be fully met as all I had to do was take away the time in milliseconds relative to one second
32	Each enemy killed must earn 1 game score point. Score must update	Fully met	This original user requirement was able to be fully met as all I had to do was add 1 to the score every time the bullet collided with the enemy
33	If the user has ran out of ammo, ammo must spawn. Ammo must update	Fully met	This original user requirement was able to be fully met as all I had to do was check if the ammo was less than 1 and if so spawn in an ammo crate
34	if the user is below a certain health, health packs must spawn. Health must update.	Fully met	This original user requirement was able to be fully met as all I had to do was check if the user was below a certain health and spawn in the health pack once
35	enemies must spawn until they meet a certain number(dependent on difficulty)	Fully met	This original user requirement was able to be fully met as all I had to do was assign the amount of enemies to a variable and spawn enemies until the array, that the enemies spawn from, is equal to the variable
36	user must use WASD or the arrow keys to move about	Fully met	This original user requirement was able to be fully met as all I had to do was make 2 events for pressing the keys up and down
37	user must use the space bar to shoot in the current direction they are in.	Fully met	This original user requirement was able to be fully met as all I had to do was make 2 events for pressing the space bar up and down
38	Help forms for all forms must play a video upon loading into the form.	Fully met	This original user requirement was able to be fully met as all I had to do was give the player the url to the video and ensure that it is a mp4 file
39	The user must see if they'd like to see easy, normal or hard player states via pressing buttons	Fully met	This original user requirement was able to be fully met as all I had to do was only display the sets of data that contained easy, if the easy button was pressed and etc.
40	Each form title must be presented at the top in a big font size	Fully met	This original user requirement was able to be fully met as all I had to do was make the font size big
41	For entering data the text for each box must be accurately labelled	Fully met	This original user requirement was able to be fully met as all I had to do was make sure that the correct label was beside the correct box. E.g. the username label beside the box for the username
42	Help buttons must be in corners	Fully met	This original user requirement was able to be fully met as all I had to do was position the help button in the corner
43	There must be sprite images for moving images	Fully met	This original user requirement was able to be fully met as all I had to do was change the image if the player/ enemy was moving right I would display the moving right image.
44	Everything must be centered	Fully met	This original user requirement was able to be fully met as all I had to do was set the start position to center screen.
45	There must be images of the enemy	Partially met	This original user requirement was able to be partially met as I had only included the 1 picture of the enemy.

46	There must be images of the player	Partially met	This original user requirement was able to be partially met as I had only included the 1 picture of the player.
47	Must be an image on the quiz for each subject	Fully met	This original user requirement was able to be fully met as all I had to do was display the image corresponding to the question subject. E.g. if math's questions were being asked I would display the math's image
48	There must be an image on the game form for the heals, ammo and bullets	Fully met	This original user requirement was able to be fully met as all I had to do was assign the image to each of the picture.
49	There must be an image for the podium on the scoreboard form	Not met	This original user requirement was not able to be as I have removed the podium and in place used the grid view for viewing the leader board
50	There must be an image of an arrow pointing to a button	Not met	This original user requirement was not able to be met as I thought that it didn't fit the aesthetic of the game
51	Simple instructions must be underlined	Fully met	This original user requirement was able to be fully met as all I had to do was enable the property for underlining text
52	There must be instructions on how to play the game	Fully met	This original user requirement was able to be fully met as all I had to do was type out how to play the game
53	There must be images of game controls	Fully met	This original user requirement was able to be fully met as all I had to do was display WASD, arrow keys and a space bar for a visual representation of how to control the game
54	There must be GIF as an image of the game	Fully met	This original user requirement was able to be fully met as all I had to do was display the gif
55	There must be difficulty images	Fully met	This original user requirement was able to be fully met as all I had to do was display images relative to the difficulty level
56	There must be description of difficulty	Fully met	This original user requirement was able to be fully met as all I had to do was describe what each difficulty contained
57	There must be correct naming of buttons	Fully met	This original user requirement was able to be fully met as all I had to do was make sure the buttons name was accurate to what it is meant to do.

User Evaluation Degree of fulfilment 51/57 = 89.47%

The overall degree of fulfilment has shown to have achieved a score of 51/57 leading to a 89.47% degree of fulfilment, this shows that I have 51 fully met original user requirements. I believe that this percentage demonstrates my plan and execution throughout my whole project. In my evaluation of my user requirements I had 3 partially met user requirements consisting of images and username uniqueness, reflecting upon these degrees of fulfilment I should've allocated more time towards ensuring more strict conditions for the usernames to satisfy the degree of fulfilment. Finally I had 3 original user requirements not met these consisted of changing the way the leaderboard is displayed and an image not displayed. I had removed the podium as I felt it would be an improvement If I had displayed all of the users scores

with their respective placement and as for the arrow I had removed I felt that it didn't fit the aesthetic I had set for my program, reflecting upon this it shows that I was able to make consideration about the user experience.

Strengths

The strengths of my game and quiz have been highlighted and are able to be split into 3 main categories : **Design** – looks, positioning, and audio, **Functionality** – playing the game and quiz, and gameplay loop and Finally **Educational Element**.

One of my programs strength is the **functionality**, this part I am most proud of is the ability to play the game on a loop but limited this was able to be possible by using a decrement system with the lives and continuing as long as they still had more than 0 lives, this to me feels like a big strength that has been achieved and has improved the functionality of my program greatly, this part of the task wasn't asked by me but I instead did it and thus shows my ability to think creatively, and anticipate the needs of the user. Another part of the functionality I am proud of is the ability to play the game and quiz, starting with the game this functionality was able to be achieved by detecting WASD, arrow keys and space bar input and moving/ shooting the character in the correct way as well as having a scoring system implemented, this to me shows me that I have built the core of the functionality well as well as that from the users' perspective they found the game fun and playable as the clients had confirmed that my program I developed would be a useful tool in their classroom, now with the quiz this functionality was able to be achieved by implementing a multiple choice answering system along with a typed response for math's questions only this again wasn't asked of me as referring back to what I planned was having a multiple choice for all subjects but I changed it to only be a typed response for the math's subject this shows my ability to go beyond what I had tasked myself as well as showing my ability to adapt and evolve my ideas as the development of my program continued.

Another one of my programs strengths would be the **design** of my program. Reflecting back to my hand drawn UI's I had added in elements that have shown to not exist in my digital UI's these could consist of the positioning of elements or the addition or removal of elements as this was done to better suit the needs of the user from the feedback they had given me, this shows my ability to iterate and refine my final design as well showing my ability to respond to the feedback I had received.

Another one of my programs strengths would obviously be the **Educational Element** of my quiz. I am very proud that this category was able to become a strength in my program, as referring back to the client summary they had said that the educational aspect of my program has show to be very successful evident by the high or majority scores from the questionnaire, from the users' end they had also thought that the educational aspect was successful whilst others did not, but the positive outweighed the negative. This strength shows my ability to have the time and dedication to create a educational game.

Weaknesses

The weaknesses of my game and quiz have been highlighted and can be split into 3 main categories : **Design**, **Functionality** and **Educational Element**

One of my programs weaknesses is the design aspect of it, particularly the hand drawn UI designs as from the users' feedback I was told about the colour scheme that I had used to design them, reflecting upon this feedback I had made these changed in the digital UI designs, although I had received this feedback and changed my UI's this was still weakness that was found. Along with the weakness of the hand drawn designs the feedback I had received from them where shown to be a mixed set of feedbacks with some users saying that I had added an important function to the UI which they liked, whilst others had said that I didn't implement the functions they had though of and thus notified me of the weakness in my UI. Although negative comments have been in the summarize I have been given I have been able to correct these and this shows my ability critically evaluate my own designs, iterate them and refine them after receiving the feedback.

Another one of my programs weaknesses is the functionality, as reflecting upon what the history client had said they have told me that my timer element I had implemented into my quiz hindered their students learning experience as reflecting upon this decision now it wasn't the best appropriate choice I could've made to the subject as after completing my research I found out that the history subject requires a lot of reading and thinking, and I especially know this for a fact as I studied GCSE History. Reflecting upon the users' response for how they found the difficulty is a key detail I had found out as it tells me that my difficulty functionality isn't a completely robust system, and thus additional question(s) would've needed to be asked to have obtained a completely robust system. Unfortunately unlike the previous weakness I was not able to correct this weakness as I didn't have enough time during development.

Another one of my programs weaknesses is the educational element of my quiz, as reflecting upon what the history client had stated about the educational element it shows my program IS suitable for a long-term teaching aid, but this client has said that my program isn't suitable for them which tells me that I haven't been able to make all the subjects an equal learning experience, I believe the reason for this lies within history itself as history is always happening and updated reflecting the reason why the educational element has a weakness, as if the curriculum changes at Saint Patrick's Grammar School then only 3 of my other subjects will be usable

Improvements

An improvement I would add to my design would be changing what my whole program is based upon, as reflecting upon my program is can be seen that it is based off the show 'Doctor Who' I realized that is isn't the best choice for my targeted users' as they would be from the ages of 11 – 14 and this show is more know over in England rather than Northern Ireland, so reflecting upon the feedback I had been given by the client a more suited show to have used would be the Irish show Derry Girl's as this is a show set in time of the troubles in Northern Ireland. I believe this improvement would've made the design of my program more successful achieving better question answers and more positive feedback.

Another improvement I would make is the functionality of the quiz as reflecting back to the weaknesses I had identified the history client had said my timer element had hindered the learning experience, if I was given more time during development I would add in a function that would check the subject chosen and if it was history it would double their time, so if easy was chosen originally they would have 30 seconds but instead they would have 60 seconds this is the same for normal and hard resulting in times with 120 seconds and 180 seconds. I believe this would fix this weakness but also give an unfair advantage towards the history subject.

Another improvement I would make is towards the functionality of my quiz again, as reflecting back on what the history client said I believe that the multiple choice quiz could be changed/ improved, I would do this by having history has a typed responses as well as multiple choice as this would give the user more flexibility in how they answer as I now see that the original format didn't suit the nature of the history subject.

Another improvement I would make would be the educational aspect of my program as from looking at the users' answers for question 3, 2 of the users had said that it was sort of helpful this shows that I can improve my educational content as well as that I believe that my questions may become outdated in the near future for learning wise and so I would improve this issue by being able to select what content of history to learn from as I believe this will ensure the effectiveness of my educational content for history.

Another improvement I would add to my program would be ensuring that visuals of my program match what is expected of the user as in question 4 of the users questions 1 of the users had said my game wasn't visually appealing without knowing why they had said this I can only assume that it was the placement of my elements in my game.

Self-Evaluation

Analysis

When I was first tasked with creating this project I knew I had a long road ahead of me so because of this I wanted to get all of the necessary things correct around the first time as this would not only save time when it came to later stages of the development, but it would also ensure me that the foundations of my program were as strong as they possibly could. Also when I was tasked with this project I was made sure I was starting it with a clear mind set with a goal that I set to achieve.

To start this project off I began by interviewing the 4 main clients, them being history, geography, math's and science, I had asked them what problems their students where having in their class and they had all said that they were struggling with the content in their class, I then had asked them what year their students where in some said year 10, whilst others said year 8 so to account for this I researched information for the clients subjects along with gathering questions from years 8, 9 and 10. After completing my research I had talked with the clients to ensure that I had collect the correct information for the questions along with how I would approach the educational aspect of the program, after a while we all came to a conclusion that I would approach the educational aspect with a game that once you lose a life you get prompted with questions that are relevant to the subject you are struggling with, this would be consisting of a game and quiz within my program.

For the requirements of my game I sat with all of the clients each and asked what they wanted my program to do all of the clients had given me great user requirements although some could've been best suited together, this is evident in the 1st draft requirements 1 and 2, after gathering all of the requirements I sat down and began refining them to suit the program I planned to built and after I had refined them to the point I was happy with I sat down with the clients again and showed them the refined requirements and they were all happy with the out come of the requirements.

A skill I have obtained during this stage in my development has been to understand how a game can be developed from only using my imagination, this skill I don't have any experience with as it is the first time I am using this skill. A skill I have used once again is obtaining information effectively and correctly as this skill was previously obtained last year whilst researching for my coursework for technology and design. These skills that I have obtained or used have taught me that I am capable of thinking creatively as well as independently, and that I am able to take an idea starting with the concept of it to a fully developed program without having to rely on external inspiration.

Design

Starting with the design stage I believe it is the most critical as this is how everything in my program was going to be based off of this included the positioning of elements, the use of elements and the overall look of the designs. This part of program will consisted of 4 stages which includes: the initial design, feedback, wireframe and the improved solutions. These stages helped me reduce the possibilities of having major parts that are scraped.

Firstly the initial designs, I sat with clients and asked them how they would all have the layout of my program, I had given the clients some time to do research of their own so they could find layouts of programs that they liked and after 3 – 5 days they had all reported back to me showing me their findings, what they liked most and what they thought I could do to improve the layout that they had found. After I had found out and gathered all of the information I needed I began to work on a 1st draft set of storyboards of my program using pen, paper and colours to best illustrate the requirements I had previously been given to the best of my abilities. Although this part took a week to complete this part had resulted in the creation of the foundations of the design of my program, which has thus gave me a clear and structured starting point to build upon as I moved into later stages of my development.

After completing my initial designs I had shown them to the clients and all of them had went through each of the story boards and after they had all reviewed the story boards some or all of them had given me feedback on what I could add, what I could remove or what they liked or disliked. After I had received the feedback, me and the clients had went through my original list of user requirements and using the feedback I created a new list of requirements to better suit my program by adding new requirements or removing some old requirements that were not needed, after completing this, it left me with a complete list of updated finalized user requirements and after I had sorted these requirements into an importance list, not so I can prioritize development but show the client what I believe to be most important in my program.

After receiving my feedback I had went and created a wireframe with the clients to ensure that they were happy with where each of the forms would bring you. This had resulted in a complete overview in what way my program will function path wise.

Using the feedback I had received and the finalized user requirements, I created a final set of story board designs. With these the efficiency of completing them was much faster this time round as I not only had a clearer vision of what I wanted my design to look like but understood what was asked of me by the clients.

After finalizing the final draft story boards I then went on to write pseudocode for all of my forms as it explains to the clients how my program functions, as well as this it can tell me if I made any issues during the development process.

Development

The development part of my program was definitely the most time consuming, intriguing, intensive, learning experience I've had as up until this point in my coding hobby I had only experienced procedural programming in C# as with this task I had been set it taught me about a new way of coding called object-oriented programming.

When first beginning this project I was very unsure of how I would be able to use this new way of coding as well as that the size of the project gave me a fright, although this wasn't as big as a fright due to my previous skill in learning procedural programming in C#. When it came time to using this new way of coding I began to look online on how to use this way of coding and due to this learning period it took me 1 to 2 weeks to fully understand the concept of object-oriented programming.

After my learning period I began to use the contents of what OOP contained such as classes and objects which I began to be more fond of as I greatly enjoyed implementing this new way of coding in my project, with using OOP it made my code able to be reusable which allowed me to save time. Where I began development of my program was with the log in form as this is where I first started using object-oriented programming in my program and did this to further build repetitiveness for coding in object-oriented programming.

During the development of my program a good portion of the time was spent working on the game and quiz as I had to integrate the classes for the difficulty which gave me a lot of trouble when doing so. This was able to be sorted as with the help of research and YouTube tutorials I was able to figure out how I could implement the objects in my classes into the quiz, but I found after how exhausted I was so instead for the game I broke up the coding time as to give my self a break and always come back with a fresh clean mindset. Early on in the development I had put the time into ensuring that the login and sign in had the correct barriers in place to ensure the username and password typed matched my idea of a username and password.

After completing this I moved onto the core game and quiz mechanics, as for the enemies I had to store them into a list of pictureboxes and give them an amount and if a random number set in the game wasn't equal to the enemy then 1 would be added on and a enemy would spawn, initially I had this coded as a forloop inside of my restart game method which worked how ever when I introduced a variable it didn't work as no enemies had spawned, but this was later fixed by making it a while loop inside of the game timer event. This approach was able to be completed flawlessly with no other issues arising because of it.

Whilst coding my game I differed away from what was originally tasked of me, make a game that when I lose a life a question appears and if you get it wrong you lose another life and a new question appears. Instead I took the approach of if you lose a

life a set of questions pop up and you will not know if you got them right or wrong as this to me feels like a good learning experience, as in a test you would answer questions and hand up your finished test to receive the result.

A slight issue I encountered was with the game as in it I have a custom background the issue with this is that every time a picture box moved it would have to update the entire form like a CRT does, thankfully this was able to fixed quite easily by enabling double buffering and optimizing the game form for it.

Within my program I have added in several amounts of error trapping techniques which consist of exceptions and custom exceptions. I had used custom exceptions in my form for choosing the difficulty as if it is left null then the game and quiz will not work according to standards that had been set, and I had used exceptions to ensure a text file is present.

Finally when I came to finding any bugs that had been missed most if not all of them had been found by the time my development of my program had been finished. This was able to be done by reviewing my code after making any new changes to it, running it and ensuring that it does what I have coded it to do.

Testing

Comparing testing to designing or development it was a lot more relaxed for me as I was checking all of requirements to ensure that they met the standards that they were designed to be. Going into testing I knew I would have a few failed tests but the amount of tests that had failed shocked me as I previously though I had debugged and caught all of the issues before hand. After completing it shows me that it isn't some waste of time but instead ensures you had a program that does exactly what you have designed it to do.

To carry out my test plan effectively I began my listing all of my user requirements in a table, this would be done so I would be able to test all of my requirements including my functional requirements and my non functional requirements. All of my functional requirements were tested against white box testing, whilst the non functional requirements were tested against black box testing. Whilst testing my requirements I purposefully put in incorrect or extreme data and anytime I would positive results would occur. This would usually help expose and errors that may have gone unnoticed originally.

The results of my test plan had shown to have 60 successful tests out of the 68 tests that had been conducted, how ever 8 of those tests had failed due to small mistakes made whilst coding, property not set correctly or being brought to new forms. These issues that were found during testing were able to be fixed almost immediately, along with that I documented what the code of the test looked like and had shown what the new code looked like including what made it fail the test and how I fixed the issue so it would pass another test again.

With a test plan it teaches me that not all code is correct and robust that there will be issues that aren't noticed until a program has ran. This test plan has also been able to further ensure me that I am able to debug my code when an issue arises as this will be a useful skill to have and develop upon in later years

Overview

Upon reflection of the development of my program I feel that this project I had been tasked with had been a challenge yet it has been a rewarding experience that I had undertaken in my hobby to code. I took each stage proudly and with a clear, focused mind set, I was able to design and develop a program that not only met the requirements of my clients but also it had achieved a number of goals I had set my self independently. Reflecting upon the skills I have gained throughout this project has now made me more capable and more confident in my ability to design and develop a piece of code, I am also well satisfied that I have received great amounts of feedback from clients and users and it shows a reflection of the time and effort I had invested into this project.